Business as (Un)usual: Immigrant Storefronts and Native Backlash*

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Does immigrant business creation cause natives to become more accepting of immigrants or trigger backlash? I argue that although they display evidence of economic contributions, storefront businesses also make local immigrant presence and cultural change visible. To estimate their effect, I leverage French administrative records covering more than twenty million businesses in over 34,000 French municipalities. Using a shift-share instrumental-variable design combining national trends in immigrant business creation with municipalities' industrial composition, I find that an increase in storefront exposure between 2009 and 2019 significantly increases the local vote share of the far right in national elections. Descriptive results suggest that far-right support rises more with exposure to storefront businesses associated with immigrants from Muslim-majority countries. I find no statistically significant evidence that these businesses harm natives economically. These findings suggest that for immigrants at the center of cultural conflict, even explicit economic contributions can lead to electoral backlash.

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1 Introduction

The political rise of far-right populist parties, from fringe groups to vastly successful movements winning elections in some of the world's most powerful countries, is one of the major transformations of our time. This change, which scholars and politicians still dedicate considerable energy to understanding, has not only produced lasting policy consequences, including Brexit (Algan et al. 2017; Guriev and Papaioannou 2022), but also accelerated the erosion of

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democratic norms (Levitsky and Ziblatt 2018) and deepened polarization (Bischof and Wagner 2019). Few issues have become as closely tied to this political revolution in Western democracies as immigration and the attitudes natives have displayed toward it (Ford and Jennings 2020; Cools, Finseraas, and Rogeberg 2021; Pupaza and Wehner 2023; Margalit, Raviv, and Solodoch 2025; Alrababah et al. 2025).

Scholars have devoted substantial attention to explaining natives' attitudes toward immigrants (Hainmueller and Hopkins 2014), emphasizing both economic (Scheve and Slaughter 2001; Hanson, Scheve, and Slaughter 2007; Dancygier and Donnelly 2013) and cultural factors (Kinder and Kam 2009; Hainmueller and Hangartner 2013; Maxwell 2019). These theories typically focus on the presence of immigrants and their characteristics: skills, wealth, race, language, or religion. In comparison, the decisions of immigrants have received little attention in political science. One of the fundamental economic choices facing migrants is whether to create a business instead of, or alongside, working in a salaried job. Research has shown that immigrants create businesses at significantly higher rates than natives (Azoulay et al. 2022). Millions of immigrant-owned businesses operate in Western democracies, sometimes serving as a routine and central point of exposure to immigration for native residents. Moreover, immigrant-associated businesses have been recurrent targets of far-right hostility, from political campaigns targeted at kebab shops in France to outright bans in some Italian cities. Yet, their political impact remains unexamined by social scientists.

In this paper, I argue that by shaping how natives experience immigration, immigrant businesses matter politically. The direction of their effect is ambiguous. On the one hand, to natives, immigrant businesses may represent a desirable occupation for immigrants: business owners are less likely to rely on welfare benefits and to compete with natives over salaried jobs, and they actively create consumption and labor opportunities. Exposure to businesses associated with immigrants could therefore improve how natives perceive immigrants by showcasing their contributions to host economies. On the other hand, by signaling immigrant presence and cultural change, and sometimes competing with natives economically, immigrant businesses could trigger backlash. How does local exposure to immigrant-associated businesses affect voters?

I answer this question using comprehensive business-level administrative data from France, a major economy with a long history of immigration and a rising far-right party. I classify the owners of more than 20 million businesses by probable origin from their first and last names, using an algorithm that I validate against recorded countries of birth for 118,759 individuals from national death records and against classifications by French MTurk workers. Because natives may respond to businesses they perceive as immigrant-associated rather than strictly to immigrant ownership, I interpret predicted owner origin as a proxy for perceived association with immigrants. I geocode every business and time-stamp its opening and closing dates to construct the first systematic spatial measure of local exposure to immigrant-associated businesses, covering 34,802 municipalities annually between 2008 and 2021.

I use these data to test whether and how exposure to immigrant-associated storefronts affects support for the National Rally (RN), France's main far-right party. The central empirical challenge of this paper is the endogeneity of business presence: immigrant businesses are created in response to immigrant settlement, local demand, and labor-market conditions, each of which may independently affect local support for the far right. Addressing this challenge requires finding a source of variation in exposure to immigrant businesses that is unrelated to these confounders. I do so with a shift-share instrumental variable (SSIV) design (Bartik 1991; Borusyak, Hull, and Jaravel 2022) that leverages the component of change in local immigrant-business exposure predicted by national, industry-specific trends in immigrant business creation (the *shifts*), weighted by each municipality's predetermined industrial composition (the *shares*). Under the assumption that these national trends are exogenous, conditional on baseline covariates, this approach recovers the effect of induced change in municipal exposure to immigrant storefront businesses on change in the local far-right vote share. Diagnostic checks show that the instrument is not related to pre-period change in RN support and does not significantly predict growth in native businesses, in the local foreign-born population, or in non-storefront immigrant businesses.

The estimates are consistent with a backlash effect: an approximate doubling in a municipality's exposure to immigrant-associated storefronts between 2009 and 2019 increases the RN vote share by approximately 6.6 percentage points, with a 95% confidence interval of

[3.6, 11.6] percentage points. Exposure increases the RN's vote share among all registered voters, but has no net effect on turnout and produces no statistically significant shift for other major parties. Results are robust to alternative choices of inference strategy, elections analyzed, instrument construction, and adjustment choices.

I introduce two sets of mechanisms that could explain this backlash. First, I test mechanisms based on visibility: businesses are typically visible to all residents who share a public space and often persist over time. Immigrant-associated businesses may look visually distinct and signal the presence of specific immigrant groups. Exposed natives may update their beliefs about the presence of an immigrant group, which may trigger demographic anxieties if they hold negative priors about that group. They may also simply resent the change to their physical environment. These mechanisms jointly generate two predictions: first, backlash should be stronger for storefront businesses than for less visible, non-storefront businesses. Second, it should be stronger for businesses associated with groups at the center of pre-existing cultural conflict than for businesses associated with other immigrant groups. The second set of mechanisms is about the local economic impact of these businesses. As producers, immigrant businesses may divert customers from native-owned firms or displace them. As employers, they may disproportionately hire co-ethnics, threatening natives' employment and income. If backlash reflects economic self-interest, I predict that growth in immigrant businesses should be associated with an observable degradation of local economic conditions.

To evaluate predictions from the visibility-based mechanisms, I test whether some businesses are more strongly associated with the RN vote share than others. The SSIV design does not provide enough power and specificity to test for heterogeneous effects between different endogenous exposure measures. Instead, I estimate the relationship between vote share and exposure to different types of businesses using a panel analysis of all elections between 2008 and 2021. These estimates are descriptive, since the timing and location of business creation are endogenous. I find that within-municipality change in RN support is more strongly associated with exposure to storefronts associated with Muslim-majority countries, the group at the center of cultural conflict in France. The relationship is positive and significant for Muslim-associated storefronts but negative and not significant at the 5% level for Muslim-associated

non-storefront businesses, which generally provide less public visibility. The relationship is null or negative for other immigrant groups. These descriptive patterns are consistent with the visibility mechanisms. To test the economic mechanism, I implement the SSIV design on local economic outcomes. Although I cannot rule out economic degradation, I find no consistent evidence of material harm: estimates of the effect on native businesses, income, industry diversity, and property values are not statistically distinguishable from zero. Unemployment among non-immigrant residents, if anything, declines. The results are therefore more consistent with the visibility mechanisms, though the evidence cannot rule out distributional losses, perceived economic threat, or damage that is not captured by the economic measures I have available.

This paper makes two primary contributions. First, it advances the literature on the rise of far-right populist parties (Golder 2016; Berman 2021). Prior work shows that immigrant presence and demographic shocks tend to increase far-right support (Hopkins 2010; Pupaza and Wehner 2023; Alrababah et al. 2025; Margalit, Raviv, and Solodoch 2025). I show that the presence and characteristics of immigrants explain only part of their political impact: their economic activity, a product of their own economic decisions and opportunities, also influences natives' attitudes and political behavior. Business creation, in particular, produces variation in visibility that can affect far-right voting even conditional on immigrant presence. To my knowledge, this is the first paper to systematically measure local exposure to immigrant economic activity, rather than residential settlement or professional exposure, and to estimate its specific electoral effect across an entire country. More broadly, this paper highlights how routine economic activity at the local level can shape aggregate political outcomes.

Second, the paper contributes to theories about how natives respond to immigration by showing that the mode of exposure can shape its political consequences. The literature on immigration attitudes has emphasized labor-market competition (Scheve and Slaughter 2001; Mayda 2006), fiscal threat (Facchini and Mayda 2009; Hanson, Scheve, and Slaughter 2007; Cavaillé and Ferwerda 2023), sociotropic concerns (Dancygier and Donnelly 2013), and cultural concerns (Hainmueller and Hiscox 2007; Kinder and Kam 2009; Schildkraut 2014; Hainmueller and Hangartner 2013; Tabellini 2020), while work on intergroup contact has asked

when exposure to out-groups reduces or intensifies prejudice (Allport 1954; Hopkins 2010; Ellison, Shin, and Leal 2011; Enos 2014; Paluck, Green, and Green 2019; Mousa 2020; Nathan and Sands 2023). Immigrant businesses sit at the intersection of these perspectives: they expand locally available goods and services, but also make immigrant presence visible and tangible in public space. My results suggest that when storefronts are associated with culturally salient immigrant groups, even in the absence of observable economic degradation, backlash may dominate.

2 Theory

2.1 Attitudes Toward Immigrants and Immigrant Businesses

Research on the determinants of native attitudes toward immigrants has distinguished between a set of economic factors, including skills (Scheve and Slaughter 2001; Mayda 2006), income and concerns over fiscal costs (Hanson, Scheve, and Slaughter 2007; Facchini and Mayda 2009; Cavaillé and Ferwerda 2023), and sociotropic beliefs about the impact of immigration (Dancygier and Donnelly 2013), and a set of cultural factors, including concerns over national identity, ethnocentrism, and cultural distance between natives and immigrants (Kinder and Kam 2009; Hainmueller and Hangartner 2013; Schildkraut 2014). These attitudes also vary with the education of natives (Hainmueller and Hiscox 2007; Cavaillé and Marshall 2019) and the relative political power of native and immigrant groups (Dancygier 2010).

Both primary sets of explanations focus on the number of immigrants in a territory and their individual or group characteristics such as income, education, race, or religion. Scholarship has paid less attention to the economic decisions made by immigrants across these groups, though these decisions shape their lives and roles in their host country. One such decision is whether to earn an income by creating a business rather than, or alongside, working in a salaried job. While economists and sociologists have studied the individual determinants of this decision, including risk profiles, labor-market discrimination, and ethnic networks (Lucas 1978; Aldrich and Waldinger 1990; Fernandez and Fernandez-Mateo 2006; Fairlie and Lofstrom 2015; Auer et al. 2019; Kerr and Mandorff 2023), less is known about its political

impact. Yet, millions of immigrant-owned businesses exist in Western democracies. Scholars have in fact shown that immigrants create businesses at higher rates than natives, contributing to economic activity and innovation in their host countries (Portes, Haller, and Guarnizo 2002; Azoulay et al. 2022; Kerr 2017; Sequeira, Nunn, and Qian 2020). Despite this, immigrant businesses are recurrent targets of far-right hostility, as seen in local campaigns against kebab shops in France (Laurent 2015), denunciations by national politicians in Italy (Robinson and Mezzofiore 2018), and riots in Britain and Germany (Shuster 2016; Gill 2024).

I argue that business creation matters politically because it shapes how natives experience immigration. For natives, contact with a given immigrant group will be experienced differently depending on whether its members work in salaried jobs or operate storefront businesses, which tend to be more persistent and more visible to the whole community. The relationship studied in this paper is therefore distinct from the effect of residential exposure to immigrants and demographic change, which have been examined extensively in prior work (Hopkins 2010; Ellison, Shin, and Leal 2011; Enos 2014; Paluck, Green, and Green 2019; Nathan and Sands 2023). Conditional on immigrant presence among residents, how does variation in local exposure to immigrant-associated businesses affect the political behavior of native residents?

Natives form imperfect beliefs about whether a particular business is associated with immigrants. Natives may successfully infer ownership when business owners serve customers directly, when businesses display identity markers, or when employees are hired through co-ethnic networks (Elliott 2001; Fernandez and Fernandez-Mateo 2006). However, even when natives know who owns a business, immigrant association is a function of natives' own biases and beliefs. Second- or third-generation business owners may be perceived as immigrants because of their race or ethnicity, whereas foreign owners from neighboring countries who share ascriptive characteristics with the local majority may be perceived as natives. The argument developed in this paper relies on the perception that a business is immigrant-associated rather than on formal foreign or foreign-born ownership.

Exposure to immigrant-associated businesses can affect natives in opposite directions. On the one hand, the goods, services, and jobs these businesses provide could improve natives'

views of immigrants, and commercial interactions could reduce prejudice as predicted by the contact hypothesis (Allport 1954). On the other hand, their visibility and their competition with native firms could heighten perceptions of cultural or economic threat. I predict that backlash is likely to dominate in aggregate for two reasons. First, potential economic benefits from these businesses are likely to be perceived primarily by their customers and employees, whereas storefronts are visible to all residents, including those who never shop there. Second, brief commercial encounters are unlikely to meet the conditions under which contact reduces prejudice, and casual exposure to an out-group can instead increase exclusionary attitudes (Enos 2014).

Hypothesis 1 (backlash) *An increase in exposure to immigrant-associated storefront businesses increases the local vote share of far-right parties.*

Note that the outcome of both this hypothesis and the ones below is variation in the far-right vote share. This serves as a substantively important and, importantly, observable implication of anti-immigrant sentiment. This may work in at least two ways. First, backlash rooted in day-to-day exposure to immigrant-associated businesses may raise the salience of immigration. Among some voters, this should activate latent restrictionist attitudes that did not previously form the basis of vote choice (Hainmueller and Hiscox 2007; Hopkins 2010; Maxwell 2019), which is likely to benefit far-right parties that have historically owned the issue (Meguid 2005). Second, backlash may cause natives previously less attentive to immigration policy to favor restrictive policies aligned with far-right platforms (Key 1955; Ansell et al. 2022; Gidron 2022). In the two sections below, I present specific mechanisms that may generate backlash, and derive testable empirical implications from each.

2.2 Visibility Mechanisms

Exposure to storefront businesses generates visibility in a way that differs from exposure to immigrant residents in three respects. First, whereas immigrant residents come in repeated contact mostly with members of their own networks, a storefront business occupies commercial premises visible to all natives who share a public space. Second, storefront businesses

often persist over the years and affect the visual identity of a community. Finally, they often carry identity markers that natives may interpret as revealing the presence of immigrant residents or their specific origin. These identity markers can be the goods sold by a business, its signage, or the language or ascriptive characteristics of its owner or staff. Because immigrant-associated businesses can be found in native-dominated neighborhoods and not solely immigrant enclaves, they may be, for many natives, one of the most routine points of contact with immigration.

That visibility can then shift natives' attitudes in at least two ways. First, businesses signal otherwise unseen immigrant presence: natives exposed to these signals may revise their beliefs about the size, or growth, of local immigrant communities. This may trigger feelings of group threat, as existing work shows about contact and demographic change (Green, Strolovitch, and Wong 1998; Enos 2014). This belief update may be stronger when prior perceptions of immigrant presence are noisy, as Alesina, Miano, and Stantcheva (2023) have shown is often the case in Western countries. Second, storefront businesses affect the visual environment of a community. Independently of demographic change, natives who draw value from an environment visually reflective of their preferred identity may resent this physical change. The literature has previously shown that voters can act electorally on other instances of undesirable changes to their environment (Stokes 2016; Schildkraut 2014; Faury 2024).

Although these two visibility-based mechanisms stem from different micro-foundations, they jointly generate two predictions about which businesses I expect to cause electoral backlash. First, because these mechanisms operate through visibility, backlash should be stronger for businesses operating as storefronts than for less visible, non-storefront businesses operating from offices, homes, or industrial facilities.

Hypothesis 2 (visibility) *Exposure to immigrant-associated storefront businesses increases far-right support more than exposure to immigrant-associated non-storefront businesses.*

Second, natives are more likely to react to signals that an immigrant group is present when that group is at the center of a broader cultural conflict, or when their prior perceptions of that group are negative. Although natives may blame changes to their visual environment on less-salient groups, businesses associated with such groups are less likely to trigger backlash.

Hypothesis 3 (group salience) *Exposure to businesses associated with immigrants at the center of cultural conflict increases far-right support more than exposure to businesses associated with other immigrant groups.*

2.3 Economic Mechanisms

An alternative explanation for backlash would be that immigrant-associated businesses cause economic damage to natives. As producers of goods and services, immigrant businesses may reroute customers away from native businesses. In the extreme, this may cause the displacement or bankruptcy of some native businesses. Native producers may therefore support anti-immigration parties out of self-interest, and this desire may extend to members of their network or community. As employers, some immigrant business owners may predominantly hire co-ethnics (Fernandez and Fernandez-Mateo 2006). Natives may perceive this as a threat to their employment and income. This mechanism would be consistent with political economic theories of attitudes toward immigration (Scheve and Slaughter 2001; Hanson, Scheve, and Slaughter 2007; Cavaillé and Ferwerda 2023). As in other contexts, natives exposed to immigrant-associated businesses may turn to the far right in the belief that restricting immigration would reduce these economic threats.

This set of economic mechanisms generates a distinct prediction: if backlash is explained by economic self-interest, exposure to immigrant-associated businesses should be associated with a degradation of observable economic conditions for natives. This could include the displacement of native businesses, an increase in native unemployment, a decrease in income, or a decline in property values.

Hypothesis 4 (economic harm) *An increase in exposure to immigrant-associated storefront businesses is associated with negative average changes in observable economic outcomes for natives.*

Note that there are reasons to be cautious about such economic mechanisms. First, existing evidence that economic self-interest explains immigration attitudes is limited (Hainmueller and Hopkins 2014). Second, research has shown that immigrant entrepreneurs often

specialize in industries and products distinct from natives' (Piore 1979; Massey et al. 1993; Kerr and Mandorff 2023). Immigrant-associated businesses may therefore expand the diversity of locally available goods and services without posing economic threats.

A null or negative estimate of the effect of exposure to immigrant-associated businesses would suggest, respectively, that exposure is too weak a factor to affect voting behavior, or that contact and material benefits dominate in aggregate. In Section 5, I first estimate the net effect of exposure to immigrant-associated storefronts on far-right support. In Section 6, I then examine whether relationships between far-right vote and business visibility, owner origin, and local economic outcomes are more consistent with the visibility-based or economic mechanisms. Before doing so, I introduce the context and the data in the next two sections.

3 Context

France's immigration history includes substantial migration from other European countries as well as from its former colonies. Over the twentieth century, recurrent waves of immigration saw residents of French colonies move to Metropolitan France from North and West Africa and from Southeast Asia. These trends continued after former colonies obtained independence and have since been complemented by immigration from other regions in Africa and Asia, in particular since the mid-2010s refugee crisis.

Debates relating to the integration or assimilation of immigrants have been salient for several decades and become recurrent agenda items for conservative politicians. Particular attention has been paid to Muslim immigrants. Most notably, in 2004, President Chirac's right-wing government passed a law prohibiting students from wearing conspicuous religious signs or clothing in public primary and secondary schools, including Islamic headscarves (Abdelgadir and Fouka 2020). Right-wing parties have since regularly introduced bills attempting to expand this ban.[1] The National Rally, France's main far-right party, champions opposition to immigration and to the perceived threat posed by the growing prevalence of Islam in French society. It has achieved increasingly strong electoral performance over the period, as shown

1. Recent unsuccessful attempts include a 2019 headscarf ban for school outings, a broader ban under the 2021 "Anti-Separatism" bill, and 2022 and 2025 bans of headscarves in sports competitions.

in Appendix B.2. Because debates over national identity, secularism, and integration have disproportionately focused on immigration from Muslim-majority countries, I operationalize Hypothesis 3 as a contrast between businesses associated with Muslim-majority countries and businesses associated with other immigrant groups.

In the context of this larger debate, immigrant businesses have recurrently been mentioned as cultural disturbances by voters in the news media (Bronner 2022; Makooi 2024) and have generated specific political attention. In 2015, Mayor Robert Ménard announced that he wanted to rid his municipality of kebab shops (Laurent 2015), an initiative that was followed by similar initiatives elsewhere (Calatayud 2024b, 2024a). These episodes suggest that some political entrepreneurs explicitly attempt to connect immigrant-associated businesses to cultural anxieties.

Although its history and politics make France an interesting setting to study the impact of immigrant-associated businesses, the theory discussed above should be relevant in other contexts. Its scope applies in particular to countries where businesses are often operated by immigrant groups whose origins are recognizable from storefronts, where immigration by these groups is politically salient, and where an anti-immigration party provides an outlet for potential backlash against it.

4 Data and Measurement

4.1 Data Sources

Electoral Outcomes. To measure electoral support for the far right, I use the vote share of the National Rally (RN) at the municipal level in national elections.[2] I calculate the vote share as the proportion of valid votes cast going to the RN and do not include minor far-right parties. For two-round elections, I only include results from the first round, even when the RN is present in the runoff, to avoid capturing patterns of vote transfer between rounds that vary between elections. Appendix B.2 shows that the national vote share of the RN has substantially increased over time, and with significant heterogeneity between regions.

2. See the list of elections in Appendix B.1.

Location of Businesses. Data on businesses comes from the National Registry of Enterprises (RNE), a government database that is updated automatically every time companies formally report corporate milestones such as incorporation and cessation of activities. I extract all available data on individual businesses as of July 2024 using the Application Programming Interface (API) of the French National Institute of Industrial Property.[3] All private commercial, artisanal, agricultural, and sole-proprietorship companies are listed.[4] Each company includes a list of "establishments", which are economic entities operated by the company.[5] I use establishments as the empirical units of business activity and refer to them as "businesses". In the majority of cases, a company operates only one business. Larger companies may operate several businesses.[6] For each business, RNE data records the days on which activities began and ended, the exact address, and a code referencing the industry of economic activity. I geocode all businesses using the National Address Bank API of the French government.

Industries. The activity of each business is identified by an APE code. There are a total of 732 unique APE codes. I classify businesses as "storefront" or "non-storefront" at the industry level. Storefront industries include retail and hospitality activities such as shops, restaurants, hotels, and customer-facing personal services. Businesses in these industries typically operate in street-facing premises that are visible in public space. Non-storefront businesses include back-end activities such as wholesale trade, and services typically operated from offices or industrial facilities. Appendix B.5 reports the full classification of APE codes. The main estimation strategy focuses on storefront businesses, but Section 6 reports descriptive heterogeneity estimates by visibility. The classification is not based on the physical characteristics of individual businesses, which I cannot observe. An unmarked hardware store is therefore coded as storefront, and a wholesaler with visible premises as non-storefront.

3. The registry keeps entries for companies and businesses that have ceased activity, so the July 2024 extraction recovers the population of businesses active at any point during the study period, not just those operating in 2024. Some entries may have been removed if individuals requested it for data privacy reasons.

4. Omitted entities include public-sector companies and non-profit organizations.

5. See illustration in Appendix B.3.

6. Approximately three quarters of all companies operate one business only. Less than 1% operate five or more.

Origins of Business Owners. The central argument of this paper concerns businesses associated by natives with immigration and immigrants. This association depends on natives' priors and perceptions, and on idiosyncratic characteristics of each business, including signage, business names, products offered, and the ascriptive characteristics of owners and employees. I use the business owner's origin as a proxy. I interpret it as the latent characteristic probabilistically driving, with error, these observable markers.

Although France provides uniquely rich data on businesses, identifying the origins of business owners presents a central challenge in the French context because systematic data collection on origins is restricted and data on self-reported race and ethnicity are not available at scale. Citizenship is reported for a small subset of entrepreneurs in the RNE but missing in a majority of observations.[7] A more general measure is therefore needed. The register does not provide any measure of the perceived group association, the construct of interest. Appendix B.6 briefly discusses the legal and statistical context for research on immigrant origins in France.

To address this challenge, I predict the origin of every business owner from their first and last names, using a commercial name-origin classifier supplied by *Namsor* and evaluated in prior academic research (Sebo 2023; Bursztyn et al. 2024).[8] I classify as immigrant any individual whose predicted origin is outside of Europe, the US, or Canada, as the national discourse on immigration has focused on immigrants from non-Western countries.[9] This coarse categorization alleviates concerns over within-Europe classification error. 90.8% of all businesses in the data have at least one non-missing owner name. In Appendix C.4, I show that the pattern of missing names does not seem to be of concern for the analysis.

To test Hypothesis 3, I distinguish Muslim-associated businesses from other immigrant-associated businesses. I classify a business as Muslim-associated when the owner's predicted

7. Citizenship, even if available at scale, would not be the appropriate construct: French citizens who are second-generation immigrants may be perceived as immigrants and are frequently targeted by anti-immigration rhetoric (Bherer 2023).

8. The algorithm relies on machine learning and linguistic analysis, and accounts for host-country naming patterns: a second-generation immigrant in France may carry a French first name but retain their parents' last name. *Namsor*'s "diaspora" model (used in fall 2024) weights names accordingly. Appendix C provides more information about the algorithm.

9. Businesses with more than one owner are coded as immigrant-associated if 50% or more of their owners are classified as immigrants. The company-level classification is assigned to each of its associated businesses.

country of origin has a majority-Muslim population according to Pew Research Center (2011), as detailed in Appendix C.3. Muslim-associated businesses represent approximately 78% of immigrant-associated businesses, consistent with France's immigration history.

I validate the classification in two ways, detailed in Appendix C. First, I compare the *Namsor* classification to recorded countries of birth for 118, 759 foreign-born individuals in France's national death records. This paper aggregates country-specific classifications up to two binary variables: immigrant-associated or not, and Muslim-associated or not. The latter is a subset of the former. At these two levels of aggregation, accuracy of the algorithm is high when compared against the death records: the algorithm distinguishes immigrant from native origin with a precision of 0.97 and recall of 0.77 (with an F_1 score of 0.86), and identifies Muslim-majority origin correctly with similar accuracy (precision = 0.93, recall = 0.89, F_1 = 0.91). Second, because the theory operates through perception, I compare the *Namsor* classification of a sample of names to classifications of the same names made by 11 online raters based in France. This serves as a direct benchmark of how a name is perceived by a French observer. Agreement is comparably high (F_1 = 0.90 for the immigrant origin, 0.84 for Muslim-majority origin). Overall, the analysis relies on the assumption that the classification error of different company owners is unrelated to geographic characteristics that could confound the causal estimates (Egami et al. 2024).

Throughout the rest of this paper, I refer to businesses whose owners are classified as immigrants, native, Muslim, and non-Muslim immigrants as "immigrant businesses", "native businesses", "Muslim businesses", and "non-Muslim immigrant businesses," respectively. Figure 1 displays the ten storefront industries that show the largest difference between immigrant ownership and native ownership. At one end of the spectrum, 15% of immigrant businesses are fast food shops, compared with only 4.8% of native businesses. At the other end, 9% of native businesses are real-estate agencies, compared to only 3.5% of immigrant businesses. Appendix B.7 provides more detail on the industries most represented among immigrant businesses (Figure B8) and on the most prevalent origins among immigrant business owners (Figure B9).

The presence of immigrant businesses is not simply a function of the presence of immigrant residents. Figure B6 in Appendix B.4 shows that while the proportion of immigrant-

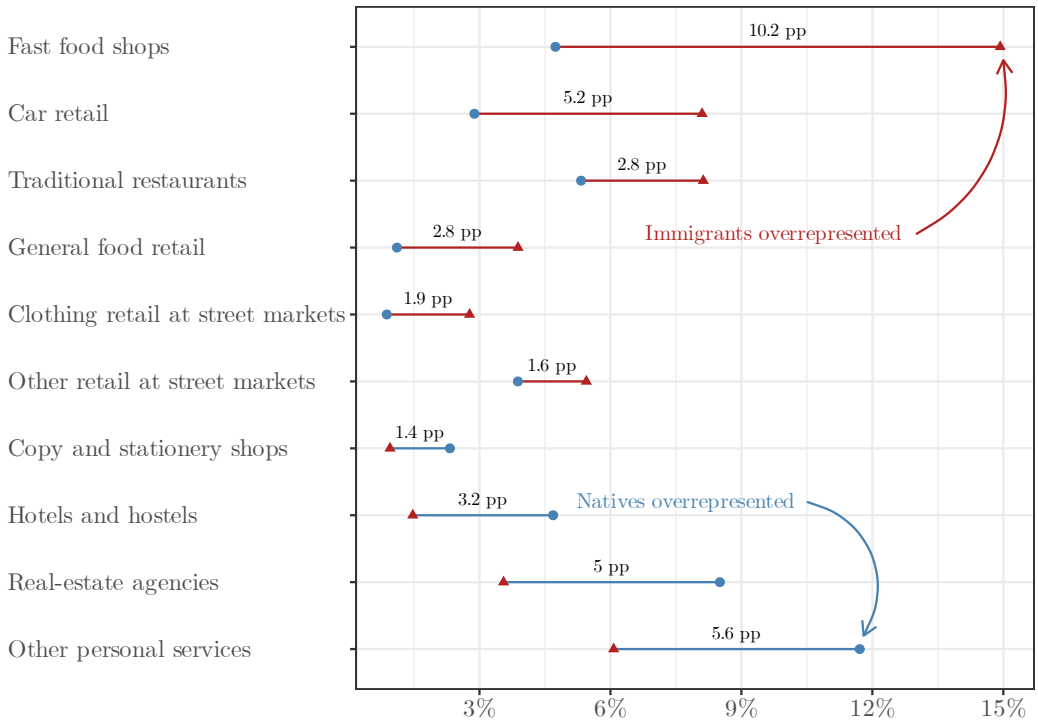


Figure 1: Ten Industries with Largest Differences between Immigrants and Natives

Note: Data from January 2024. Blue circles represent the proportion of all native storefront businesses operating in a given industry in January 2024. Red triangles represent the proportion of all immigrant-associated storefront businesses in that same industry. The segment between the two points illustrates the difference in specialization.

associated businesses is strongly associated with the foreign-born share of the population, substantial variation exists. Across immigrant residency levels, there is a wide range of immigrant-associated business presence.[10] Figure B7 shows that the share of immigrant-associated businesses has grown at a significantly faster rate than the share of foreign-born residents. Some, but not all, of this variation is a mechanical result of the fact, mentioned above, that some immigrant-associated businesses are owned by second- or third-generation residents.

Other Variables. Data for additional municipal-level variables, including employment, education, population, death records, and the share of foreign-born residents, come from the French National Institute for Statistics and Economic Studies (Insee). Geographic data on municipal borders, infrastructure, and roads come from the French National Institute of Geo-

10. The literature on ethnic entrepreneurship suggests that while immigrant businesses may find larger markets in ethnic enclaves, they may also find fiercer competition (Fairlie and Lofstrom 2015; Kerr and Mandorff 2023). Moreover, entrepreneurs in the most represented storefront industries are likely to seek markets beyond their own enclaves, given native demand for these goods and services.

graphic and Forest Information (IGN).

4.2 Measure of Exposure

Contact with immigrant-associated businesses is the treatment of interest. I predict that natives should update their attitudes by *being exposed to* (seeing, purchasing from, hearing about) businesses in their community. I conceptualize a municipality's exposure to immigrant-associated businesses, the construct of interest, as the number of businesses in and around that municipality for which 50% or more of business owners are classified as immigrants, with a distance decay as businesses get farther from the border of the municipality's downtown area.[11] I do not restrict exposure to businesses located within a municipality's boundaries, because residents are not confined to their municipality of residence and often interact with businesses in neighboring municipalities, in particular when few or no businesses are operating in their own municipality. The weights capture the idea that being exposed to businesses that are closer to a municipality matters more to this municipality's residents. I use the log of this sum of weights to account for the right skew of the distribution.

For all storefront businesses $i \in 1, \ldots, K_{m,t}$ within a 10 km distance of municipality m at time t, where $G_i = 1$ if business i is classified as immigrant-associated, exposure is formally defined at the level of municipality m at time t as:

$$\text{Exp}_{m,t} = \log\left(\sum_{i=1}^{K_{m,t}} w_{m,i} \mathbb{1}\{G_i = 1\} + c\right) \quad \text{with} \quad w_{m,i} = \frac{1}{1 + d_{m,i}} \tag{1}$$

where c is an offset for the logarithm such that zero-exposure municipalities are not dropped. The baseline specification uses d measured in units of 100 meters and $c = 0.1$, but Sections F.3 and F.4 of the appendix examine these choices.

I compute the measure of exposure for every year between 2008 and 2021 to create a municipality-year dataset for 34,802 municipalities and over 20 million businesses. Figure 2 illustrates how businesses are accounted for in the calculation of exposure for a given munic-

11. The downtown area is estimated with density-based clustering using the municipality's buildings. This ensures that while every business in the center of the municipality is equally weighted, businesses located in the outskirts of a municipality and away from residential buildings receive lower weights.

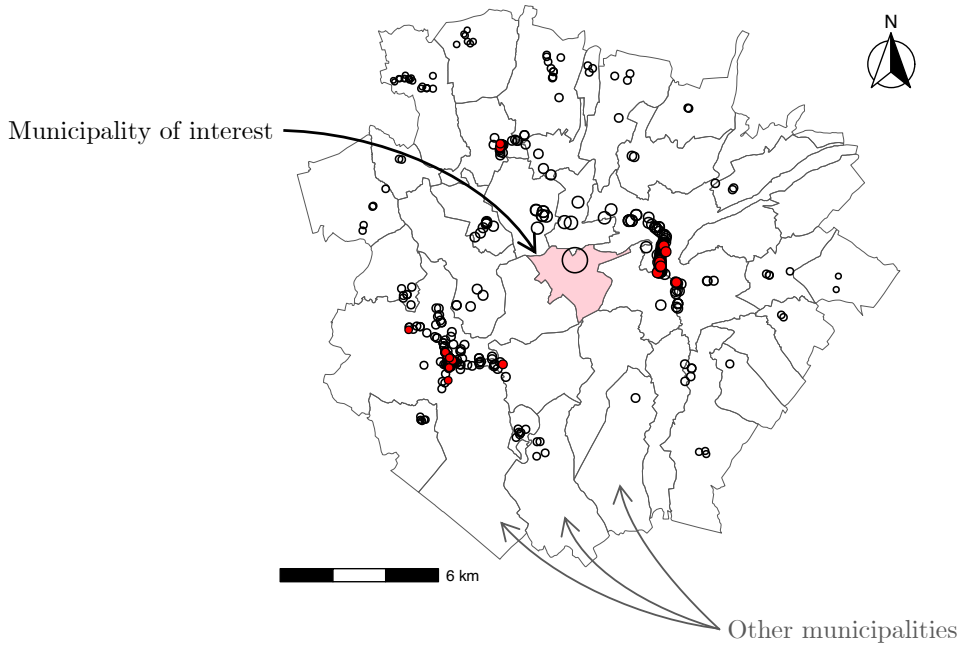


Figure 2: Contribution of Each Business to the Measure of Exposure

Note: Contribution of each business to the exposure measure for one municipality (Pretin, Jura), shaded in pink, in 2024. The borders of surrounding municipalities are displayed around it. Each symbol represents a business. Its area is proportional to that business's weight $w_{m,i}$ and thus smaller for businesses located farther away from the municipality. Black circles represent native businesses, and red dots represent immigrant-associated businesses.

ipality: $\text{Exp}_{m,t}$ is equal to the logged sum of the red weights and the offset c. The mean and median values of exposure (unlogged) are respectively 16.5 and 1.16 immigrant businesses, reflecting the right skew of the data. Average total storefront exposure, including native businesses, is 82.4.

The descriptive analysis in Section 6 includes specific exposure variables such as exposure to Muslim-associated businesses. These measures are constructed similarly to Equation 1, but with different subsets of businesses. For instance, exposure to Muslim-associated businesses replaces $\mathbb{1}\{G_i = 1\}$ with $\mathbb{1}\{M_i = 1\}$, where M_i takes a value of one if the business is classified as Muslim associated. Another variable is exposure to non-storefront businesses. This measure replaces all storefront businesses $i \in 1, \ldots, K_{m,t}$ by all non-storefront businesses $j \in 1, \ldots, K_{m,t}^{\text{non-storefront}}$.

5 Research Design

5.1 Identification Strategy and Instrument Construction

The main empirical goal of this paper is to estimate the effect of an increase in exposure to immigrant-associated storefront businesses on support for the far right at the municipal level. However, observed changes in exposure are not exogenous: immigrant entrepreneurs create businesses in response to immigrant settlement, local demand for goods and services, labor-market conditions, and other pre-existing economic and political forces. These factors may in turn affect support for the far right independently of businesses.

Addressing this challenge requires identifying an exogenous source of variation in exposure to immigrant businesses. I do so by leveraging a shift-share instrumental-variable (SSIV) design that uses the component of local change in immigrant-business exposure generated by national industry-specific trends and municipalities' pre-existing industrial composition. The specific quantity I estimate is the effect of change in exposure between 2009 and 2019 induced by these national trends on the local change in the RN vote share between the 2009 and 2019 elections. In this section, I describe the construction of the instrument and the identification assumptions.

A shift-share instrument (Bartik 1991; Borusyak, Hull, and Jaravel 2022) is the weighted average of a set of shocks (the *shifts*) common to all units, where the unit-specific weights (the *shares*) measure differential exposure to those shocks. In my application, the units are municipalities, the shocks are national changes in immigrant-business activity across industries, and the shares capture municipalities' predetermined industrial composition. I first define the instrument formally, and then propose an intuitive interpretation. For municipality m, I construct the instrument as

$$Z_m = \sum_{j=1}^{J} S_{m,j} \widetilde{G}_j^{immig}. \tag{2}$$

$S_{m,j}$ is the share of municipality m's pre-treatment exposure accounted for by businesses in industry j. These shares are measured in 2005, before the treatment and outcome periods, and

calculated across all local storefront businesses (native and immigrant). They are therefore not immigrant-specific. Shares are adjusted to reflect distance to the municipality, like exposure in Equation 1, and sum up to one within each municipality. For industry j in municipality m:[12]

$$S_{m,j} = \frac{\sum_{i=1}^{K_{m,j,2005}} w_{m,i}}{\sum_{i=1}^{K_{m,2005}} w_{m,i}} \tag{3}$$

The shocks $\widetilde{G}_j^{immig}$ are the residualized national growth between 2009 and 2019 in the number of immigrant-associated businesses in each industry j. I first measure raw growth G_j^{immig} using the Davis-Haltiwanger-Schuh symmetric growth rate (Davis, Haltiwanger, and Schuh 1996).[13] I then residualize and recenter the shocks to enhance the plausibility of the identifying assumptions described below (Borusyak, Hull, and Jaravel 2022). First, because macroeconomic changes affect businesses owned by both immigrants and natives, I regress national immigrant-business growth on national native-business growth as follows:

$$G_j^{immig} = \alpha + \beta G_j^{native} + u_j. \tag{4}$$

I use the estimated residual shocks $\widehat{u}_j$ rather than raw immigrant-business growth G_j^{immig} to remove the linear component shared with native-business growth. This reduces the extent to which the shocks reflect general industry growth or decline and makes them more specific to immigrant-associated businesses. Second, I recenter these residual shocks within quintiles of baseline industry size:

$$\widetilde{G}_j^{immig} = \widehat{u}_j - \overline{\widehat{u}}_{q(j)}, \tag{5}$$

where $\overline{\widehat{u}}_{q(j)}$ is the average residual growth among industries in a given quintile (defined by 2009 number of businesses). The resulting shocks $\widetilde{G}_j^{immig}$ therefore draw identifying variation

12. The denominator sums over every eligible storefront business, whereas the numerator is restricted to businesses in industry j.

13. The Davis-Haltiwanger-Schuh (DHS) growth rate formula is $\frac{X_{it}-X_{it-1}}{\frac{1}{2}(X_{it}+X_{it-1})}$. The DHS growth rate is bounded between -2 and +2 and is better suited for comparing growing, declining, entering, and exiting industries without dropping data or distorting the sample.

from immigrant-specific business growth across industries of similar initial size.

The instrument relies on two intuitive empirical regularities: (i) immigrant business creation and native business creation are distributed differently across industries, and (ii) industries are persistently concentrated in particular locations. Immigrant industrial specialization may reflect historical barriers to entry in some industries, as well as ethnic networks that facilitate access to industry-specific knowledge, suppliers, labor, or capital (Bonacich 1973; Kerr and Mandorff 2023). Businesses in the same industry tend to cluster geographically because they draw on common suppliers, infrastructure, specialized labor, and local demand (Ellison and Glaeser 1997; Glaeser, Kerr, and Ponzetto 2010). In Appendix D.3, I validate geographic persistence empirically by showing that the business composition of municipalities in 2000 predicts the types of businesses created a decade later. The logic of the instrument's relevance can therefore be summarized as follows: I expect exposure to immigrant-associated businesses to grow more in municipalities historically specialized in industries that happen to experience a greater trend in immigrant-specific business creation between 2009 and 2019.

I draw on the identification approach formalized by Borusyak, Hull, and Jaravel (2022). Causally interpreting the estimates requires two major identifying assumptions. First, conditional on some shock-level characteristics and municipal covariates, national industry shocks are independent of unobserved determinants of municipalities' political trajectories (*conditional shock exogeneity*). Second, the shocks affect the outcome through induced change in exposure and not through other changes caused by the same industry shocks (*exclusion restriction*). Importantly, the shares (baseline business mix) do not need to be exogenous. Inference additionally requires there to be a sufficient number of independent industry shocks. I assess the identifying assumptions using balance tests, placebo estimates, measures of shock concentration, estimates with industry exclusions, and sensitivity to additional shock-level conditioning. I summarize the key findings in this section and report full results in the appendix.

In Appendix E.1.2, I evaluate the plausibility of the conditional exogeneity assumption by estimating "pretrends" and show no significant relationship between the instrument and preceding local trends in RN support. In Appendix E.2, I show that the instrument does not

predict change in the local foreign-born population, in exposure to native businesses, or in exposure to non-storefront immigrant businesses. This suggests that the first stage is specific to the treatment of interest, which supports the plausibility of the exclusion restriction. Inference requires the number of effective shocks to be large. In Appendix E.3, I show the exposure-weighted inverse Herfindahl index recommended by Borusyak, Hull, and Jaravel (2022) implies an effective number of approximately 31.7 shocks.[14] In Appendix E.4, I show that removing food, hospitality, personal services, or automobile-related industries individually leaves the point estimate broadly stable.[15] Similarly, omitting any of the 20 industries contributing the most, one at a time, leaves the estimate stable. In Appendix F.1, I also show that results remain significant when using a finite-sample randomization inference approach.

In Appendix E.1.2, I show that the instrument is associated with some baseline municipality characteristics (unemployment, population, and distance to highways). These variables are adjusted for in the main specification, so significant relationships do not themselves threaten identification, but they highlight the importance of the conditional exogeneity assumption. In Appendix E.1.1, I show that industries with larger positive immigrant-business shocks have a lower level of immigrant ownership at baseline. This pattern is consistent with a saturation or regression to the mean dynamic: industries in which immigrant ownership was already high are less likely to see further immigrant-specific growth. This could pose a concern if industries with a higher proportion of immigrant-associated businesses at baseline are distributed geographically in a way that is correlated with municipalities' political outcomes, which would violate exogeneity. The absence of a significant relationship between the value of the instrument and the pre-period change in RN vote provides some reassurance about the plausibility of the exogeneity assumption. In Appendix E.1.1, I show that conditioning directly on industries' baseline immigrant share preserves the sign and order of magnitude of the point estimate but weakens the first stage to the point that the main test becomes inconclusive. This suggests that part of the first stage works through the component of the instrument that is correlated

14. An analysis of Rotemberg weights (Goldsmith-Pinkham, Sorkin, and Swift 2020) suggests an effective number of shocks of 15 (more details in Appendix E.4). The HHI-analogous number of Borusyak, Hull, and Jaravel (2022) is more appropriate for the conditional shock exogeneity approach.

15. Removing all consumer-facing industries jointly leaves insufficient variation in the instrument and collapses the first stage, so we know that the identifying variation is concentrated in consumer-facing sectors, though no single industry explains the result. Removing automobile industries weakens the first stage.

with industries' baseline immigrant share. The conditional exogeneity assumption therefore requires that no other variable correlated with baseline immigrant ownership confound the results. Reassuringly, the alternative Muslim-oriented instrument introduced in Section 5.3 and described in more detail in Appendix G produces similar estimates in magnitude and significance, and remains strong when conditioning on the variables on which this instrument is imbalanced, though it comes at the cost of exclusion restriction concerns.

5.2 Estimation and Inference

I estimate the effect of immigrant-business exposure using two-stage least squares. The endogenous treatment is the change, between 2009 and 2019, in the exposure of municipality m to immigrant-associated storefront businesses. The outcome is the municipal-level change in RN vote share between the 2009 and 2019 European Parliament elections. I use these two European elections to leverage a long between-election period while maximizing the availability of control variables.[16] Because European elections have been described as "second-order" elections with different logics of participation, potentially encouraging protest votes (Reif and Schmitt 1980; Hix and Marsh 2007; Hobolt and Wittrock 2011), I show in Appendix F.2 that the results are similar if I use the 2007 and 2017 presidential elections instead. The first and second stages are

$$\Delta \text{Exp}_m = \alpha_{d(m)} + \mu_{c(m)} + \pi Z_m + X_m' \lambda + \nu_m, \tag{6}$$

$$\Delta \text{RN}_m = \alpha_{d(m)} + \mu_{c(m)} + \beta \widehat{\Delta \text{Exp}}_m + X_m' \theta + \varepsilon_m, \tag{7}$$

where $\alpha_{d(m)}$ and $\mu_{c(m)}$ denote department and municipality-type fixed effects. These fixed effects adjust for department and municipality-type trends, while the first-difference specification already absorbs the fixed characteristics of municipalities. Municipality types are seven categories established by Insee that characterize municipalities' degree of urbanization and population density.[17] X_m is a vector of pre-treatment covariates, measured in 2009,

16. Several of which are only available after 2008.

17. The seven categories are: large urban centers, intermediary urban centers, small cities, urban peripheries, rural towns, dispersed rural habitat, very dispersed rural habitat.

including population in the municipality (logged) and around the municipality (logged and with a distance decay), exposure to native businesses (logged), foreign-born population in the municipality (logged) and in neighboring municipalities (logged and with a distance decay), proportion of residents unemployed, proportion holding a high school diploma, proportion holding a college degree, income per capita (logged), and distance to the nearest highway (logged). I also adjust for the 2009 RN vote share and for change in the RN vote share between the 2004 and 2009 European elections. I weight municipalities by their 2006 population, giving greater weight to exposure changes affecting municipalities with more residents. The same weights enter the first and second stages. The estimand is therefore a resident-weighted municipal effect.

Identifying variation comes from a set of national industry shocks. Inference needs to account for the fact that different municipalities are exposed (to different degrees) to this same set of shocks, creating dependencies. I therefore use the exposure-robust[18] procedure developed by Adão, Kolesár, and Morales (2019) and Borusyak, Hull, and Jaravel (2022), which is consistent when allowing for dependencies between municipalities affected by the same shocks. I report these AKM standard errors, as well as null-imposed AKM0 95% confidence sets and the exposure-robust first-stage F-statistic. In Appendix F.1, I also report Conley standard errors allowing spatial dependence between municipalities up to 50 kilometers (Conley 1999). In practice, in this setting, AKM0 inference is typically more conservative since the key dependency is at the common shock level. I also report additional randomization-inference results in Appendix F.1.

5.3 Results and Robustness

Table 1 displays the main estimates.[19] Column 1 shows that the instrument strongly predicts growth in immigrant-storefront exposure. The first-stage coefficient is approximately 0.92, and the AKM first-stage F-statistic exceeds the conventional relevance threshold of 10.

The 2SLS estimate reflects the electoral response in municipalities whose exposure grows

18. Here, "exposure" is understood in the statistical analysis sense and refers to units' exposure to the shifts in the SSIV, not to my measure of business exposure.

19. The final sample size is 34,659 because some observations are missing covariates (missing income accounts for 118 small municipalities).

Table 1: Shift-Share Instrumental Variable Estimates

	Δ Exposure (log)	Δ RN Vote Share	
	First stage	Reduced form	2SLS
	(1)	(2)	(3)
Instrument	0.917***	0.094**	
	(0.244)	(0.029)	
Δ Exposure (log)			0.102**
			(0.020)
AKM0 95% confidence set		[0.029, 0.157]	[0.055, 0.180]
Controls	✓	✓	✓
Department FE	✓	✓	✓
Municipality-type FE	✓	✓	✓
Population weights	✓	✓	✓
Outcome mean	0.635	0.205	0.205
Observations	34659	34659	34659
AKM first-stage F	14.07		

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). In Columns 2 and 3, stars are from AKM0 tests. The instrument refers to the shift-share instrument, as described in Section 5.1. Exposure (log) refers to exposure to immigrant-associated storefront businesses, as described in Section 4.2. Conley standard errors allowing for spatial dependence are reported in Table F1 of the appendix.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

as a result of their historical industrial composition. I find that it is positive and significant: a one-unit increase in instrumented logged immigrant-business exposure increases the RN vote share by 0.102. The magnitude of the estimate implies that doubling exposure to immigrant storefronts increases RN vote share by approximately 6.6 percentage points, with an AKM0 95% confidence set of 3.6 to 11.6 percentage points.[20] Unlogged exposure can be approximately understood as equivalent to the effective number of businesses in the downtown area of a municipality.[21]

In Appendix F.1, I show that the results are robust to using Conley standard errors and to an alternative permutation inference approach. In Appendix F.2, I show that the results are robust to using the 2007 and 2017 presidential elections instead of the 2009 and 2019 European elections. In Appendix F.3, I show that the results are robust to using a measure of exposure constructed with different decay weights for distance to the municipality and, in

20. With the offset value for zero logarithms, the magnitude for doubling is obtained as $\beta \times \log\left(\frac{2+c}{1+c}\right) = \beta \times \log\left(\frac{2.1}{1.1}\right)$.
21. For instance, an unlogged exposure value of 3 may mean three businesses downtown or two businesses downtown and a few in neighboring municipalities.

Appendix F.4, to a range of different offset values accounting for zeros inside the logarithm. In Appendix F.5, I show that the results are robust to reconstructing the instrument using leave-one-department-out national industry shocks, to reduce concerns that the value of the instrument in each municipality would be explained by endogenous business creation around that same municipality (Borusyak, Hull, and Jaravel 2022). In Appendix F.6, I show that the results are robust to an additional adjustment for the presence of immigrant residents in the municipalities, which uses the name-based classifier on municipal death records. This test provides reassurance that the results are not driven by the omission of second-generation immigrants in the controls.

This instrument uses the growth in all immigrant businesses as the shift entering the construction of the shift-share design. In Appendix G, I replicate the analysis with a Muslim-oriented instrument, which uses the growth in Muslim-associated businesses only, residualized against the growth of both native businesses and non-Muslim-immigrant-associated businesses. That alternative instrument's 2SLS estimates are consistent in size and significance with the main analysis above. This re-analysis serves two purposes. First, I show that this version of the instrument retains enough power to condition on industries' baseline immigrant share, and that the main conclusion is robust to this additional adjustment. This helps attenuate potential concerns about the imbalance identified in Section 5.1, though Table G7 of the appendix shows that this comes at the cost of some exclusion restriction concerns. Second, I show that a Muslim-oriented instrument cannot be used to directly test Hypothesis 3: the instrument is predictive of both Muslim and non-Muslim immigrant-business exposure, likely because Muslim and non-Muslim immigrants tend to create businesses in similar industries. As a result, the exclusion restriction for a heterogeneity analysis would not hold and the effect of exposure only to businesses associated with one specific group cannot be identified that way.[22] The following section instead examines whether descriptive panel data evidence is consistent with Hypothesis 3 as well as the other mechanisms introduced in Section 2.

Two additional electoral outcomes help understand this effect. First, in Appendix H.1, I show that exposure also increases the RN's share of registered voters, as opposed to the

22. The issue is similar for a non-Muslim-oriented instrument.

RN's share of valid votes cast in the main analysis, but does not significantly affect turnout. Together, these estimates suggest that RN gains may come from vote switching rather than from the mobilization of new voters, though I cannot rule out that the null average effect on turnout could conceal changes in turnout by different groups offsetting each other. This picture is consistent with theoretical expectations: the beliefs and preferences of some voters become more aligned with the RN platform and the salience of immigration increases, activating anti-immigration attitudes among some voters and affecting vote choice. Second, in Appendix H.2, I report estimates for other major political parties. None of the estimates of interest reach statistical significance at the 5% level, but the coefficients for the Far Left and Socialist Party, which promote relatively more inclusionary immigration policies, are negative. This points to no clear single loser: RN votes may instead come from a variety of left-wing parties, as well as other minor parties.

6 Mechanisms

The estimates from Table 1 are consistent with a net backlash effect of increased exposure to immigrant-associated businesses (Hypothesis 1). In this section, I test observable implications of the visibility and economic mechanisms introduced in Section 2. First, I use a municipal-level panel dataset to examine, descriptively, whether the relationship between exposure and RN vote is stronger for visible businesses (Hypothesis 2) and for businesses associated with Muslim-majority origins (Hypothesis 3). Second, I use the SSIV instrument to test whether increased exposure worsens local economic conditions (Hypothesis 4).

6.1 Which Businesses Are Associated with Electoral Backlash?

Storefront businesses make immigrant economic activity more observable than businesses operating from offices, homes, or industrial facilities. Hypothesis 2 therefore predicts that backlash should be concentrated among storefront businesses. In France, immigration from Muslim-majority countries has been particularly salient in political debates over national identity, secularism, and cultural integration. Hypothesis 3 accordingly predicts a stronger

association for Muslim-associated businesses. To examine whether changes in RN support are more strongly associated with exposure to visible businesses associated with Muslim-majority origins, I estimate the following equation using a municipal-level panel dataset including all elections that took place in France between 2008 and 2021. I estimate:

$$RN_{m,t} = \alpha_m + \lambda_t + \boldsymbol{\beta}' \mathbf{Exp}_{m,t}^{1,2,3} + X'_{m,t}\gamma + \varepsilon_{m,t}, \tag{8}$$

where $\boldsymbol{\beta}$ is a vector of the coefficients of interest, and $\mathbf{Exp}_{m,t}^{1,2,3}$ are the respective exposure variables for the following specifications, indexed 1, 2, and 3. In specification 1, $\mathbf{Exp}_{m,t}$ includes two variables: exposure to immigrant storefront businesses and exposure to immigrant non-storefront businesses. In specification 2, it includes four variables: storefront and non-storefront exposure, respectively for Muslim and other immigrant-associated businesses. Finally, specification 3 includes five variables: the same as specification 2 but with Muslim-associated storefront exposure broken down between food businesses and non-food businesses. α_m and λ_t denote municipality and election fixed effects. The vector $X_{m,t}$ includes the time-varying municipal versions of the covariates used in the SSIV analysis. Like the IV analysis, this panel is weighted by 2006 municipality population. I use Conley standard errors allowing for spatial dependence up to 50 kilometers.

The municipality fixed effects absorb stable differences across locations, while the election fixed effects absorb national changes common to all municipalities. The coefficients therefore describe whether within-municipality changes in business exposure are associated with changes in RN support. Because the timing and location of business creation remain endogenous, I interpret these estimates descriptively.

Figure 3 displays the results. In specifications pooling immigrant origins, the relationship between total immigrant-storefront exposure and RN support is not statistically distinguishable from zero. However, the storefront and non-storefront coefficients differ significantly: a Wald test of $\beta_{\text{storefront}} = \beta_{\text{non-storefront}}$ in specification 1 rejects equality (p=0.013), consistent with Hypothesis 2. Decomposing exposure by origin reveals a significant positive relationship for Muslim-associated storefronts, whereas the coefficient for Muslim-associated non-storefront businesses is negative but imprecisely estimated. A Wald test comparing Muslim-

and non-Muslim-associated storefront coefficients in specification 2 also rejects equality (p=0.006), consistent with Hypothesis 3. Taken together, these patterns are consistent with the positive association being concentrated among visible, Muslim-associated storefront businesses.

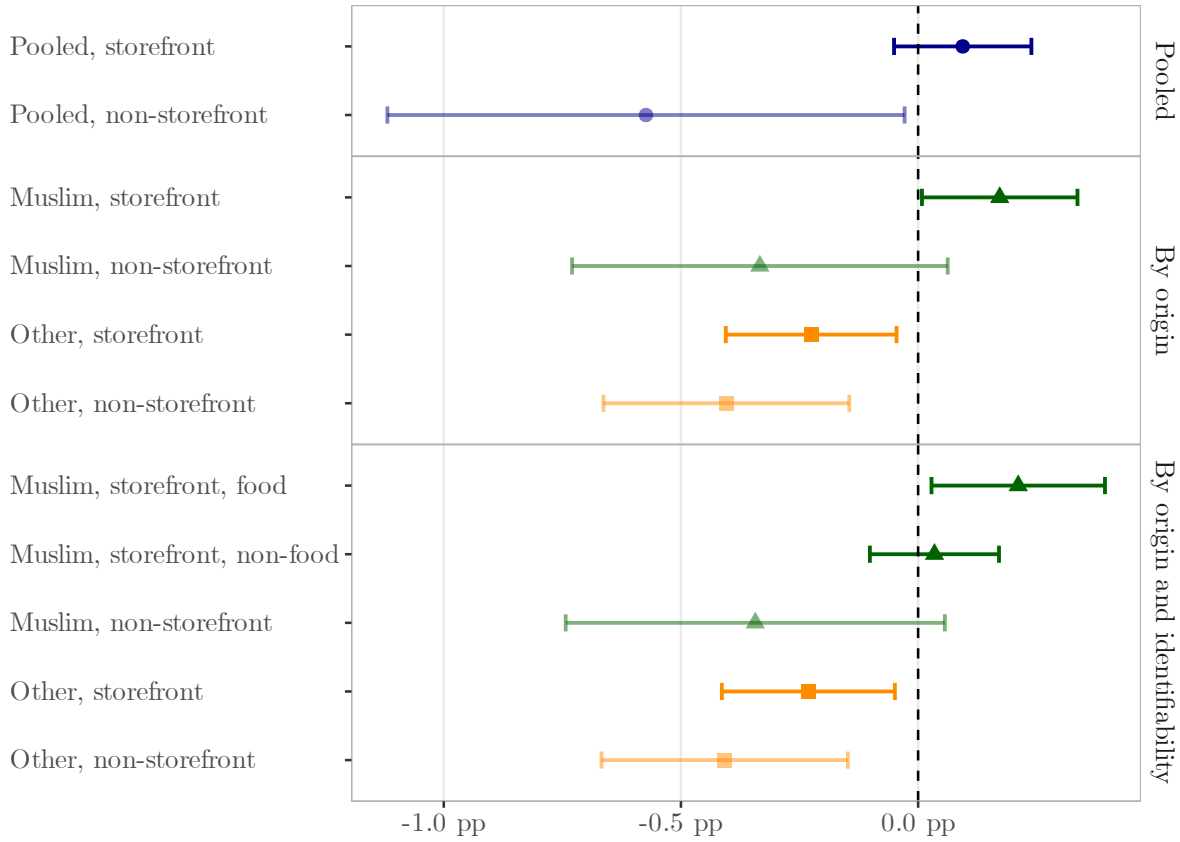


Figure 3: Estimates from the Descriptive Panel Analysis

Note: Error bars represent 95% confidence intervals estimated with a Conley (50 km) standard error estimator allowing for spatial dependence. The magnitude of the coefficients is scaled to correspond to a doubling of exposure. The circles are coefficients for businesses pooled across origin groups. The triangles are coefficients for businesses whose owners' names are associated with Muslim-majority countries. The squares are coefficients for other immigrant-associated businesses. Detailed estimates are displayed in Table I1 of the appendix.

The third panel of Figure 3 assesses heterogeneity within the broader category of Muslim-associated storefront businesses. Group identity can more easily be inferred for certain kinds of businesses. Food businesses, such as fast-food shops or butchers, plausibly signal origin more legibly than storefronts selling goods that are not culturally marked. Consistent with this, the positive association is clearest among Muslim food storefronts. The non-food estimate is smaller and not statistically distinguishable from zero. Across specifications, the point estimates for non-Muslim storefronts and for non-storefront businesses are negative. Visibility, group salience, and the identifiability of group markers therefore all point in the same

direction. Although they do not causally identify heterogeneity between Muslim and non-Muslim businesses, these results support an interpretation of the backlash effect as at least partly grounded in a visibility mechanism, particularly when exposure makes visible groups toward which natives hold pre-existing hostility.

Note that these coefficients are considerably smaller than the SSIV estimates. Although the two designs estimate different quantities, this difference deserves examination. Three factors plausibly push the panel estimates toward zero. First, exposure is measured with error, since the name classification has imperfect recall (Appendix C) and the registered opening and closing dates may not be perfectly measured. The within-municipality analysis compounds the resulting attenuation bias, since municipality fixed effects absorb persistent exposure but not the noise (Griliches and Hausman 1986). Second, the panel variation is endogenous in a direction that could plausibly bias estimates downward: entrepreneurs have incentives to avoid municipalities turning toward the far right (Lueders 2026), inducing a negative correlation between business entry and RN trajectories that the instrument removes. Finally, 2SLS identifies a weighted average of causal responses concentrated on municipalities whose exposure is shifted by the national industry shocks (Angrist and Imbens 1995), whereas the panel averages over all variation, including growth that just follows immigrant settlement. The contribution of this panel analysis is therefore a description of the contrast between origins and business types, while the IV estimates provide a better estimate of the significance and magnitude of the net effect.

6.2 Observable Local Economic Consequences

An alternative explanation for the observed electoral backlash is that it reflects economic degradation that gets blamed on immigrant businesses (Hypothesis 4). Immigrant-associated businesses could displace native businesses, increase competition over workers and customers, or contribute to a broader deterioration in local economic conditions. I examine this possibility by re-estimating the instrumental variable specification from Equation 7 with a set of local economic outcomes. Note that the absence of a significant result would not prove definitively that no economic mechanism explains backlash. Not only could there be economic damage

beyond the variables I measure, but even in the absence of any degradation, voters could inaccurately blame immigrant businesses for unrelated hardships or perceived economic decline.

The outcomes, detailed in Appendix I, capture distinct forms of economic harm. Native-business exposure measures whether immigrant-storefront growth leads to net changes in native-business presence, which may reflect displacement. Unemployment and income per capita capture local labor-market and income effects. Industry diversity, measured as one minus the concentration of businesses across industries, reflects changes in the composition of local commercial activity. A decrease would suggest that immigrant businesses reduce the diversity of goods and services locally available. I also examine change in property values, measured as median housing prices per square meter, between 2014 and 2019.[23]

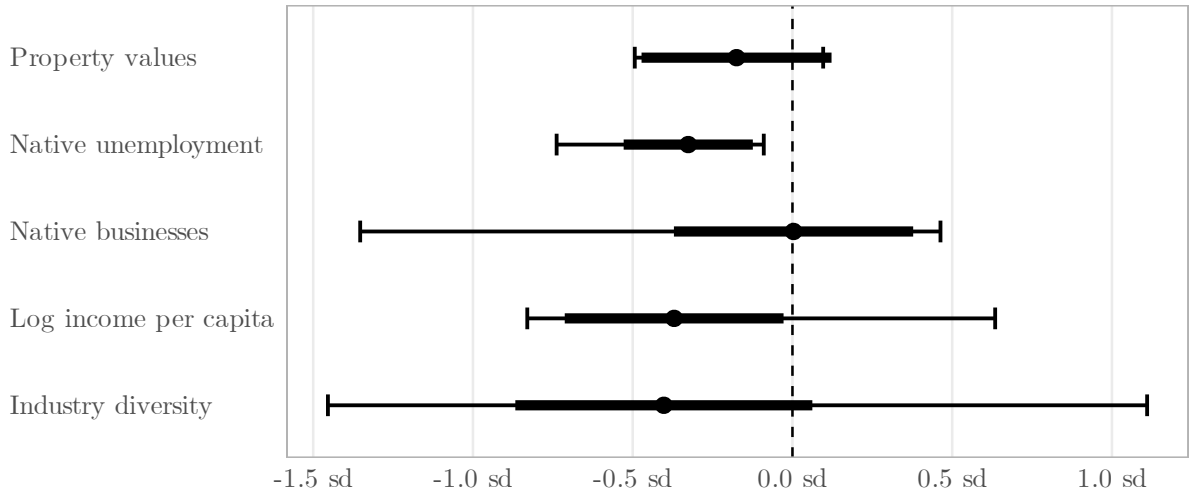


Figure 4: Second-Stage Estimates for the Economic Outcomes

Note: Thick bars represent Conley (50 km) confidence intervals and thin bars represent AKM0 95% confidence sets. The magnitude of the coefficients is scaled to correspond to a doubling of exposure. Each outcome is standardized to have a mean of zero and a standard deviation of one. All specifications include the controls, fixed effects, and population weights. The housing-price outcome covers 2014-2019 and municipalities with sufficient recorded transactions in both years. The corresponding tables are displayed in Appendix I.

Figure 4 reports the 2SLS estimates with both Conley and AKM0 95% confidence intervals. The effect is statistically distinguishable from zero only for unemployment: an approximate doubling in exposure produces a decrease of 0.326 standard deviations in the share of non-immigrant residents aged 15 or older who are unemployed. This suggests that growth in immigrant businesses may, if anything, improve local employment opportunities for non-immigrants. None of the other coefficients are statistically significant. The estimate for native-

23. Property data is only available starting in 2014. See more details in Appendix I.

business exposure is close to zero, though its 95% AKM0 confidence set includes displacement as well as expansion. The estimate for industry diversity is also imprecise, with a negative point estimate. The coefficient for income is negative and excludes zero under Conley inference but not under exposure-robust AKM0, which is more appropriate to reflect the common shock exposure of different municipalities (Adão, Kolesár, and Morales 2019). The negative income estimate could reflect a negative labor-market effect, though the decrease in unemployment makes this less likely. Alternatively, it could indicate compositional changes in population: Faury (2024) suggests that higher-income native residents unhappy with change may migrate out, whereas lower-income residents who cannot afford to move will vote in protest. The municipal average itself cannot distinguish income loss among residents from changes in residential composition. The property-value estimate is also negative but not significant.[24] Declines in average income and in property value are economic consequences but could be downstream effects of the visibility mechanisms.

Taken together, these outcomes do not establish empirical patterns consistent with Hypothesis 4, though they cannot rule out an economic explanation of backlash: the imprecise intervals include negative values and the economic effect could be null on average but harm specifically the voters who are most likely to turn out. Moreover, voters could perceive an economic threat even in the absence of actual harm or blame unrelated economic degradation on immigrant-associated businesses.

6.3 Discussion

Overall, the analyses above are more consistent with a visibility-based explanation for the backlash: growth in RN support is most clearly associated with growth in visible businesses, as opposed to non-storefront ones, and with businesses associated with immigrants from Muslim-majority countries, which have been at the center of cultural conflict in France independently of the economic impact of immigration. Importantly, the panel analysis should only be interpreted as descriptive evidence and cannot distinguish between mechanisms grounded

24. It is the only outcome for which the Conley interval is wider than the AKM0 confidence set. The two approaches are robust to different dependence structures: geographic proximity for Conley and common shock exposure for AKM0. For the spatially clustered subsample used for property values, the geographic component may dominate.

in signaling the presence of immigrants and those grounded in changing the visual environment of natives. Both of these points deserve further attention by researchers. The economic outcomes suggest no consistent pattern of local economic harm: immigrant businesses do not appear to displace native businesses, to increase unemployment and industry concentration, or to decrease property prices. The decrease in income is not statistically significant with an inference approach robust to common shock exposure. This analysis cannot rule out distributional losses, perceived economic threat (Dancygier and Donnelly 2013), or economic consequences beyond this set of available outcomes. More research is therefore needed to further test economic channels, or rule them out entirely.

I also conducted a pre-registered survey experiment designed to test these mechanisms. It did not provide robust evidence that brief experimental exposure replicates the longer-term relationships observed in the observational analysis. Details on the design and results are reported in Appendix J. The observational results may reflect repeated exposure, cumulative changes in the perceived identity of a municipality, or political narratives that tie visible economic activity to salient immigration debates.

7 Conclusion

Millions of immigrant businesses operate across Western democracies. In this paper, I show that growth in exposure to immigrant-associated storefronts increases support for the far right: in France, a doubling of exposure between 2009 and 2019 increased the National Rally's vote share by approximately 6.6 percentage points. Descriptive evidence of mechanisms is more consistent with an interpretation grounded in the visibility of immigration: the association between exposure and far-right support is stronger for storefront businesses associated with Muslim-majority countries, the group at the center of cultural conflict in France, and is clearest for food businesses, whose group association may be more easily identifiable. In contrast, I find null or negative relationships for non-storefront businesses and businesses associated with non-Muslim immigrant origins. I find no consistent evidence that immigrant businesses damage local economies: results do not suggest that they displace native businesses

and native unemployment, if anything, declines. Overall, the analysis is more consistent with visibility-based mechanisms than with observable economic harm.

These results have several implications. First, they complement research on how local demographic change shapes immigration attitudes and far-right support (Hopkins 2010; Hangartner et al. 2019; Cools, Finseraas, and Rogeberg 2021; Alrababah et al. 2025): even conditional on the local immigrant population, changes in the visibility of immigration can affect voting behavior. Second, the backlash against businesses suggests a potential vicious cycle. Immigrants often create businesses because discrimination limits their access to salaried employment (Fairlie and Lofstrom 2015). If those businesses then fuel negative attitudes toward immigrants, they may further worsen labor-market prospects, which may make self-employment even more desirable, sustaining backlash. This cycle makes rhetoric against immigrant businesses attractive for far-right political entrepreneurs, explaining anecdotal evidence of politicians denouncing immigrant businesses (Laurent 2015). Third, the findings speak to the tensions of globalization (Algan et al. 2017; Colantone and Stanig 2018; Autor et al. 2020): policies fostering entrepreneurship and openness may simultaneously support local economic activity and trigger backlash rooted in cultural conflict. Although perceived economic threat may operate even absent measurable harm, this is consistent with scholarship showing that cultural factors, rather than self-interest, drive attitudes toward immigration (Hainmueller and Hopkins 2014).

The findings suggest at least three avenues for future research. First, researchers should seek to distinguish mechanisms within the visibility family: do natives react to updated beliefs about the presence of a group toward which they hold prejudice, or to divergence from their preferred physical environment? Extending this research to contexts where immigration is salient for different groups, and where immigrants' occupations differ, would help separate these channels. Second, researchers should investigate the effects of immigrant entrepreneurship on elite-level political behavior and strategies. Elite rhetoric shapes perceptions of immigrant businesses, and businesses may thus contribute to the success of political entrepreneurs who make them focal points, as in France (Laurent 2015) and in Italy, where a movement of mayors launched a "Yes to Polenta, No to Couscous" campaign in 2009 (Hill

2025). Studying how politicians select these targets, and with what electoral returns, would connect the bottom-up exposure channel documented here to top-down accounts of anti-immigrant mobilization. Finally, researchers should examine the political determinants of immigrant entrepreneurship itself. Immigrants' residential decisions respond to local political attitudes (Lueders 2026), and their business location decisions likely do as well. Hostile environments may deter immigrant entrepreneurship and the economic activity it generates, feeding perceptions of local decline that themselves fuel nativism. Alternatively, lower rents may draw immigrant businesses into already-declining areas, making them one of the main points of contact between natives and immigrants precisely where economic grievance is sharpest. Either dynamic would further entangle the economic and cultural politics of immigrant businesses introduced in this paper.

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A Ethics and AI Disclosures

Ethics. I have no conflict of interest to report. The survey protocol (see Appendix J) was determined exempt by the Institutional Review Board of the researcher's home institution. Respondents provided prior informed consent and were compensated, on average, at a rate equivalent to $12.7 per hour. The responses were entirely anonymized, no deception was used, and no risk to the respondents was identified. The Amazon Mechanical Turk workers (Appendix C.2) were compensated $0.30 per batch of twenty names to classify and were not asked to provide any information beyond this immediate task.

Use of Artificial Intelligence. I used ChatGPT 4o in summer 2025 to suggest initial language for some components of the pre-analysis plan for the survey described in Appendix J. I then re-wrote and adjusted all of these components. I used ChatGPT 5 for consistency checks of the robustness analyses from the appendix and for copy-editing of the final manuscript in fall 2026. I used Claude Opus 4.8 in summer 2026 to check for consistency and errors in the data cleaning and analysis code and to suggest initial ways to implement the randomization inference test (in Appendix F.1), and to improve the validation of the classification (in Appendix C.2). I revised both of these codes substantially. I remain responsible for all of the code and the writing.

Data availability. I will make the replication code and data publicly available for the municipal-level data, the validation data, and the survey experiment. The classified business-level information that enters the exposure measure links identifiable owners to predicted origins and is therefore subject to data-protection restrictions under the French National Commission on Informatics and Liberty. I will make these data available without limitation to the extent permitted by French regulation.

B Data and Context

B.1 Election Data

Figure B1 displays all the elections for which data were available at the time of the primary data collection of this project. Cantons are electoral districts within departments. Before 2015, cantonal elections elected members of departmental councils (then *conseils généraux*), with approximately half of cantons voting every three years. Beginning in 2015, these elections were replaced by departmental elections, in which the full departmental council is renewed every six years. Senate elections are not included because senators are elected indirectly by an electoral college composed primarily of other elected officials.

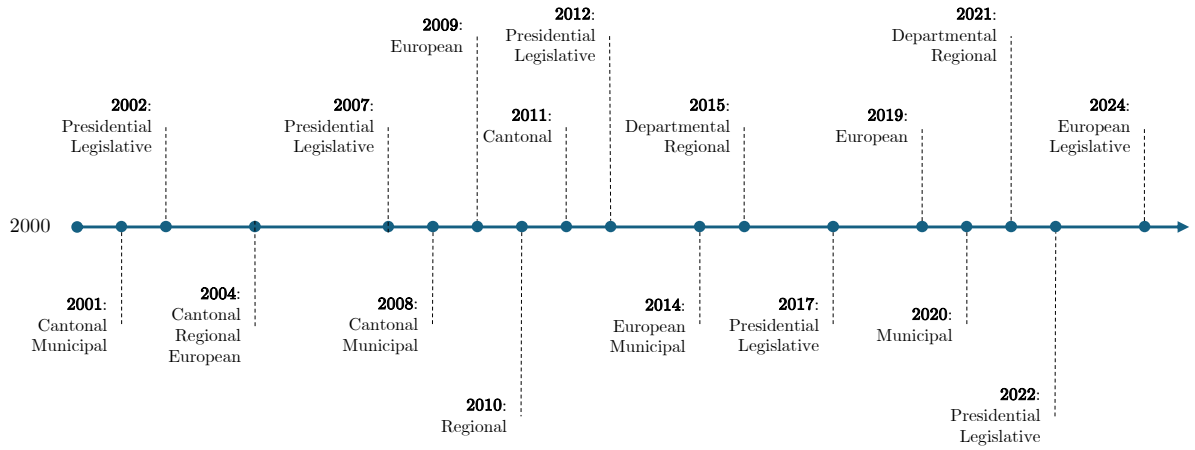


Figure B1: Electoral Results Available since 2000

Note: Chronology of all elections that have taken place in France and for which data were available at the time of the primary data collection of this project. Cantonal election coverage only includes a rotating subset of municipalities. Municipal election data is only available with candidate party labels for municipalities over 3,500 residents. Because data is available as early as 2001, the constraints on the paper's analysis come from other data sources, primarily on the treatment (businesses) and covariates.

B.2 Far-Right Vote

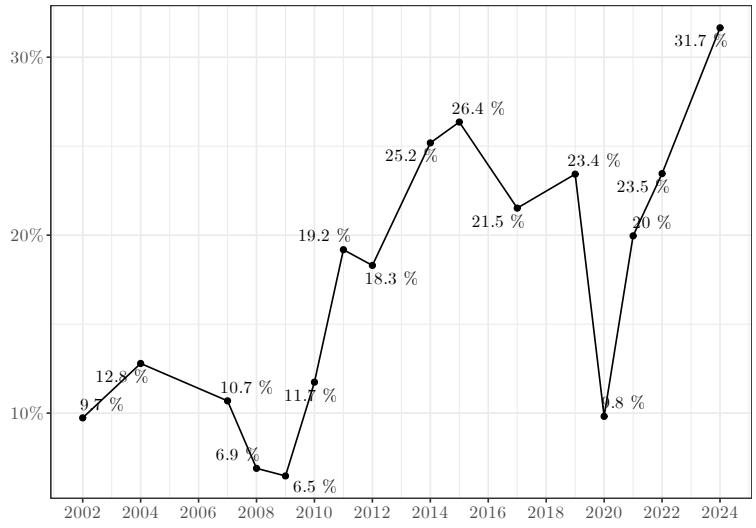


Figure B2: Electoral Support for the Far Right, 2002-2024

Note: National average of the RN vote share over time. Municipal elections only include municipalities with over 3,500 residents. The 2020 drop can be explained by the subsample of municipalities whose results are available for municipal elections: only results for municipalities with over 3,500 residents are reported, and these more urban municipalities tend to show lower rates of support for the far right.

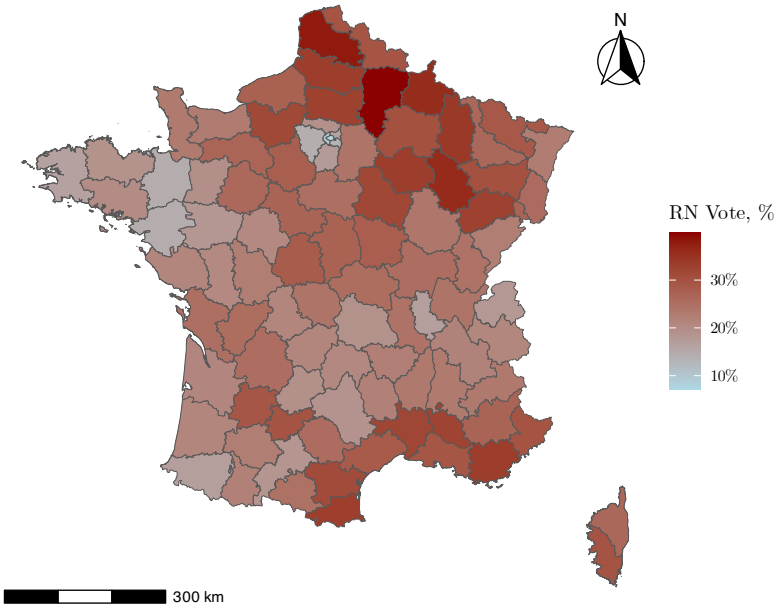


Figure B3: RN Vote by Department, 2019

Note: RN vote share by department in the 2019 European elections. Support for the far right is unevenly distributed: it is lowest in the Paris region (small light blue cell in the Central-Northern part of the map), and stronger in territories that have now emerged as RN strongholds, including Southeastern and Northern France.

B.3 Businesses: Creation and Data Structure

Figure B4 shows the steps generally required to create a business in France. Note that each business structure type (SARL, SA, "microentrepreneurship", etc.) requires specific steps. For example, micro-entrepreneurs, a category often selected by small businesses, including in the gig economy, operate in their own name and do not need company statutes or share capital, but must declare their activity and complete the applicable registration formalities. Every enterprise must be registered with the *Registre National des Entreprises* (RNE). Registration assigns the enterprise a SIREN identifier, while each establishment (which I use as the "business" construct of interest) operated by the enterprise has its own SIRET identifier.

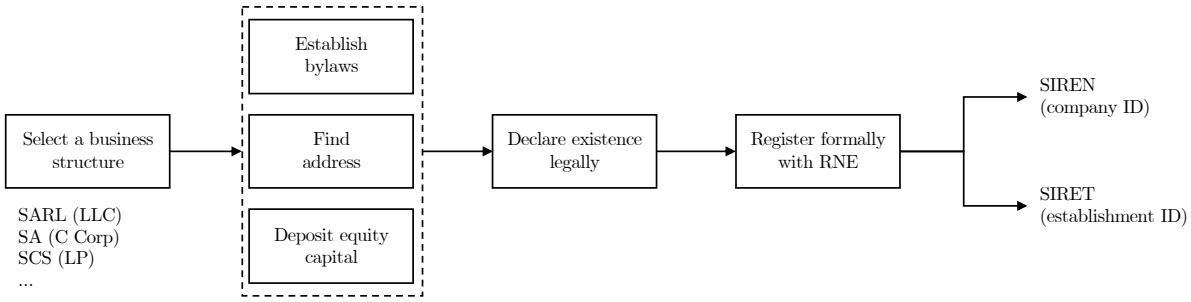


Figure B4: Business Creation Process in France

Note: This figure displays the steps required to create and register a business in France. Importantly, a SIREN unique identifier is assigned to each company, and a SIRET unique identifier to each business. Classification is done at the company level, since owners are common across businesses, as described in Figure B5.

My data rely on the information stored in the *Registre National des Entreprises* and have the structure described in Figure B5. I predict the probable origin of business owners at the *company* level, while I geocode and time-code each *business*. Personal names are recorded at the company (SIREN) level. When possible, I use the registered entrepreneur's name. For legal entities, I use names in the "beneficial-owner" records when that field is present. When it is not, I use individuals listed in the company's representation records. The "owner" variable therefore includes information not only on owners and registered entrepreneurs, but also sometimes on company officials. I treat it as a consistent proxy for the perceived origin associated with a business.

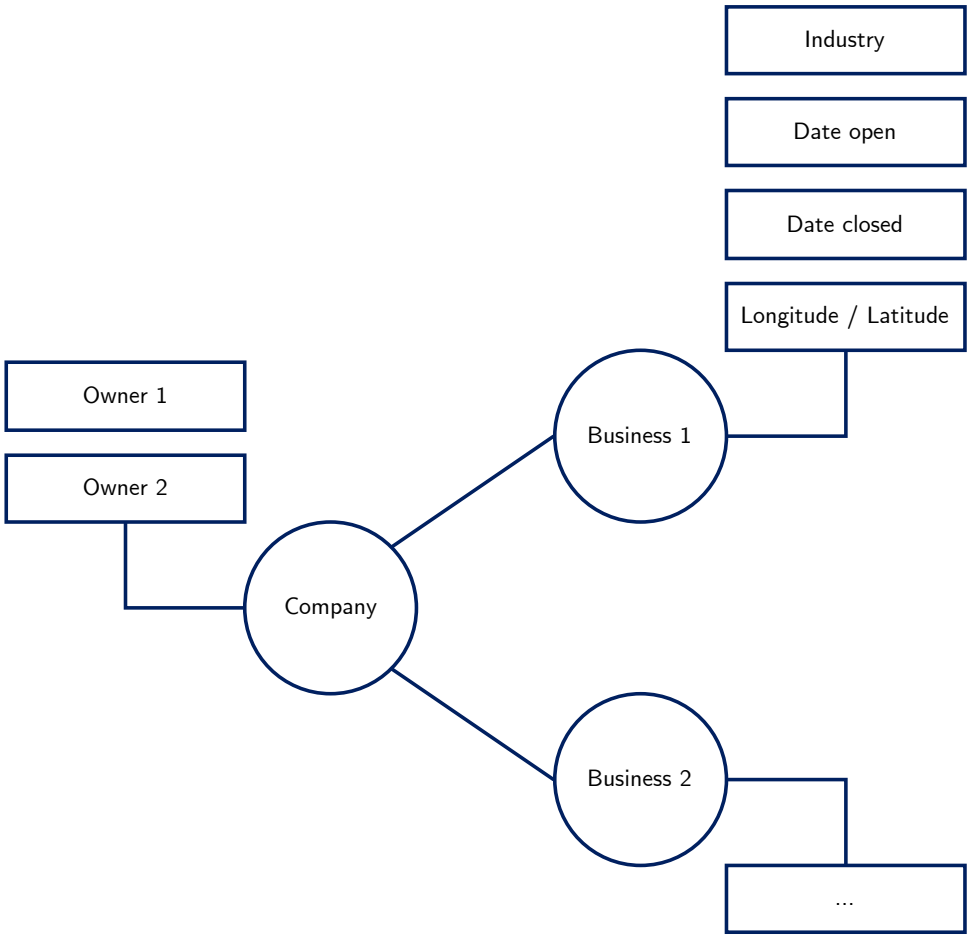


Figure B5: Nested Structure of the Individual Business Data

Note: Stylized description of the business data. Each company's owner is classified as described in Appendix C. A company can include several businesses (establishments). The data include detailed information on each business: location, dates of opening and closing, and industry. The exposure measure captures how many such businesseseach municipality is exposed to over time.

B.4 Immigrant Businesses and Immigrant Residents

The two plots below describe the relationship between immigrant businesses, the theoretical and empirical specificity of this paper, and immigrant residents. Figure B6 shows that the proportion of municipal storefront businesses that are associated with immigrants varies widely across values of the proportion of the municipal population that is foreign-born. Figure B7 shows that the share of immigrant-associated businesses has grown at a significantly faster rate than the share of foreign-born residents.

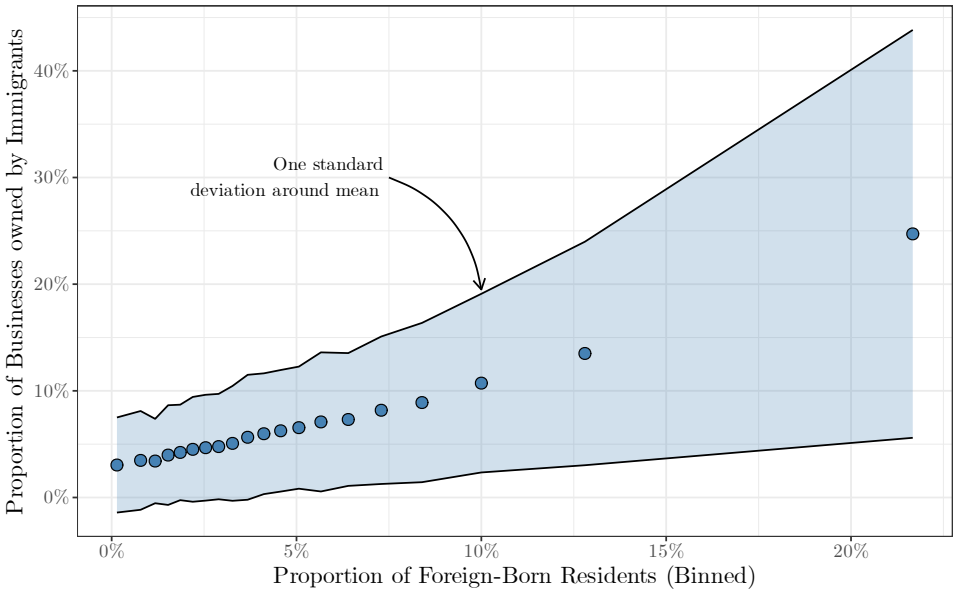


Figure B6: Immigrants among Residents and Business Owners, January 2022

Note: Each dot represents a bin including one twentieth of the data in January 2022. The band represents one standard deviation around each bin, interpolated between them. This shows that the distribution in the proportion of immigrant-associated businesses is wide for any given proportion of foreign-born individuals among municipal residents.

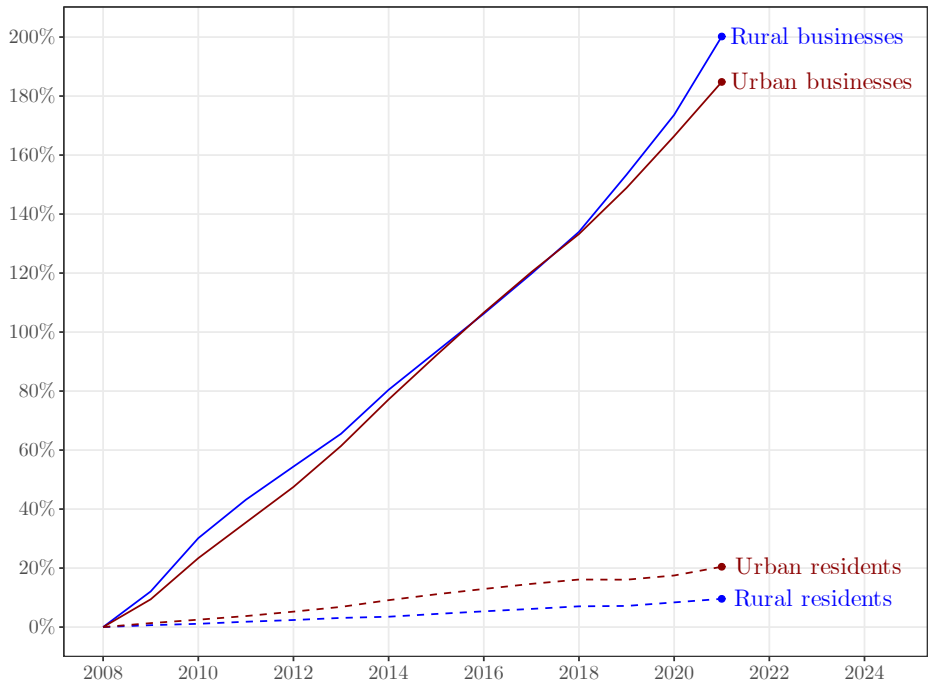


Figure B7: Growth in the Proportion of Immigrants

Note: Each line describes the cumulative growth among units of the specified group, relative to the baseline year of 2008. The solid lines represent growth in the shares of immigrant-associated businesses out of all storefronts in rural and urban areas, respectively. The dashed lines represent growth in the shares of residents who are foreign-born in rural and urban areas, respectively. Note that a value of 200% does not mean a proportion larger than one but a tripling relative to baseline.

B.5 Industry Classification

Table B1: Industry Storefront (SF) Classification

APE	Industry	SF	APE	Industry	SF
01.11Z	Growing of cereals, legumes & oilseeds		01.12Z	Growing of rice	
01.13Z	Growing of vegetables, roots & tubers		01.14Z	Growing of sugar cane	
01.15Z	Growing of tobacco		01.16Z	Growing of fibre crops	
01.19Z	Growing of other non-perennial crops		01.21Z	Growing of grapes	
01.22Z	Growing of tropical & subtropical fruits		01.23Z	Growing of citrus fruits	
01.24Z	Growing of pome & stone fruits		01.25Z	Growing of other tree fruits & nuts	
01.26Z	Growing of oleaginous fruits		01.27Z	Growing of beverage crops	
01.28Z	Growing of spices & aromatic/medicinal crops		01.29Z	Growing of other perennial crops	
01.30Z	Plant propagation		01.41Z	Raising of dairy cattle	
01.42Z	Raising of other cattle & buffaloes		01.43Z	Raising of horses & other equines	
01.44Z	Raising of camels & camelids		01.45Z	Raising of sheep & goats	
01.46Z	Raising of swine/pigs		01.47Z	Raising of poultry	
01.49Z	Raising of other animals		01.50Z	Mixed farming (crops & livestock)	
01.61Z	Support activities for crop production		01.62Z	Support activities for animal production	
01.63Z	Post-harvest crop activities		01.64Z	Seed processing for propagation	
01.70Z	Hunting, trapping & related services		02.10Z	Silviculture & other forestry activities	
02.20Z	Logging		02.30Z	Gathering of wild non-wood forest products	
02.40Z	Support services to forestry		03.11Z	Marine fishing	
03.12Z	Freshwater fishing		03.21Z	Marine aquaculture	
03.22Z	Freshwater aquaculture		05.10Z	Mining of hard coal	
05.20Z	Mining of lignite		06.10Z	Extraction of crude petroleum	
06.20Z	Extraction of natural gas		07.10Z	Mining of iron ores	
07.21Z	Mining of uranium & thorium ores		07.29Z	Mining of other non-ferrous metal ores	
08.11Z	Quarrying of stone, gypsum, chalk & slate		08.12Z	Gravel, sand, clay & kaolin extraction	
08.91Z	Mining of chemical & fertiliser minerals		08.92Z	Extraction of peat	
08.93Z	Extraction of salt		08.99Z	Other mining & quarrying n.e.c.	
09.10Z	Support activities for petroleum/gas extraction		09.90Z	Support activities for other mining	
10.11Z	Processing & preserving of meat		10.12Z	Processing & preserving of poultry meat	
10.13A	Industrial production of meat products		10.13B	Charcuterie (cured/deli meats)	✓
10.20Z	Processing & preserving of fish & seafood		10.31Z	Processing & preserving of potatoes	
10.32Z	Manufacture of fruit & vegetable juice		10.39A	Other processing/preserving of vegetables	
10.39B	Processing & preserving of fruit		10.41A	Manufacture of crude oils & fats	
10.41B	Manufacture of refined oils & fats		10.42Z	Manufacture of margarine & edible fats	
10.51A	Manufacture of liquid milk & fresh products		10.51B	Manufacture of butter	
10.51C	Manufacture of cheese		10.51D	Manufacture of other dairy products	
10.52Z	Manufacture of ice cream & sorbets		10.61A	Grain milling (flour)	
10.61B	Other grain milling activities		10.62Z	Manufacture of starches & starch products	
10.71A	Industrial bread & fresh pastry making		10.71B	Baking of bakery products	

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Table B1 continued from previous page

APE	Industry	SF	APE	Industry	SF
10.71C	Bakery & bread-pastry shops	✓	10.71D	Pastry making (patisserie)	✓
10.72Z	Manufacture of biscuits & preserved pastry		10.73Z	Manufacture of pasta & noodles	
10.81Z	Manufacture of sugar		10.82Z	Manufacture of cocoa, chocolate & confectionery	
10.83Z	Processing of tea & coffee		10.84Z	Manufacture of condiments & seasonings	
10.85Z	Manufacture of prepared meals & dishes		10.86Z	Manufacture of homogenised & dietetic food	
10.89Z	Manufacture of other food products n.e.c.		10.91Z	Manufacture of farm animal feeds	
10.92Z	Manufacture of pet food		11.01Z	Distilling & blending of spirits	
11.02A	Manufacture of sparkling wines		11.02B	Manufacture of wine (vinification)	
11.03Z	Manufacture of cider & fruit wines		11.04Z	Other non-distilled fermented beverages	
11.05Z	Manufacture of beer		11.06Z	Manufacture of malt	
11.07A	Manufacture of mineral & bottled waters		11.07B	Manufacture of soft drinks	
12.00Z	Manufacture of tobacco products		13.10Z	Preparation & spinning of textile fibres	
13.20Z	Weaving of textiles		13.30Z	Finishing of textiles	
13.91Z	Manufacture of knitted & crocheted fabrics		13.92Z	Manufacture of made-up textile articles	
13.93Z	Manufacture of carpets & rugs		13.94Z	Manufacture of cordage, rope & netting	
13.95Z	Manufacture of non-wovens (excl. apparel)		13.96Z	Manufacture of technical textiles	
13.99Z	Manufacture of other textiles n.e.c.		14.11Z	Manufacture of leather clothes	
14.12Z	Manufacture of workwear		14.13Z	Manufacture of other outerwear	
14.14Z	Manufacture of underwear		14.19Z	Manufacture of other apparel & accessories	
14.20Z	Manufacture of articles of fur		14.31Z	Manufacture of knitted hosiery	
14.39Z	Manufacture of other knitted apparel		15.11Z	Tanning & dressing of leather & fur	
15.12Z	Manufacture of luggage, handbags & saddlery		15.20Z	Manufacture of footwear	
16.10A	Sawmilling & planing of wood		16.10B	Wood impregnation	
16.21Z	Manufacture of veneer & wood-based panels		16.22Z	Manufacture of assembled parquet floors	
16.23Z	Manufacture of builders' carpentry & joinery		16.24Z	Manufacture of wooden containers	
16.29Z	Manufacture of other wood, cork & basketware		17.11Z	Manufacture of pulp	
17.12Z	Manufacture of paper & paperboard		17.21A	Manufacture of corrugated paper & board	
17.21B	Manufacture of cartonboard containers		17.21C	Manufacture of paper packaging	
17.22Z	Manufacture of household & sanitary paper		17.23Z	Manufacture of paper stationery	
17.24Z	Manufacture of wallpaper		17.29Z	Manufacture of other paper & board articles	
18.11Z	Printing of newspapers		18.12Z	Other printing	
18.13Z	Pre-press & pre-media services		18.14Z	Bookbinding & related services	
18.20Z	Reproduction of recorded media		19.10Z	Manufacture of coke oven products	
19.20Z	Manufacture of refined petroleum products		20.11Z	Manufacture of industrial gases	
20.12Z	Manufacture of dyes & pigments		20.13A	Enrichment & reprocessing of nuclear fuel	
20.13B	Manufacture of other inorganic basic chemicals		20.14Z	Manufacture of other organic basic chemicals	
20.15Z	Manufacture of fertilisers & nitrogen products		20.16Z	Manufacture of plastics in primary forms	
20.17Z	Manufacture of synthetic rubber		20.20Z	Manufacture of pesticides & agrochemicals	
20.30Z	Manufacture of paints, varnishes & inks		20.41Z	Manufacture of soap, detergents & cleaners	
20.42Z	Manufacture of perfumes & toilet preparations		20.51Z	Manufacture of explosives	

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Table B1 continued from previous page

APE	Industry	SF	APE	Industry	SF
20.52Z	Manufacture of glues		20.53Z	Manufacture of essential oils	
20.59Z	Manufacture of other chemical products n.e.c.		20.60Z	Manufacture of man-made fibres	
21.10Z	Manufacture of basic pharmaceutical products		21.20Z	Manufacture of pharmaceutical preparations	
22.11Z	Manufacture & retreading of tyres		22.19Z	Manufacture of other rubber products	
22.21Z	Manufacture of plastic plates, sheets & tubes		22.22Z	Manufacture of plastic packaging goods	
22.23Z	Manufacture of builders' plastic ware		22.29A	Manufacture of technical plastic parts	
22.29B	Manufacture of other plastic consumer goods		23.11Z	Manufacture of flat glass	
23.12Z	Shaping & processing of flat glass		23.13Z	Manufacture of hollow glass	
23.14Z	Manufacture of glass fibres		23.19Z	Manufacture of other & technical glass	
23.20Z	Manufacture of refractory products		23.31Z	Manufacture of ceramic tiles & flags	
23.32Z	Manufacture of bricks, tiles & baked clay		23.41Z	Manufacture of ceramic household/ornament ware	
23.42Z	Manufacture of ceramic sanitary fixtures		23.43Z	Manufacture of ceramic insulators	
23.44Z	Manufacture of other technical ceramics		23.49Z	Manufacture of other ceramic products	
23.51Z	Manufacture of cement		23.52Z	Manufacture of lime & plaster	
23.61Z	Manufacture of concrete building products		23.62Z	Manufacture of plaster building products	
23.63Z	Manufacture of ready-mixed concrete		23.64Z	Manufacture of mortars & dry concrete	
23.65Z	Manufacture of fibre cement articles		23.69Z	Manufacture of other concrete/plaster articles	
23.70Z	Cutting, shaping & finishing of stone		23.91Z	Manufacture of abrasive products	
23.99Z	Manufacture of other non-metallic minerals		24.10Z	Manufacture of basic iron & steel	
24.20Z	Manufacture of steel tubes & pipes		24.31Z	Cold drawing of bars	
24.32Z	Cold rolling of narrow strip		24.33Z	Cold forming or folding of steel	
24.34Z	Cold drawing of wire		24.41Z	Precious metals production	
24.42Z	Aluminium production		24.43Z	Lead, zinc & tin production	
24.44Z	Copper production		24.45Z	Other non-ferrous metal production	
24.46Z	Processing of nuclear fuel		24.51Z	Casting of iron	
24.52Z	Casting of steel		24.53Z	Casting of light metals	
24.54Z	Casting of other non-ferrous metals		25.11Z	Manufacture of metal structures & parts	
25.12Z	Manufacture of metal doors & windows		25.21Z	Manufacture of radiators & boilers (heating)	
25.29Z	Manufacture of metal tanks & containers		25.30Z	Manufacture of steam generators	
25.40Z	Manufacture of weapons & ammunition		25.50A	Forging, stamping & powder metallurgy	
25.50B	Metal cutting & stamping		25.61Z	Treatment & coating of metals	
25.62A	Bar turning (machining)		25.62B	Industrial machining	
25.71Z	Manufacture of cutlery		25.72Z	Manufacture of locks & hinges	
25.73A	Manufacture of moulds & patterns		25.73B	Manufacture of other tools	
25.91Z	Manufacture of steel drums & containers		25.92Z	Manufacture of light metal packaging	
25.93Z	Manufacture of wire products, chains & springs		25.94Z	Manufacture of fasteners & screw products	
25.99A	Manufacture of metal household articles		25.99B	Manufacture of other fabricated metal products	
26.11Z	Manufacture of electronic components		26.12Z	Manufacture of loaded electronic boards	
26.20Z	Manufacture of computers & peripherals		26.30Z	Manufacture of communication equipment	
26.40Z	Manufacture of consumer electronics		26.51A	Manufacture of navigation instruments	

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APE	Industry	SF	APE	Industry	SF
26.51B	Manufacture of measuring/testing instruments		26.52Z	Manufacture of watches & clocks	
26.60Z	Manufacture of medical irradiation equipment		26.70Z	Manufacture of optical & photographic instr.	
26.80Z	Manufacture of magnetic & optical media		27.11Z	Manufacture of motors & transformers	
27.12Z	Manufacture of electricity distribution gear		27.20Z	Manufacture of batteries & accumulators	
27.31Z	Manufacture of fibre optic cables		27.32Z	Manufacture of other wires & cables	
27.33Z	Manufacture of wiring devices		27.40Z	Manufacture of electric lighting equipment	
27.51Z	Manufacture of electric domestic appliances		27.52Z	Manufacture of non-electric domestic appliances	
27.90Z	Manufacture of other electrical equipment		28.11Z	Manufacture of engines & turbines	
28.12Z	Manufacture of fluid power equipment		28.13Z	Manufacture of other pumps & compressors	
28.14Z	Manufacture of taps & valves		28.15Z	Manufacture of gears & transmission parts	
28.21Z	Manufacture of ovens, furnaces & burners		28.22Z	Manufacture of lifting & handling equipment	
28.23Z	Manufacture of office machinery (excl. PCs)		28.24Z	Manufacture of power-driven hand tools	
28.25Z	Manufacture of ventilation & cooling equipment		28.29A	Manufacture of packing & weighing machinery	
28.29B	Manufacture of other general-purpose machinery		28.30Z	Manufacture of agri & forestry machinery	
28.41Z	Manufacture of metal-forming machine tools		28.49Z	Manufacture of other machine tools	
28.91Z	Manufacture of machinery for metallurgy		28.92Z	Manufacture of mining & construction machinery	
28.93Z	Manufacture of food & beverage machinery		28.94Z	Manufacture of textile & apparel machinery	
28.95Z	Manufacture of paper & board machinery		28.96Z	Manufacture of plastics & rubber machinery	
28.99A	Manufacture of printing machinery		28.99B	Manufacture of other special-purpose machinery	
29.10Z	Manufacture of motor vehicles		29.20Z	Manufacture of vehicle bodies & trailers	
29.31Z	Manufacture of auto electrical equipment		29.32Z	Manufacture of other parts for motor vehicles	
30.11Z	Building of ships & floating structures		30.12Z	Building of pleasure & sporting boats	
30.20Z	Manufacture of railway locomotives & stock		30.30Z	Manufacture of air & spacecraft	
30.40Z	Manufacture of military fighting vehicles		30.91Z	Manufacture of motorcycles	
30.92Z	Manufacture of bicycles & invalid carriages		30.99Z	Manufacture of other transport equipment n.e.c.	
31.01Z	Manufacture of office & shop furniture		31.02Z	Manufacture of kitchen furniture	
31.03Z	Manufacture of mattresses		31.09A	Manufacture of seats & upholstered furniture	
31.09B	Manufacture of other furniture		32.11Z	Striking of coins	
32.12Z	Manufacture of jewellery & related articles		32.13Z	Manufacture of imitation jewellery	
32.20Z	Manufacture of musical instruments		32.30Z	Manufacture of sports goods	
32.40Z	Manufacture of games & toys		32.50A	Manufacture of medical & dental instruments	
32.50B	Manufacture of spectacles & eyewear		32.91Z	Manufacture of brooms & brushes	
32.99Z	Other manufacturing n.e.c.		33.11Z	Repair of fabricated metal products	
33.12Z	Repair of machinery		33.13Z	Repair of electronic & optical equipment	
33.14Z	Repair of electrical equipment		33.15Z	Repair & maintenance of ships & boats	
33.16Z	Repair & maintenance of aircraft & spacecraft		33.17Z	Repair of other transport equipment	
33.19Z	Repair of other equipment		33.20A	Installation of metal structures & piping	
33.20B	Installation of industrial machinery		33.20C	Assembly of industrial process control systems	

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APE	Industry	SF	APE	Industry	SF
33.20D	Installation of electrical/electronic equipment		35.11Z	Production of electricity	
35.12Z	Transmission of electricity		35.13Z	Distribution of electricity	
35.14Z	Trade of electricity		35.21Z	Manufacture of gas	
35.22Z	Distribution of gaseous fuels via mains		35.23Z	Trade of gas through mains	
35.30Z	Steam & air conditioning supply		36.00Z	Water collection, treatment & supply	
37.00Z	Sewerage		38.11Z	Collection of non-hazardous waste	
38.12Z	Collection of hazardous waste		38.21Z	Treatment & disposal of non-hazardous waste	
38.22Z	Treatment & disposal of hazardous waste		38.31Z	Dismantling of wrecks	
38.32Z	Recovery of sorted materials		39.00Z	Remediation & other waste management	
41.10A	Development of residential building projects		41.10B	Development of office building projects	
41.10C	Development of other building projects		41.10D	Legal vehicles for building projects	
41.20A	Construction of individual houses		41.20B	Construction of other buildings	
42.11Z	Construction of roads & motorways		42.12Z	Construction of railways & underground rail	
42.13A	Construction of bridges & structures		42.13B	Construction & maintenance of tunnels	
42.21Z	Construction of utility projects for fluids		42.22Z	Construction of electricity & telecom networks	
42.91Z	Construction of water & maritime projects		42.99Z	Construction of other civil engineering n.e.c.	
43.11Z	Demolition		43.12A	Site preparation & earthworks	
43.12B	Specialised or large-scale earthworks		43.13Z	Test drilling & boring	
43.21A	Electrical installation in buildings		43.21B	Electrical installation on public roads	
43.22A	Plumbing & gas installation		43.22B	Heating & air conditioning installation	
43.29A	Insulation work		43.29B	Other construction installation n.e.c.	
43.31Z	Plastering		43.32A	Wood & PVC joinery installation	
43.32B	Metal joinery & locksmithing		43.32C	Fitting-out of retail premises	
43.33Z	Floor & wall covering		43.34Z	Painting & glazing	
43.39Z	Other building completion & finishing		43.91A	Roof framework construction	
43.91B	Roof covering & roofing		43.99A	Waterproofing work	
43.99B	Erection of metal structures		43.99C	General masonry & building shell work	
43.99D	Other specialised construction n.e.c.		43.99E	Rental of construction equipment with operator	
45.11Z	Sale of cars & light motor vehicles	✓	45.19Z	Sale of other motor vehicles	✓
45.20A	Maintenance & repair of light vehicles	✓	45.20B	Maintenance & repair of other vehicles	✓
45.31Z	Wholesale of motor vehicle parts		45.32Z	Retail of motor vehicle parts	✓
45.40Z	Sale & repair of motorcycles & parts	✓	46.11Z	Agents: farm produce, livestock & textiles	
46.12A	Fuel purchasing groups		46.12B	Agents: fuels, metals, minerals & chemicals	
46.13Z	Agents: timber & building materials		46.14Z	Agents: machinery, ships & aircraft	
46.15Z	Agents: furniture, household & hardware		46.16Z	Agents: textiles, clothing & footwear	
46.17A	Food purchasing groups		46.17B	Agents: food, beverages & tobacco	
46.18Z	Agents specialising in other products		46.19A	Non-food purchasing groups	
46.19B	Agents in sale of a variety of goods		46.21Z	Wholesale of grain, seeds & animal feed	
46.22Z	Wholesale of flowers & plants		46.23Z	Wholesale of live animals	
46.24Z	Wholesale of hides, skins & leather		46.31Z	Wholesale of fruit & vegetables	
46.32A	Wholesale of meat		46.32B	Wholesale of meat products	

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APE	Industry	SF	APE	Industry	SF
46.32C	Wholesale of poultry & game		46.33Z	Wholesale of dairy, eggs & edible oils	
46.34Z	Wholesale of beverages		46.35Z	Wholesale of tobacco products	
46.36Z	Wholesale of sugar, chocolate & confectionery		46.37Z	Wholesale of coffee, tea, cocoa & spices	
46.38A	Wholesale of fish & seafood		46.38B	Wholesale of other specialised food	
46.39A	Wholesale of frozen food		46.39B	Non-specialised wholesale of food & drink	
46.41Z	Wholesale of textiles		46.42Z	Wholesale of clothing & footwear	
46.43Z	Wholesale of electrical household appliances		46.44Z	Wholesale of china, glassware & cleaning	
46.45Z	Wholesale of perfume & cosmetics		46.46Z	Wholesale of pharmaceutical goods	
46.47Z	Wholesale of furniture, carpets & lighting		46.48Z	Wholesale of watches & jewellery	
46.49Z	Wholesale of other household goods		46.51Z	Wholesale of computers & software	
46.52Z	Wholesale of electronic & telecom parts		46.61Z	Wholesale of agricultural machinery	
46.62Z	Wholesale of machine tools		46.63Z	Wholesale of mining & construction machinery	
46.64Z	Wholesale of textile industry machinery		46.65Z	Wholesale of office furniture	
46.66Z	Wholesale of other office machinery		46.69A	Wholesale of electrical equipment	
46.69B	Wholesale of industrial supplies		46.69C	Wholesale of supplies for trade & services	
46.71Z	Wholesale of fuels & related products		46.72Z	Wholesale of metals & metal ores	
46.73A	Wholesale of wood & building materials		46.73B	Wholesale of sanitary & decor products	
46.74A	Wholesale of hardware & ironmongery		46.74B	Wholesale of plumbing & heating supplies	
46.75Z	Wholesale of chemical products		46.76Z	Wholesale of other intermediate products	
46.77Z	Wholesale of waste & scrap		46.90Z	Non-specialised wholesale trade	
47.11A	Retail of frozen food	✓	47.11B	Retail of general food (grocery)	✓
47.11C	Small grocery stores (superettes)	✓	47.11D	Supermarkets	✓
47.11E	Multi-retail general stores	✓	47.11F	Hypermarkets	✓
47.19A	Department stores	✓	47.19B	Other non-specialised retail stores	✓
47.21Z	Retail of fruit & vegetables	✓	47.22Z	Retail of meat & meat products	✓
47.23Z	Retail of fish & seafood	✓	47.24Z	Retail of bread, pastry & confectionery	✓
47.25Z	Retail of beverages	✓	47.26Z	Retail of tobacco products	✓
47.29Z	Other retail of food in specialised stores	✓	47.30Z	Retail of automotive fuel	✓
47.41Z	Retail of computers & software	✓	47.42Z	Retail of telecom equipment	✓
47.43Z	Retail of audio & video equipment	✓	47.51Z	Retail of textiles	✓
47.52A	Retail of hardware & paint (small stores)	✓	47.52B	Retail of hardware & paint (large stores)	✓
47.53Z	Retail of carpets & wall/floor coverings	✓	47.54Z	Retail of household appliances	✓
47.59A	Retail of furniture	✓	47.59B	Retail of other household equipment	✓
47.61Z	Retail of books	✓	47.62Z	Retail of newspapers & stationery	✓
47.63Z	Retail of music & video recordings	✓	47.64Z	Retail of sporting equipment	✓
47.65Z	Retail of games & toys	✓	47.71Z	Retail of clothing	✓
47.72A	Retail of footwear	✓	47.72B	Retail of leather goods & luggage	✓
47.73Z	Dispensing chemists (pharmacies)	✓	47.74Z	Retail of medical & orthopaedic goods	✓
47.75Z	Retail of cosmetics & toiletries	✓	47.76Z	Retail of flowers, plants & pet supplies	✓
47.77Z	Retail of watches & jewellery	✓	47.78A	Retail of optical goods	✓
47.78B	Retail of coal & fuels	✓	47.78C	Other specialised retail n.e.c.	✓

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APE	Industry	SF	APE	Industry	SF
47.79Z	Retail of second-hand goods	✓	47.81Z	Retail of food via stalls & markets	✓
47.82Z	Retail of textiles & clothing via markets	✓	47.89Z	Retail of other goods via stalls & markets	✓
47.91A	Mail-order retail (general catalogue)		47.91B	Mail-order retail (specialised catalogue)	
47.99A	Door-to-door retail		47.99B	Retail via vending machines & other non-store	
49.10Z	Interurban passenger rail transport		49.20Z	Freight rail transport	
49.31Z	Urban & suburban passenger transport		49.32Z	Taxi operation	
49.39A	Scheduled road passenger transport		49.39B	Other road passenger transport	
49.39C	Cable cars & ski lifts		49.41A	Interurban road freight transport	
49.41B	Local road freight transport		49.41C	Truck rental with driver	
49.42Z	Removal services		49.50Z	Transport via pipeline	
50.10Z	Sea & coastal passenger transport		50.20Z	Sea & coastal freight transport	
50.30Z	Inland passenger water transport		50.40Z	Inland freight water transport	
51.10Z	Passenger air transport		51.21Z	Freight air transport	
51.22Z	Space transport		52.10A	Refrigerated storage & warehousing	
52.10B	Non-refrigerated storage & warehousing		52.21Z	Support activities for land transport	
52.22Z	Support activities for water transport		52.23Z	Support activities for air transport	
52.24A	Port cargo handling		52.24B	Non-port cargo handling	
52.29A	Courier & express freight		52.29B	Freight forwarding & transport organisation	
53.10Z	Postal activities under universal obligation		53.20Z	Other postal & courier activities	
55.10Z	Hotels & similar accommodation	✓	55.20Z	Holiday & short-stay accommodation	✓
55.30Z	Camping grounds & caravan parks	✓	55.90Z	Other accommodation	✓
56.10A	Full-service restaurants	✓	56.10B	Cafeterias & self-service restaurants	✓
56.10C	Fast-food & takeaway	✓	56.21Z	Event catering	
56.29A	Contract food-service (canteens)		56.29B	Other food service activities	
56.30Z	Beverage serving (bars & cafes)	✓	58.11Z	Book publishing	
58.12Z	Publishing of directories & mailing lists		58.13Z	Publishing of newspapers	
58.14Z	Publishing of journals & periodicals		58.19Z	Other publishing activities	
58.21Z	Publishing of computer games		58.29A	Publishing of system & network software	
58.29B	Publishing of development-tool software		58.29C	Publishing of application software	
59.11A	Film & TV programme production		59.11B	Corporate & advertising film production	
59.11C	Motion picture production		59.12Z	Film, video & TV post-production	
59.13A	Motion picture distribution		59.13B	Video publishing & distribution	
59.14Z	Motion picture projection		59.20Z	Sound recording & music publishing	
60.10Z	Radio broadcasting		60.20A	General TV channel broadcasting	
60.20B	Thematic TV channel broadcasting		61.10Z	Wired telecommunications activities	
61.20Z	Wireless telecommunications activities		61.30Z	Satellite telecommunications activities	
61.90Z	Other telecommunications activities		62.01Z	Computer programming activities	
62.02A	Computer systems & software consultancy		62.02B	Third-party IT systems maintenance	
62.03Z	Computer facilities management		62.09Z	Other IT & computer service activities	
63.11Z	Data processing, hosting & related		63.12Z	Web portals	
63.91Z	News agency activities		63.99Z	Other information service activities n.e.c.	

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APE	Industry	SF	APE	Industry	SF
64.11Z	Central banking		64.19Z	Other monetary intermediation (banking)	
64.20Z	Activities of holding companies		64.30Z	Trusts, funds & similar financial entities	
64.91Z	Financial leasing		64.92Z	Other credit granting	
64.99Z	Other financial service activities n.e.c.		65.11Z	Life insurance	
65.12Z	Non-life insurance		65.20Z	Reinsurance	
65.30Z	Pension funding		66.11Z	Administration of financial markets	
66.12Z	Security & commodity contracts brokerage		66.19A	Legal vehicles for asset management	
66.19B	Other activities auxiliary to finance		66.21Z	Risk & damage evaluation	
66.22Z	Insurance agents & brokers		66.29Z	Other activities auxiliary to insurance	
66.30Z	Fund management activities		68.10Z	Buying & selling of own real estate	
68.20A	Renting of own dwellings		68.20B	Renting of own land & other real estate	
68.31Z	Real estate agencies	✓	68.32A	Management of real estate	
68.32B	Legal vehicles for real estate management		69.10Z	Legal activities	
69.20Z	Accounting, bookkeeping & auditing		70.10Z	Activities of head offices	
70.21Z	Public relations & communication consultancy		70.22Z	Business & management consultancy	
71.11Z	Architectural activities		71.12A	Surveying activities	
71.12B	Engineering & technical consultancy		71.20A	Vehicle technical inspection	
71.20B	Technical testing & analysis		72.11Z	R&D in biotechnology	
72.19Z	R&D in natural sciences & engineering		72.20Z	R&D in social sciences & humanities	
73.11Z	Advertising agencies		73.12Z	Media representation for advertising	
73.20Z	Market research & opinion polling		74.10Z	Specialised design activities	
74.20Z	Photographic activities		74.30Z	Translation & interpretation	
74.90A	Construction cost estimating		74.90B	Other professional & technical activities	
75.00Z	Veterinary activities		77.11A	Short-term car rental	✓
77.11B	Long-term car leasing	✓	77.12Z	Renting & leasing of trucks	✓
77.21Z	Renting of recreational & sports goods	✓	77.22Z	Renting of video tapes & disks	✓
77.29Z	Renting of other personal & household goods	✓	77.31Z	Renting of agricultural machinery	
77.32Z	Renting of construction machinery		77.33Z	Renting of office & computer machinery	
77.34Z	Renting of water transport equipment		77.35Z	Renting of air transport equipment	
77.39Z	Renting of other machinery & equipment		77.40Z	Leasing of intellectual property	
78.10Z	Employment placement agencies		78.20Z	Temporary employment agencies	
78.30Z	Other human resources provision		79.11Z	Travel agency activities	✓
79.12Z	Tour operator activities		79.90Z	Other reservation & related services	
80.10Z	Private security activities		80.20Z	Security systems service activities	
80.30Z	Investigation activities		81.10Z	Combined facilities support activities	
81.21Z	General cleaning of buildings		81.22Z	Other building & industrial cleaning	
81.29A	Disinfection & pest control		81.29B	Other cleaning activities n.e.c.	
81.30Z	Landscape service activities		82.11Z	Combined office administrative services	
82.19Z	Photocopying & document preparation	✓	82.20Z	Call centre activities	
82.30Z	Organisation of conventions & trade shows		82.91Z	Debt collection & credit bureaus	
82.92Z	Packaging activities		82.99Z	Other business support activities n.e.c.	

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APE	Industry	SF	APE	Industry	SF
84.11Z	General public administration		84.12Z	Public admin: health, education & culture	
84.13Z	Public admin of economic activities		84.21Z	Foreign affairs	
84.22Z	Defence activities		84.23Z	Justice & judicial activities	
84.24Z	Public order & safety activities		84.25Z	Fire service activities	
84.30A	General compulsory social security		84.30B	Supplementary pension administration	
84.30C	Social income redistribution		85.10Z	Pre-primary education	
85.20Z	Primary education		85.31Z	General secondary education	
85.32Z	Technical & vocational secondary education		85.41Z	Post-secondary non-tertiary education	
85.42Z	Higher education		85.51Z	Sports & recreation education	
85.52Z	Cultural education		85.53Z	Driving school activities	
85.59A	Continuing adult education & training		85.59B	Other education n.e.c.	
85.60Z	Educational support activities		86.10Z	Hospital activities	
86.21Z	General medical practice		86.22A	Radiology & radiotherapy	
86.22B	Surgical medical practice		86.22C	Other specialist medical practice	
86.23Z	Dental practice activities		86.90A	Ambulance services	
86.90B	Medical laboratory activities		86.90C	Blood, organ & tissue banks	
86.90D	Nursing & midwifery activities		86.90E	Physiotherapy & podiatry activities	
86.90F	Other human health activities n.e.c.		87.10A	Nursing care for the elderly	
87.10B	Nursing care for disabled children		87.10C	Nursing care for disabled adults	
87.20A	Residential care for mental illness		87.20B	Residential care for substance abuse	
87.30A	Residential care for the elderly		87.30B	Residential care for physically disabled	
87.90A	Residential care for children in need		87.90B	Other residential care activities	
88.10A	Home help services		88.10B	Non-residential care: disabled & elderly	
88.10C	Sheltered work integration		88.91A	Child day-care activities	
88.91B	Non-residential care for disabled children		88.99A	Other non-residential child/youth care	
88.99B	Other social work without accommodation		90.01Z	Performing arts	
90.02Z	Support activities to performing arts		90.03A	Artistic creation (visual arts)	
90.03B	Other artistic creation		90.04Z	Operation of arts facilities	
91.01Z	Library & archive activities		91.02Z	Museum activities	
91.03Z	Historical sites & monuments		91.04Z	Botanical & zoological gardens; reserves	
92.00Z	Gambling & betting activities		93.11Z	Operation of sports facilities	
93.12Z	Activities of sports clubs		93.13Z	Fitness facilities	
93.19Z	Other sports activities		93.21Z	Amusement & theme parks	
93.29Z	Other amusement & recreation activities		94.11Z	Business & employers' organisations	
94.12Z	Professional membership organisations		94.20Z	Trade unions	
94.91Z	Religious organisations		94.92Z	Political organisations	
94.99Z	Other membership organisations n.e.c.		95.11Z	Repair of computers & peripherals	✓
95.12Z	Repair of communication equipment	✓	95.21Z	Repair of consumer electronics	✓
95.22Z	Repair of appliances & garden equipment	✓	95.23Z	Repair of footwear & leather goods	✓
95.24Z	Repair of furniture & home furnishings	✓	95.25Z	Repair of watches, clocks & jewellery	✓
95.29Z	Repair of other personal & household goods	✓	96.01A	Industrial laundry & dry-cleaning	✓

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APE	Industry	SF	APE	Industry	SF
96.01B	Retail laundry & dry-cleaning	✓	96.02A	Hairdressing	✓
96.02B	Beauty treatment	✓	96.03Z	Funeral & related activities	✓
96.04Z	Physical well-being activities	✓	96.09Z	Other personal service activities n.e.c.	✓
97.00Z	Households as employers of domestic staff		98.10Z	Households producing goods for own use	
98.20Z	Households producing services for own use		99.00Z	Extraterritorial organisations & bodies	

B.6 1978 Law

French data-protection law places special restrictions on the processing of sensitive personal information, including racial or ethnic origin and religious beliefs. The 1978 *Loi Informatique et Libertés* was adopted following concerns about computerized personal records and administrative data linkage caused by a project of systematic data collection by the government heavily criticized in the media.

In Decision No. 2007-557 DC of 15 November 2007, the Constitutional Council distinguished research using objective information about origins from classifications based on race or ethnicity. Existing surveys, for instance the Insee–Ined *Trajectoires et Origines* studies, collect information about immigrants and their descendants. The key limitation for this study is the availability of comprehensive, repeated local measures of race or ethnicity: objective measures fall under the restricted "objective information" category, and self-perceptions are not available at scale. Birthplace and citizenship do not directly capture whether a business is perceived as immigrant-associated. The name-based measure used here approximates this perceived association.

B.7 Description of Immigrant-Associated Businesses

Figure B8 displays the twenty storefront industries accounting for the largest shares of immigrant-associated storefront businesses in January 2024. Immigrant-associated businesses are spread across a wide range of customer-facing activities, although food and personal-service industries are particularly prominent. Fast-food shops account for approximately 15% of immigrant-associated storefront businesses, followed by traditional restaurants and car retail.

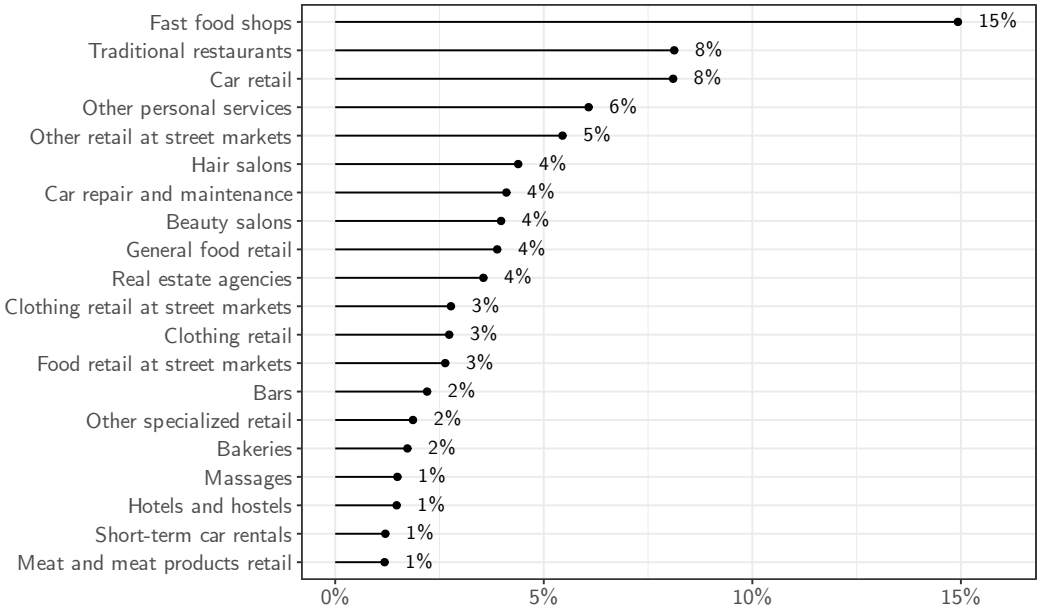


Figure B8: Industries Most Represented among Immigrant Storefronts, January 2024

Note: Each point represents the proportion of all immigrant-associated storefront businesses that operate in a given industry in January 2024.

Finally, Figure B9 displays the twenty most represented countries of origin of immigrant business owners, in January 2024, based on the classification algorithm. Immigrants from Northern and Sub-Saharan Africa are strongly represented, reflecting French colonial history. Chinese and Turkish origins reflect more recent waves of immigration.

C Measurement

C.1 Classification: Name-Based Algorithm

I classify the likely origin of each business owner from their first and last name using the fall 2024 version of Namsor's "diaspora" model. For a given name, the model estimates a probability distribution over countries and regions of origin from the distribution of first and last names in Namsor's corpus. Naming maps to specific origin countries differently in different host countries. Namsor's diaspora model accounts for these specificities. For instance, a North African last name for a person living in France is classified in the French naming context. I specify France as the host country. I use each business owner's modal predicted

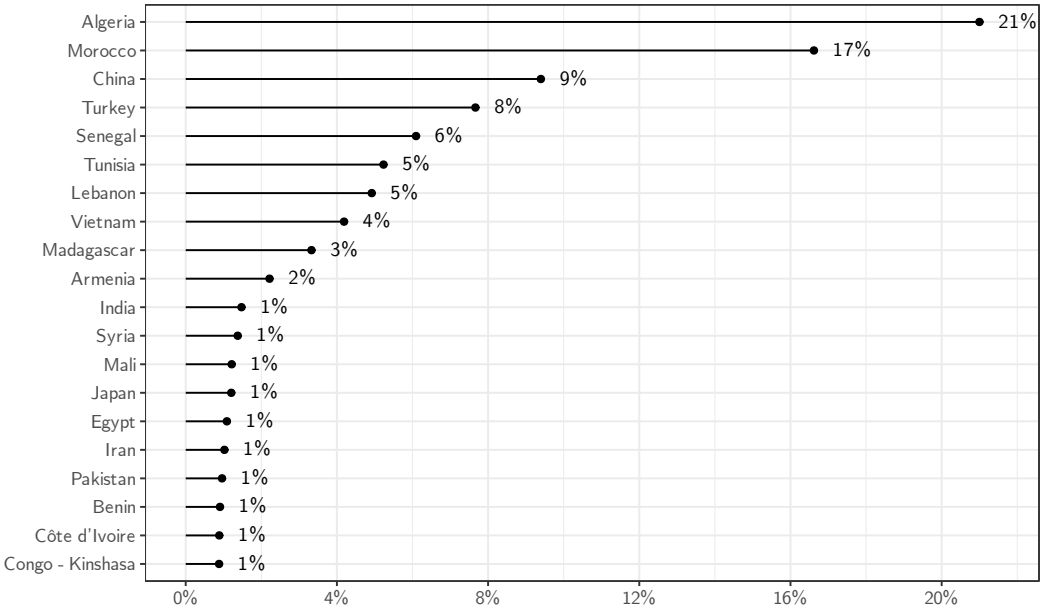


Figure B9: 20 Most Frequent Origins of Immigrant Business Owners, January 2024

Note: Each point represents the proportion of companies owned by immigrants of a given predicted origin, based on the algorithm's output, out of all immigrant businesses in January 2024. One observation corresponds to one company (which can include several businesses). Joint ownership is allocated to all the predicted origins of the owners.

origin and, on this basis, I assign to each business the two constructs used in the analysis: immigrant-associated (any origin outside Europe, the US, and Canada) versus native, and Muslim-majority origin versus other origin (see Appendix C.3 for the country list).

Table C1 illustrates the output of the Namsor algorithm for ten randomly chosen names from the RNE data. Last names are abbreviated to preserve anonymity. I use the *Origin* variable to classify the origin of each business.

C.2 Classification: Validation

I validate these predictions against two independent references. The first is the set of French administrative death records for deaths between 2000 and 2023, restricted to individuals recorded as born abroad and born in 1962 or later[25], for whom the register lists an actual country of birth. Cleaning the free-text birth country and dropping origins that I cannot match to a country produces $N = 118,759$ names for which I can compare Namsor's predicted origin to

25. I use 1962 or later to avoid classifying Algerian-born ethnically French people as having an Algerian origin. After the independence of Algeria in 1962, large numbers of ethnic French residents of Algeria migrated back to France. Removing previous years eliminates this specific classification concern.

Table C1: Examples of Output from the Namsor Algorithm

First names	Last name	Origin	Score	Alternative origins
Yannick	R.	French	0.94	Flemish,Jewish,Walloon,...
Florent	P.	French	0.95	Flemish,Swiss,Italian,...
Edwige; M.; D.	P.	French	0.96	Flemish,Jewish,German,...
Georges	J. D. F.	Portuguese	0.81	Jewish,French,German,...
Niyazi	K.	Turkish	0.96	Armenian,Azerbaijani,Georgian,...
Clement; M.; Y.	C.	French	0.96	Flemish,Jewish,British,...
Patrick	C.	French	0.96	Jewish,Flemish,British,...
Cyril	R.	Filipino	0.43	French,Hispanic,Jewish,...
Olivier; S. ;J.	D.	French	0.91	Flemish,Jewish,British,...
Thierry	G.	Hispanic	0.75	Italian,French,Jewish,...

Note: Output of the Namsor classification algorithm for 10 randomly selected names. The modal category, displayed in the "Origin" column, is used in the analysis. The last names are abbreviated to preserve anonymity and conform to French regulation. When an origin is not a country but an ethnicity (e.g., Flemish, or Catalan), I classify it as European (native) if the majority of that ethnicity lives in one or more European countries. I classify "Jewish" as native.

a recorded country of birth. The second reference is human: I drew 1,000 owner names, stratified across origin-by-religion groups, and had each name classified by two to three out of 11 raters based in France recruited on Amazon Mechanical Turk (MTurk), into one of nine categories presented in French. This produced 2,790 ratings from 140 assignments by 11 raters. I match each rating to the algorithm's label by name and, when necessary, I use the modal label across a given name's raters. Table C2 reports the confusion matrix and Table C3 reports accuracy for both references at each aggregation level.

Table C2: Validation: Confusion Matrix

Source	Binary	TP	FP	FN	TN	*N*
Insee	Immigrant/native	59874	1815	18358	38712	118759
	Muslim-majority origin	42356	3327	5364	67712	118759
	Europe/N. America	38712	18358	1815	59874	118759
MTurk	Immigrant/native	1717	351	16	706	2790
	Muslim-majority origin	912	228	112	1538	2790
	Europe/N. America	706	16	351	1717	2790

Note: Confusion matrix for the validation exercise, using 1) the Insee death records, and 2) the MTurk human-coded samples. TP, FP, FN, and TN respectively refer to True Positives, False Positives, False Negatives, and True Negatives.

Table C4 decomposes the Insee misclassification rate of the two main variables by the person's recorded origin region. The immigrant/native classification error is largest for the origins that are least represented among French immigrants: Latin America, South and Central

Table C3: Classifier Accuracy by Aggregation Level and Validation Source

Aggregation level	Metric	Insee validation	MTurk validation
Exact country	Accuracy	56.0	–
Country, excl. Europe/N. America	Accuracy	50.6	–
Region (SDG)	Accuracy	78.3	64.4
Europe/N. America (binary)	Precision	0.68	0.98
	Recall	0.96	0.67
	F1	0.79	0.79
	Accuracy	83.0	86.8
Muslim-majority origin (binary)	Precision	0.93	0.80
	Recall	0.89	0.89
	F1	0.91	0.84
	Accuracy	92.7	87.8
Immigrant/native (binary)	Precision	0.97	0.83
	Recall	0.77	0.99
	F1	0.86	0.90
	Accuracy	83.0	86.8
N (denominator)		118759	2790

Note: Insee validation uses $N = 118,759$ persons born abroad in 1962 and later recorded in the French death files for 2000–2023, comparing the Namsor origin prediction to the recorded country of birth. MTurk validation uses 1,000 names (2,790 ratings by 11 France-based raters). For the variables used in the analysis, which are binary, I report precision, recall and F_1 treating the reference (birth country for Insee, the human rating for MTurk) as truth and the algorithm as the prediction. Multiclass levels report accuracy only.

Asia outside the Muslim-majority countries, and East and Southeast Asia. These are the origins for which Namsor tends to miss immigrant origin. In contrast, the recall error is smaller for North Africa and the Middle East. The error in Muslim-majority classification is concentrated on the Muslim-origin side: it is higher for North Africa, West Africa and the Middle East (8.5-15.2%) than for Europe, East/Southeast Asia or other non-Muslim origins (3.5-6.9%). This suggests that the classification, if anything, tends to underdetect Muslim-origin.

Note that country of birth is an imperfect reference for the theoretically relevant construct of immigrant association. The theory is based on perceived association with immigration, through perceived origin. Birthplace can only capture this for foreign-born individuals. It does not speak to French-born second- or third-generation immigrants, even though a native may perceive them as being of immigrant origin. The MTurk classification is a more direct measure of the immigration-perception construct of interest. I therefore restrict the Insee val-

Table C4: Insee Misclassification Rates by True Origin Region

True origin region	N	Immigrant-binary error (%)	Muslim-binary error (%)
North Africa	30854	8.9	9.8
West Africa	11553	21.9	15.2
Middle East	6801	9.5	8.5
SE/E Asia	5188	24.0	3.5
Europe	40527	4.5	3.7
Other	23836	46.9	6.9

Note: Each cell indicates the share of individuals whose Namsor label differs from the label derived from their recorded country of birth, among the 118, 759 names in the 2000-2023 Insee death records. Origin regions are based on the recorded country of birth.

idation sample to individuals listed as being born in a foreign country. Although it evaluates the name-classification algorithm, the validation is not able to assess performance among individuals born in France. As a complement, Table C5 reports the share of individuals born in France whose names are classified as having an immigrant or Muslim origin by the Namsor algorithm. About one in ten individuals born in Metropolitan France is classified as having immigrant origins, and about 8% as Muslim-origin. These are individuals that a name-based measure but not a formal birthplace variable may classify as immigrant.

Table C5: Classification of France-Born Individuals in the Insee Records

French-born group	n	Name classified as immigrant origin (%)	Name classified as Muslim origin (%)
Metropolitan France	773395	10.0	8.3
Overseas France (DOM-TOM)	36760	10.5	5.4

Note: Classification share among people born in Metropolitan France or overseas territories, respectively.

Each MTurk worker was given a batch of twenty full names. For each name, they were asked to choose one of nine mutually exclusive categories: France, Europe or North America (other than France), North Africa and the Middle East, and Muslim/non-Muslim splits of Sub-Saharan Africa, of Central and Southern Asia, and of East and Southeast Asia. I did not include Oceania and Latin America because almost no name received that classification in the data: Australian names will typically be classified as British or American, Latin American names as Spanish or Portuguese. All the instructions were given in French. Every name was shown to

two or three raters (140 assignments in total, for 11 raters), who were paid $0.30 per twenty-name batch, with a median completion time of under three minutes. I interpret this exercise as capturing how origin reads to a French observer on the basis of name, which is similar to perceived origin, the construct of interest to the paper's mechanisms.

Because names were rated by multiple MTurk workers, I can compare Namsor's agreement with raters with raters' own agreement with one another. Table C6 reports, for each classification level, the average pairwise agreement between human raters, Krippendorff's α for the same ratings (Krippendorff 2013, p. 277), and the accuracy of the algorithm against the modal human label. I find that human raters agree on the seven-category region only 60% of the time ($\alpha = 0.51$) and on the binary constructs 83 to 86% of the time. The algorithm agrees with the modal human classification more frequently than humans agree with one another (71% on region, 90% on each analysis variable). A non-negligible part of the apparent error in Namsor classification can therefore be understood as disagreement that humans would also exhibit.

Table C6: Inter-Rater Agreement and Algorithm-vs-Human Agreement

Classification level	Human–human agreement (%)	Human–human α	Algorithm vs. modal human (%)
Region (SDG, 7-class)	60	0.51	71
Europe/NA binary	83	0.64	90
Muslim binary	86	0.69	90
Immigrant binary	83	0.64	90

Note: Agreement between human raters is the average pairwise agreement across the two or three raters of each name. α is Krippendorff's nominal alpha (Krippendorff 2013). The last column is the accuracy of the algorithm's prediction against the modal human label. Based on 1,000 names and 2,790 ratings by 11 France-based MTurk workers.

C.3 Classification: Muslim-Majority Countries and Territories

The list of countries and territories coded as having a majority-Muslim population was obtained from the Pew Research Center (2011). Countries with a Muslim proportion of the population equal to or greater than 50% were included. Although the publication dates back to 2011, the list of countries has been stable since.

Muslim-majority countries and territories: Afghanistan, Algeria, Azerbaijan, Bahrain,

Bangladesh, Brunei, Burkina Faso, Chad, Comoros, Djibouti, Egypt, Gambia, Guinea, Indonesia, Iran, Iraq, Jordan, Kazakhstan, Kuwait, Kyrgyzstan, Lebanon, Libya, Malaysia, Maldives, Mali, Mauritania, Mayotte, Morocco, Niger, Oman, Pakistan, Palestinian Territories, Qatar, Saudi Arabia, Senegal, Sierra Leone, Somalia, Sudan, Syria, Tajikistan, Tunisia, Turkey, Turkmenistan, United Arab Emirates, Uzbekistan, Western Sahara, Yemen.

Three countries, Kosovo, Bosnia and Herzegovina, and Albania, are not coded as Muslim-majority even though the report indicates proportions above 50%. This is done to enforce the rule that Muslim-associated businesses are a subset of immigrant businesses. Since these three countries are in Europe and therefore not coded as immigrant-associated, they are also not coded as Muslim-associated.

C.4 Classification: Owner-Name Coverage

I classify a business's owner origin from the owner's first and last name, as detailed in Section C. This requires the company's RNE record to include a personal name. Of the 23,146,615 establishments that were open at some point during the 2008-2021 study period, 21,007,069 include at least one classifiable owner name and therefore enter my exposure measure, and 2,139,546 do not. This represents 90.8% of all businesses. I use these businesses and drop the 9.2% that do not include a name.

The share of unclassifiable businesses falls steadily from 16.5% in 2008 to 5.8% in 2021, as displayed in Figure C1. This may reflect both the 2009 "auto-entrepreneur" reform, which simplified business creation for small sole-owner businesses, and improving name coverage in the RNE.

This could pose a threat to the validity of the results if the trend in coverage varied with the instrument. Table C7 reports regressions, across municipalities and in the main specification, of both the 2009 level and the 2009-2019 change in the local share of unclassifiable businesses on the pooled instrument. Neither coefficient is statistically distinguishable from zero. The instrument does not seem to predict change in coverage (column 1) or variation at baseline (column 2), and exposure-robust confidence sets are narrow and centered around zero. These results imply that the omission is orthogonal to the instrument, suggesting that it is non-

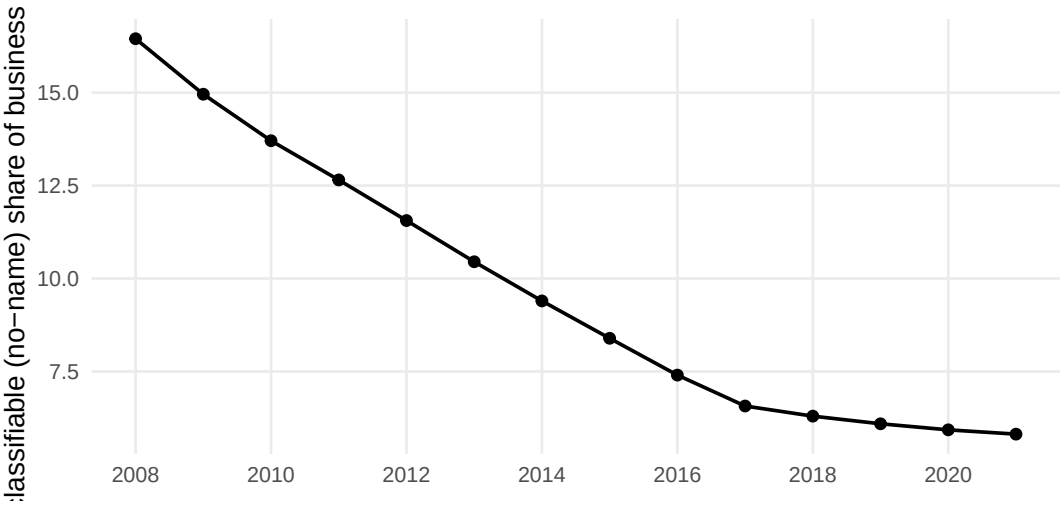


Figure C1: Share of Businesses with no Classifiable Owner Name, by Year

Note: Each point is the share of establishments open in the given year whose firm has no personal name on record and is therefore not name-classifiable. Based on the full establishment universe (23.1M businesses, 2008-2021).

systematic measurement error.

Table C7: Instrument Associations with Name Coverage

	Δ Proportion no-name	Proportion no-name (2009)
	OLS	OLS
	(1)	(2)
Instrument	−0.000	0.001
	(0.018)	(0.039)
AKM0 95% confidence set	[-0.033, 0.046]	[-0.097, 0.075]
Controls	✓	✓
Department FE	✓	✓
Municipality-type FE	✓	✓
Population weights	✓	✓
Outcome mean	−0.037	0.062
Observations	34153	34153

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. Column 1 reports a regression of the change between 2009 and 2019 in a commune's share of businesses unclassifiable because the parent company has no named owner in the RNE data on the instrument. Column 2 reports the regression of the 2009 level on the instrument. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

D Instrumental Variable: Description of the Instrument

D.1 Distribution of the Shocks (Shift)

Figure D1 displays the distribution of the industry shocks that enter the shift-share construction as the "shift." There are 87 industries coded as "storefront." The shocks are the symmetric growth rates (Davis, Haltiwanger, and Schuh 1996) of all storefront industries for businesses coded as immigrant-associated between 2009 and 2019, residualized on the corresponding symmetric growth rate for businesses coded as native, and recentered by industry size quintiles. The Davis, Haltiwanger, and Schuh (1996) symmetric growth rate uses the following formula for industry k: $g_k = \frac{n_{k,2019} - n_{k,2009}}{\left(\frac{n_{k,2019} + n_{k,2009}}{2}\right)}$. This growth rate is bounded in $[-2, 2]$ and more stable for industries with few businesses, or full entries and exits.

D.2 Concentration of the Industry Shares

The "share" component of the shift-share instrument construction is the pre-period (2005) proportion of businesses of a given municipality m belonging to each storefront industry k. Figure D2 displays the distribution of the effective number of industries across municipalities in 2005, defined as the inverse of the Herfindahl-Hirschman Index (HHI) of market concentration, where $\text{HHI} = \sum_k \left(\frac{n_k}{n_{all}}\right)^2$.

This represents the degree to which businesses are concentrated into a few industries: when the effective number of industries is low, most businesses in a municipality belong to a few industries. When it is high, businesses are more evenly distributed across a larger number of industries. Across all municipalities, the average effective number of industries is 16.6, the median is 16.3, and the weighted average using the population weights from the analysis is 23.6. This shows that the identifying variation leans on municipalities that are on average specialized in between 16 and 24 industries. The implications of a moderate number of industries for identification are detailed in the discussion of Rotemberg weights in Appendix E.4. The finite-sample implications for inference are discussed in Appendix F.1. Figure D2 shows the difference between urban and rural municipalities: unsurprisingly, businesses from a wider range of industries exist in urban areas, whereas fewer industries are represented in

rural areas.

D.3 Historical Industrial Persistence

I empirically test whether industrial agglomeration effects predicted by the literature (e.g., Ellison and Glaeser 1997; Glaeser, Kerr, and Ponzetto 2010) lead to persistence in industrial composition at the municipality level. To do so, I test whether the municipal industry composition in 2000 predicts the composition of the new businesses created between 2000 and 2010. I do so by calculating the cosine similarity between the distribution of existing 2000 industries and the distribution of 2000-2010 new businesses' industries.

To test the persistence hypothesis, I test whether the average cosine similarity (across municipalities) between existing distribution and new businesses is greater than if new businesses were instead, under a null hypothesis, drawn from the industries of similar municipalities at random. "Similar municipalities" are those in the same category of rurality or urbanity (out of seven categories) and within these categories, in the same decile for number of businesses. Results are displayed in Figure D3.

I find evidence that the observed similarity between new businesses and existing businesses in a specific municipality is stronger than it would be if businesses were just created in similar ways for similar municipalities. I can reject the null hypothesis with a p-value of 0.036.

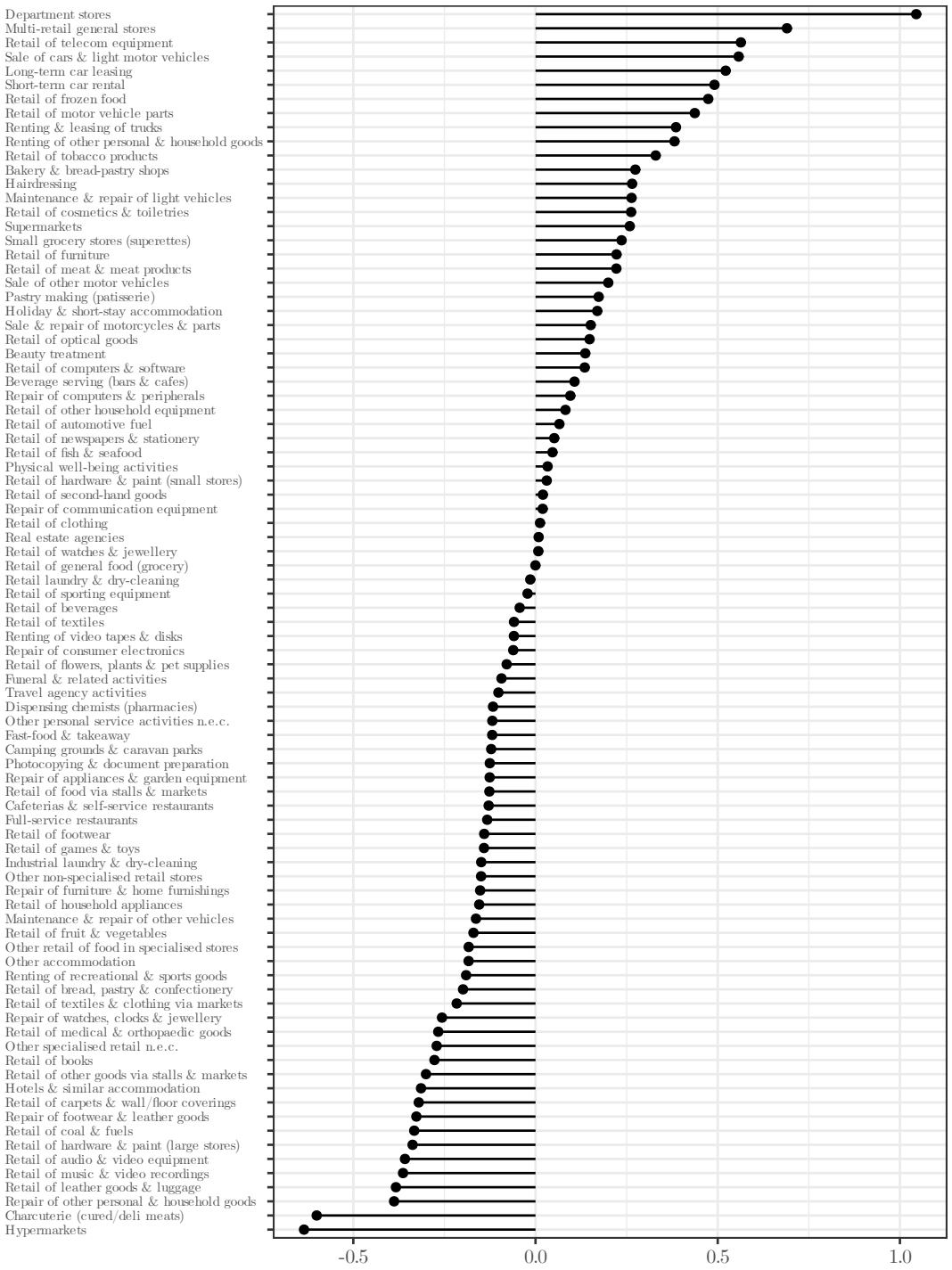


Figure D1: Distribution of the Industry Shocks

Note: Distribution of the values of the shocks that enter the shift-share instrument construction as the "shift". The shocks are the symmetric growth rates (Davis, Haltiwanger, and Schuh 1996) of all storefront industries for businesses coded as immigrant-associated between 2009 and 2019, residualized on the corresponding symmetric growth rate for businesses coded as native. The Davis, Haltiwanger, and Schuh (1996) symmetric growth rate uses the following formula for industry k: $g_k = \frac{n_{k,2019} - n_{k,2009}}{\left(\frac{n_{k,2019} + n_{k,2009}}{2}\right)}$. This value is bounded in $[-2, 2]$ and more stable for industries with few businesses, or full entries and exits. Note that these are the shocks as they enter the instruments: residualized on the corresponding industry's growth among native businesses, and recentered by quintile of baseline number of businesses.

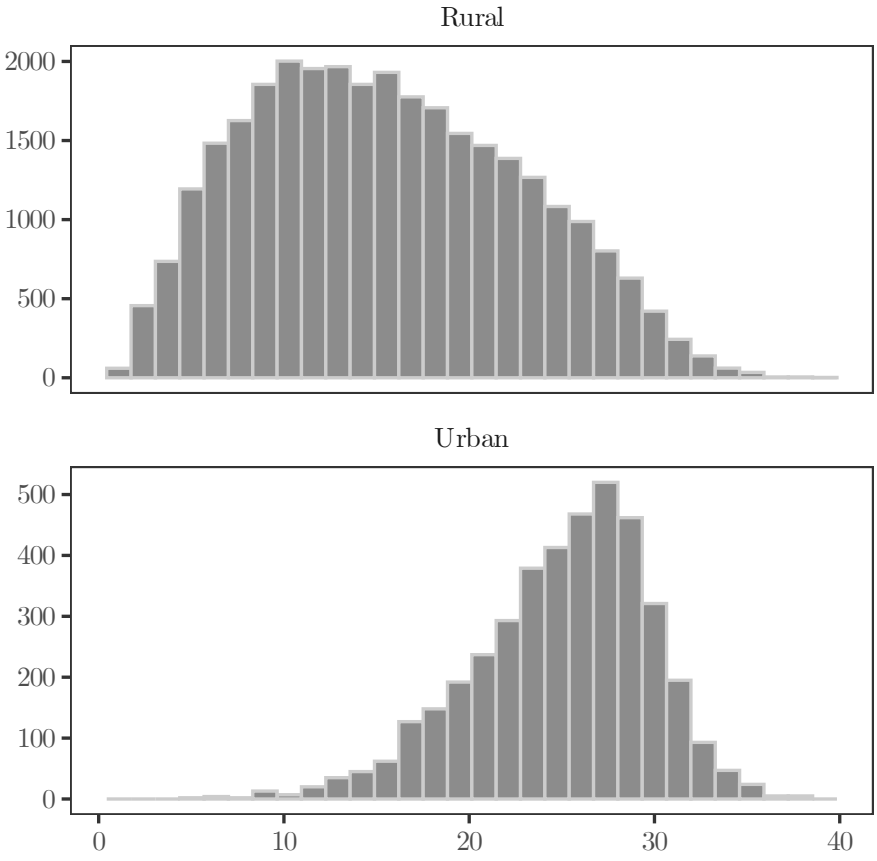


Figure D2: Concentration of the Industry Shares by Municipality

Note: Distribution of the effective number of industries across municipalities, defined as the inverse of the Herfindahl-Hirschman Index (HHI) of market concentration, where HHI $= \sum_k \left(\frac{n_k}{n_{all}}\right)^2$. When the effective number of industries is low, most businesses in a municipality belong to a few industries. When it is high, businesses are more evenly distributed across a larger number of industries.

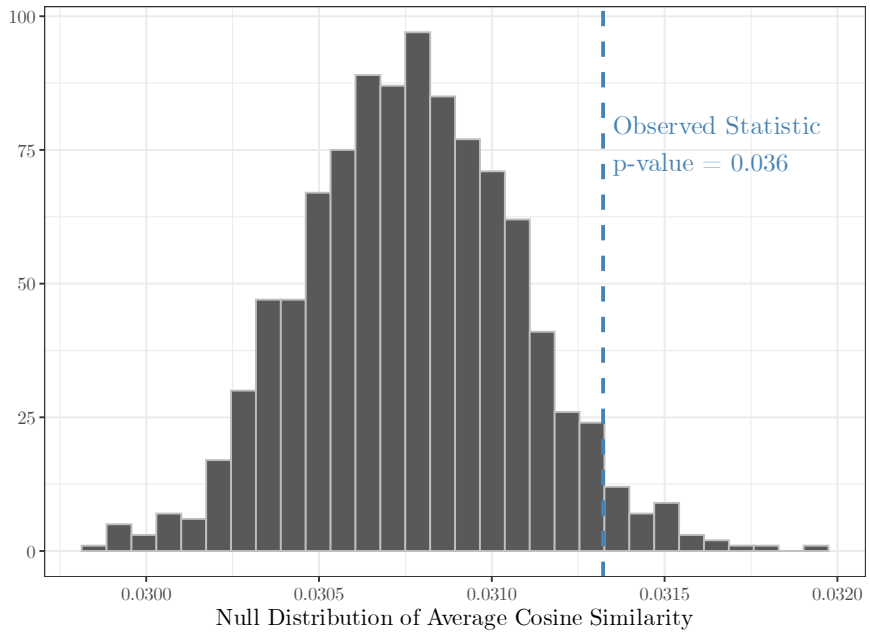


Figure D3: Test of Historical Industrial Persistence

Note: Distribution of average cosine similarity in the industrial mix of businesses across municipalities between 2000 and 2010 under the sharp null hypothesis of no industrial persistence. Under this sharp null, businesses of municipalities in the same municipality-type categories are permuted at random. The permutation p-value is consistent with industrial persistence.

E Instrumental Variable: Tests of Assumptions' Validity

This section of the appendix tests the plausibility of the identifying assumptions of the main shift-share design. Section E.1 focuses on the conditional exogeneity assumption, Section E.2 on the exclusion restriction, and Sections E.3 and E.4 on the many shocks assumption.

E.1 Balance Checks

I conduct two sets of balance tests: at the industry-shock level in Section E.1.1 and at the municipality level in Section E.1.2.

E.1.1 Shock-Level Balance

Following Borusyak, Hull, and Jaravel (2022), I regress the recentered shocks on predetermined industry characteristics, weighting each industry k by its average exposure $s_k = \sum_i e_i s_{ik}$, where e_i are the municipality population weights, and s_{ik} is the share of all pre-period businesses in municipality i operating in industry k. Estimates are displayed in Table E1.

Two characteristics are imbalanced: industries with a lower baseline rate of immigrant ownership and non-food industries receive more positive shocks. This is consistent with a regression-to-the-mean pattern: industries where immigrant ownership is already saturated at baseline may have less room to grow further. These two variables are likely capturing a single underlying imbalance, as many immigrant businesses at baseline operate in the food industry.

This imbalance could pose a challenge to exogeneity if industries with a higher baseline share of immigrant-associated businesses are located in municipalities with different potential outcomes. Aggregating this baseline characteristic to the municipality level, using industry weights (the shares of the shift-share), I find that the instrument has a correlation of -0.42 with the weighted average of immigrant-baseline shares at the municipality level. This raises the possibility that the instrument affects the vote through alternative characteristics of municipalities exposed to such industries, rather than through the exposure to immigrant-associated businesses treatment.

Table E1: Shock-Level Balance Tests

	Industry shock
	OLS
	(1)
Immigrant share	-0.084^{***}
	(0.028)
Food service	-0.133^{*}
	(0.067)
Industry size	0.024
	(0.042)
Native growth	0.050
	(0.040)
Exposure weights	✓
Observations	87
Joint F: all 4 characteristics	5.05 ($p < 0.001$)

Note: HC1 standard errors. The outcome is the recentered immigrant-business industry shock defined in Equation 5. The table reports a weighted regression of this shock on the listed predetermined industry characteristics, using industries' average exposure shares as weights. The joint Wald test tests the null that all reported coefficients are zero.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Conditioning on baseline immigrant intensity is feasible but produces weak identification: as Table E2 shows, although conditioning on the baseline share of immigrants mechanically removes the observed shock-level imbalance, it collapses the first-stage power, with the F-statistic going from approximately 14.1 to 5.9. The resulting SSIV estimate is still positive at 0.146, but its AKM0 confidence set is unbounded, so the magnitude and sign are not meaningfully identified. This suggests that the share of each industry's businesses that are immigrant-associated at baseline is important for first-stage relevance.

This helps understand the assumption required for identification: conditional on the shock-level and municipality-level controls, the baseline share of immigrants in each industry must primarily predict industries' capacity to experience immigrant-business growth rather than proxy for unobserved industry processes that independently affect subsequent RN voting. Some results enhance the plausibility of this assumption: first, the pretrend tests in Section E.1.2 are flat. Second, the instrument predicts neither the immigrant population nor native-business exposure, and the main specification adjusts for potential confounders within de-

Table E2: Shock-Level Result Adjusted for Immigrant Intensity

	Δ RN Vote Share	
	2SLS (1)	2SLS (2)
Δ Exposure (log)	0.102**	0.146**
	(0.020)	(0.040)
AKM0 95% conf. set	[0.055, 0.180]	[unbounded]
Adjusted		✓
Controls	✓	✓
Department FE	✓	✓
Municipality-type FE	✓	✓
Population weights	✓	✓
Outcome mean	0.205	0.205
Observations	34659	34659
AKM first-stage F	14.07	5.89

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. This table displays the 2SLS estimates with and without conditioning on the variable on which the industry shocks are imbalanced. Both columns show positive estimates of a similar order of magnitude but Column 2 shows that the main instrument is too weak to produce a bounded AKM0 confidence set once conditioning on the baseline share of immigrant-owned businesses in each industry. This problem does not appear for the Muslim-oriented instrument (see Table G4).
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

partment and municipality-type fixed effects.

More reassuringly, the alternative Muslim-oriented instrument specification described in Section G shows that conditioning on the baseline share of immigrants in each industry does not affect the magnitude or significance of the results. The main difference is that the stronger first-stage of the Muslim-oriented instrument allows this adjustment to be made while maintaining sufficient first-stage power to produce bounded exposure-robust confidence intervals for the 2SLS estimate.

E.1.2 Instrument-Level Balance and Pretrends

The previous section estimated balance at the level of the industry shocks. In this section, I run balance checks at the municipality level. Table E3 reports estimates from regressions of the baseline controls included in the main specification on the instrument. Each regression

adjusts for all covariates except the one being tested.

Table E3: Municipality-Level Balance Tests

	College (%, 2009)	High school (%, 2009)	Unemployment (%, 2009)
	OLS	OLS	OLS
	(1)	(2)	(3)
Instrument	−0.071*	−0.020	−0.049***
	(0.034)	(0.015)	(0.009)
AKM0 95% conf. set	[-0.142, 0.006]	[-0.057, 0.010]	[-0.066, -0.027]
Controls	✓	✓	✓
Department FE	✓	✓	✓
Municipality-type FE	✓	✓	✓
Population weights	✓	✓	✓
Outcome mean	0.189	0.158	0.049
Observations	34659	34659	34659
	Income p. cap. (log, 2009)	Dist. to highway (log, m)	Population (log, 2009)
	OLS	OLS	OLS
	(4)	(5)	(6)
Instrument	−0.033	−3.239**	0.948**
	(0.093)	(0.971)	(0.321)
AKM0 95% conf. set	[-0.250, 0.159]	[-5.142, -0.819]	[0.228, 1.646]
Controls	✓	✓	✓
Department FE	✓	✓	✓
Municipality-type FE	✓	✓	✓
Population weights	✓	✓	✓
Outcome mean	9.636	9.121	6.219
Observations	34659	34659	34659

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. Each column reports estimates from a regression analogous to Equation 6, but in which each variable of interest is substituted as the outcome instead of exposure. Variables with a % symbol in parentheses are proportions.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Columns 1, 2, and 4 show that the instrument is balanced on municipalities' proportion of college-educated residents, high-school educated residents, and income per capita. Column 3 displays a statistically significant imbalance on the proportion of municipality residents unemployed: there is a negative relationship between the instrument and unemployment. Because unemployment is typically associated with a higher level of support for far-right parties (Algan et al. 2017; Colantone and Stanig 2018), the direction of this relationship is less worrying than the opposite would be. However, unemployment could be related to unobserved changes between 2009 and 2019 that could confound the result. Similarly, the significant imbalances on distance to highway and population (in level) in Columns 5 and 6 could be correlated with unobserved 2009-2019 change confounders. None of these is itself a violation of the conditional exogeneity assumption, as they are adjusted for in the main specification

and the identifying assumption is conditional on these variables (Borusyak, Hull, and Jaravel 2022), but they help understand what violations of conditional exogeneity would look like.

Table E4 reports an estimate of the relationship between pre-period change in the outcome variable (municipal RN vote share) and the value of the instrument. This is analogous to a pretrends test in difference-in-differences designs. Under a violation of the conditional exogeneity assumption, one might expect the instrument to take a larger value in municipalities where the far right already sits on a specific trajectory, or more worryingly in municipalities where far-right support is already rising. Table E4 shows that there is no detectable relationship between pre-period change in RN support and the instrument. The coefficient is not statistically significant, and is substantively close to zero.

Table E4: Instrument: Pretrend Test

	Δ RN Vote Share (2004-09)
	OLS
	(1)
Instrument	−0.009
	(0.016)
AKM0 95% confidence set	[-0.040, 0.030]
Controls	✓
Department FE	✓
Municipality-type FE	✓
Population weights	✓
Outcome mean	−0.085
Observations	34659

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. This table shows that pre-period trends in RN support are not predicted by the instrument. Change is measured between the previous European election (2004) and the European election marking the beginning of the study period (2009).
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

E.2 Placebo First Stages

The previous section examined the plausibility of the conditional exogeneity assumption. This section turns to the exclusion restriction. Table E5 reports estimates of alternative first-stage specifications in which the endogenous treatment (exposure to immigrant-associated busi-

nesses) is replaced by variables measuring alternative causal paths.

Table E5: Instrument: Tests of Alternative Causal Channels

	Δ Native storefront exp.	Δ Foreign-born pop. (own)	Δ Foreign-born pop. (around)	Δ Immig. non-storefr. exp.
	OLS	OLS	OLS	OLS
	(1)	(2)	(3)	(4)
Instrument	0.001	−0.207	0.082	0.204
	(0.095)	(0.198)	(0.076)	(0.140)
AKM0 95% confidence set	[-0.270, 0.162]	[-0.669, 0.206]	[-0.107, 0.231]	[-0.162, 0.465]
Controls	✓	✓	✓	✓
Department FE	✓	✓	✓	✓
Municipality-type FE	✓	✓	✓	✓
Population weights	✓	✓	✓	✓
Outcome mean	0.407	0.009	0.051	0.727
Observations	34659	34659	34659	34659

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. This table shows estimates from alternative first-stage specifications where the endogenous treatment is replaced by variables measuring alternative causal channels that could suggest violations of the exclusion restriction. The estimates show that the main instrument does not predict change in exposure to native storefronts, in the log count of foreign-born residents in the municipality, in the log count of foreign-born residents around the municipality, or in the number of immigrant-associated non-storefront businesses.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Column 1 shows that the instrument does not predict change in exposure to native businesses, suggesting it is therefore immigrant-specific. Columns 2 and 3 show that the instrument does not predict change in the log count of foreign-born residents (in and around the municipality, respectively), suggesting it is business-specific and not just capturing change in exposure to immigrants. Finally, Column 4 shows that the instrument does not predict change in exposure to non-storefront businesses. This helps isolate the specificity of the instrument: growth in storefront industries is predicted by a national trend in business creation in these particular industries. "Foreign-born" is an imperfect proxy for second- or third-generation French citizens perceived as immigrants by natives. The absence of a significant relationship between immigrant-associated non-storefront businesses, which are classified on the basis of names and therefore include some of these second- or third-generation business owners, and the instrument helps alleviate potential concerns that the treatment could simply proxy for variation in immigrant-associated individuals.

E.3 Shock-Level Equivalence

Borusyak, Hull, and Jaravel (2022) show that a shift-share instrumental variable analysis where the shocks are conditionally exogenous can equivalently be expressed as a shock-level analysis, where the shocks (here, the rates of industry growth) are weighted by the exposure shares (here, the pre-period municipality-level proportion of businesses in each industry). As a sanity check, Table E6 displays the regression results. The estimate is exactly identical to the municipality-level 2SLS result from Table 1 in the main text, confirming shock-level equivalence.

Table E6: Equivalent Shock-Level Result

	Δ Shock-level outcome (RN Vote Share)
	2SLS
	(1)
Shock-level treatment	0.102***
	(0.023)
Controls (residualized)	✓
Department FE (residualized)	✓
Municipality-type FE (residualized)	✓
Population weights	✓
Outcome mean	0.205
Observations	87

Note: Standard errors assume homoskedasticity. Weighted shock-level regression equivalent to the 2SLS shift-share analysis (Borusyak, Hull, and Jaravel 2022). Observations are the number of industries. The industry weights are displayed in Figure E1. The coefficient estimate is identical to the 2SLS estimate from the main text, confirming shock-level equivalence of the shift-share analysis. The basic shock-level equivalence does not reproduce the correct standard errors. These incorrectly assume homoskedasticity.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

The exposure-share weights are displayed in Figure E1. The figure shows that the largest weight (restaurants) represents a little under 8% of the shocks, and the sum of the top ten weights represents 46% of the shocks. Despite moderate concentration of the shocks, no industry or small group of industries carries the instrument on its own.

An effective sample size for this shock-level equivalent regression can be estimated as the inverse HHI of the shocks' weights by their share across municipalities. Here, the effective number of shocks is 31.7, out of 87 nominal industries. This is not sensitive to the decision made in the analysis to weight by baseline municipality population: the effective number of

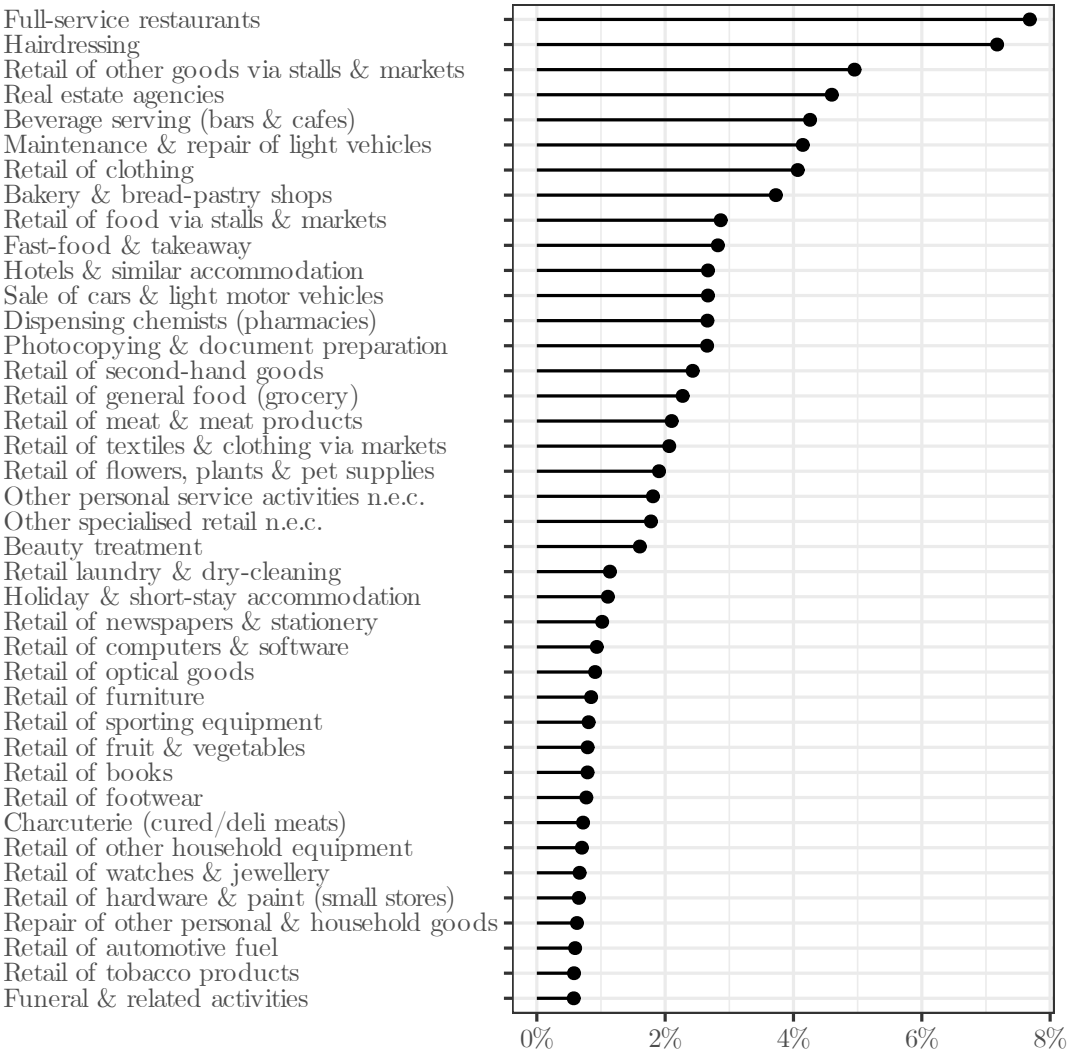


Figure E1: Weights of the Industry Shocks

Note: This figure displays the 40 industries with the largest industry-level weights as defined by the equivalence result of Borusyak, Hull, and Jaravel (2022). The largest weight represents a little under 8% of the shocks, and the sum of the top ten weights represents 46% of the shocks. Despite moderate concentration of the shocks, no industry or small group of industries carries the instrument on its own.

shocks is 28 when the municipalities are left unweighted.

Following Borusyak, Hull, and Jaravel (2022), I also estimate within-group correlation of the industry shocks. The APE industry codes are organized as nested sections containing divisions, groups, and class codes. I estimate a hierarchical random-intercept model over the industry classification. The estimated variance is concentrated at the three-digit group level, which accounts for 18.5% of total shock variance (with a parametric-bootstrap standard error of 0.11). The section, division, and class-level variance components are estimated to be zero. This suggests that inference should allow for correlation between shocks within three-

digit groups. There are 28 such groups nominally and the effective number is 12.4, compared with 31.7 for the 87 individual shocks. I replicate the analysis with exposure-robust standard errors clustered at the three-digit group level. The positive estimate remains statistically distinguishable from zero with a 95% AKM0 confidence set of [0.054, 0.193] and a p-value of 0.0097, though we should interpret these results cautiously due to the limited number of effective groups.

E.4 Rotemberg Weights

Figure E2 displays the 40 industries with the largest absolute Rotemberg weights in the shift-share IV decomposition of Goldsmith-Pinkham, Sorkin, and Swift (2020). They show that a shift-share IV estimate can be written as a weighted combination of just-identified estimates associated with the individual industry-share instruments. An industry's Rotemberg weight depends on the magnitude of its shock and on the covariance between its residualized local exposure share and the endogenous treatment. The weights sum to one but do not have to be positive. My identification argument follows the conditional shock-exogeneity framework of Borusyak, Hull, and Jaravel (2022), rather than relying on the exogeneity of the local industry shares. Rotemberg weights therefore do not directly test the identifying assumption and the discussion of the weights estimated in Section E.3 is more appropriate. Their absolute magnitudes nevertheless provide a useful complementary diagnostic of which industry components exert the greatest influence on the 2SLS estimate.

The inverse Herfindahl index of the normalized absolute Rotemberg weights is 14.7.[26] This means that identifying influence in the Goldsmith-Pinkham, Sorkin, and Swift (2020) decomposition is as concentrated as it would be if approximately 15 industries received equal absolute weights. The largest five industries account for slightly more than half of total absolute weight. This indicates meaningful concentration, although the instrument is not driven by a single industry.

To allow municipalities influenced by the same shocks to have correlated residuals, the

26. Let $q_k = |\alpha_k| / \sum_j |\alpha_j|$, where α_k denotes the Rotemberg weight of industry k. I calculate $N_\alpha = 1/\sum_k q_k^2$. This is an effective number of industries by absolute Rotemberg weight, not the effective sample size of the equivalent shock-level regression in Borusyak, Hull, and Jaravel (2022).

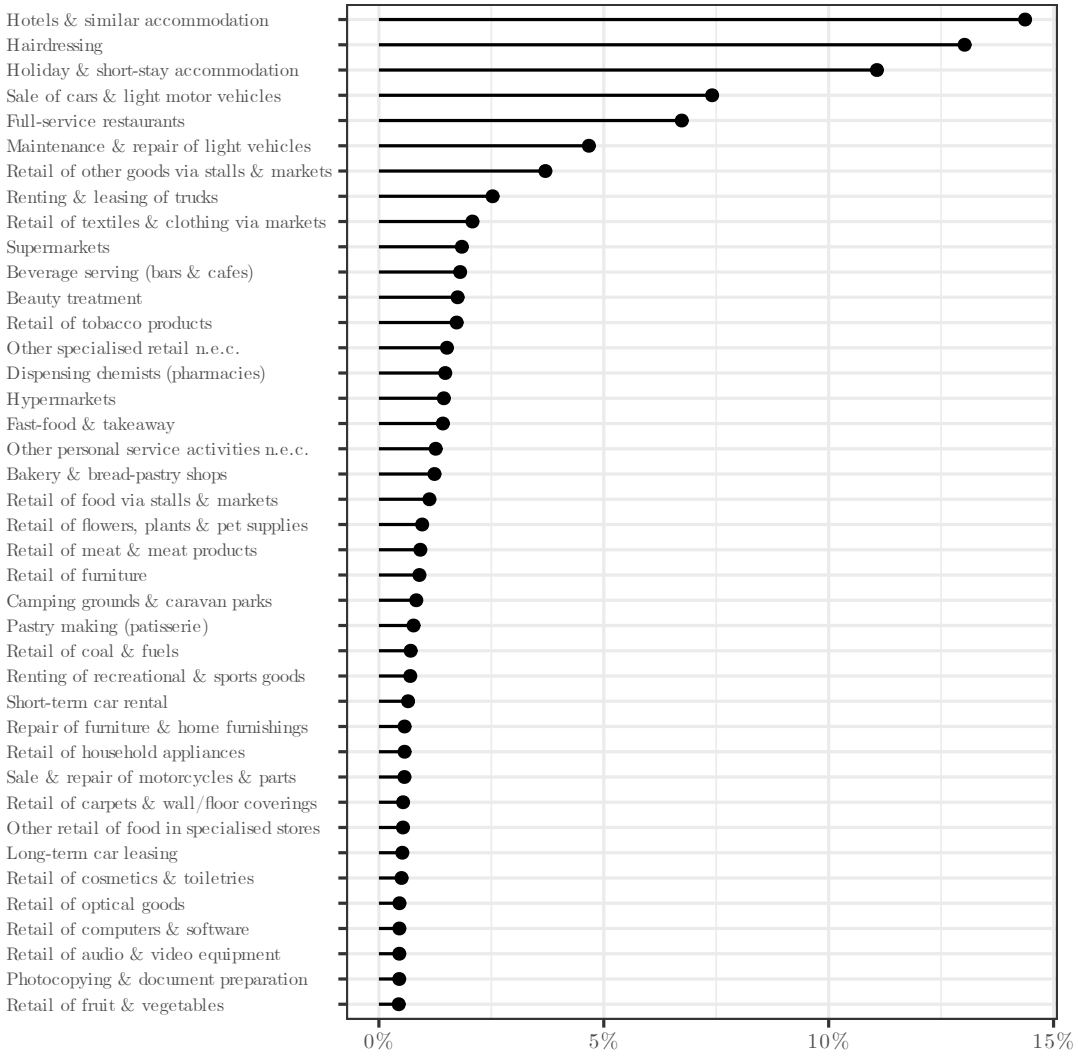


Figure E2: Absolute Rotemberg Weights

Note: This figure displays the 40 industries with the largest absolute Rotemberg weights. The ranking is substantively intuitive: hotels, hairdressing, short-stay accommodation, car sales, and full-service restaurants receive the largest weights. The weights shown are normalized by the sum of their absolute values and therefore describe concentration in absolute identifying influence rather than signed contributions to the first stage.

paper uses Adão, Kolesár, and Morales (2019) standard errors throughout the shift-share analysis. The asymptotic properties of this standard error estimator rely on the number of shocks growing large. However, the 31.7 effective shocks suggest that a finite-sample approach is appropriate, at least as a robustness test. I show in Appendix F.1 that the main estimate is robust to using the shift-share randomization inference procedure suggested by Alvarez, Ferman, and Oliveira (2022).

The concentration of absolute Rotemberg weights also raises the possibility that the estimate is disproportionately sensitive to a small number of industries. Figure E3 reports esti-

mates obtained after excluding each of the 20 industries with the largest absolute Rotemberg weights in turn. The resulting 2SLS point estimates are close to the baseline estimate, indicating that no single high-weight industry drives the result.

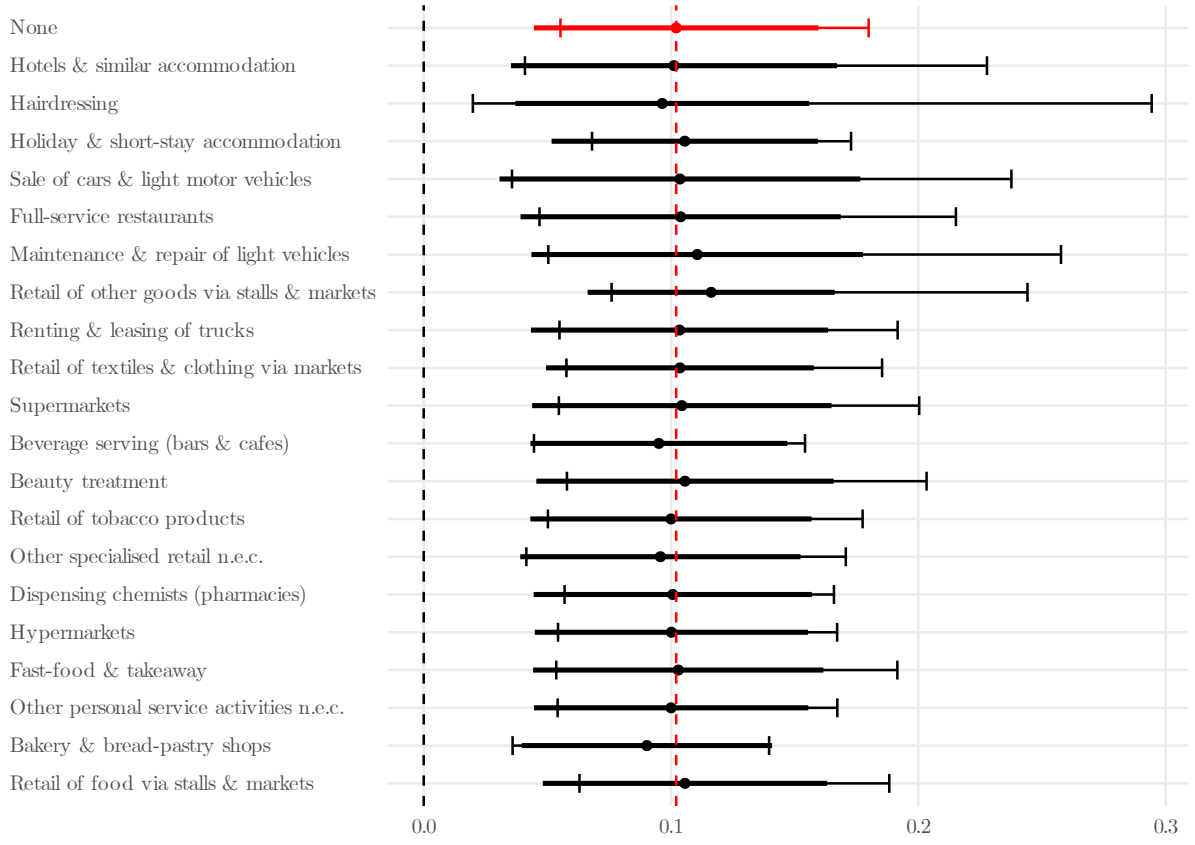


Figure E3: Leave-One-Out 2SLS Estimate for Each Industry Dropped

Note: this figure displays the main 2SLS estimate (in red) as well as 2SLS estimates re-estimated after excluding each of the top-20-Rotemberg-weight industries one at a time. The thick confidence intervals use Conley standard errors (50 km) and the thin confidence intervals are the AKM0 95% confidence intervals (Adão, Kolesár, and Morales 2019, p. 20). The latter are valid when the instrument is weak, which it becomes when dropping some of the shocks.

The results are similarly robust when dropping substantively important groups of industries one at a time, as displayed in Figure E4. The 2SLS point estimates are stable when hospitality, food service, personal service, or automobile industry groups are respectively dropped, where the groups are defined by the two-digit APE codes. All estimates remain significant using AKM0 inference except for the automobile exclusion: the first stage weakens to an extent where the second stage AKM0 95% confidence set explodes. The estimate remains significant when using Conley standard errors. The power of the first stage collapses when all of these consumer-facing industries are jointly dropped, as variation in these shocks is providing the

lion's share of the identifying variation.

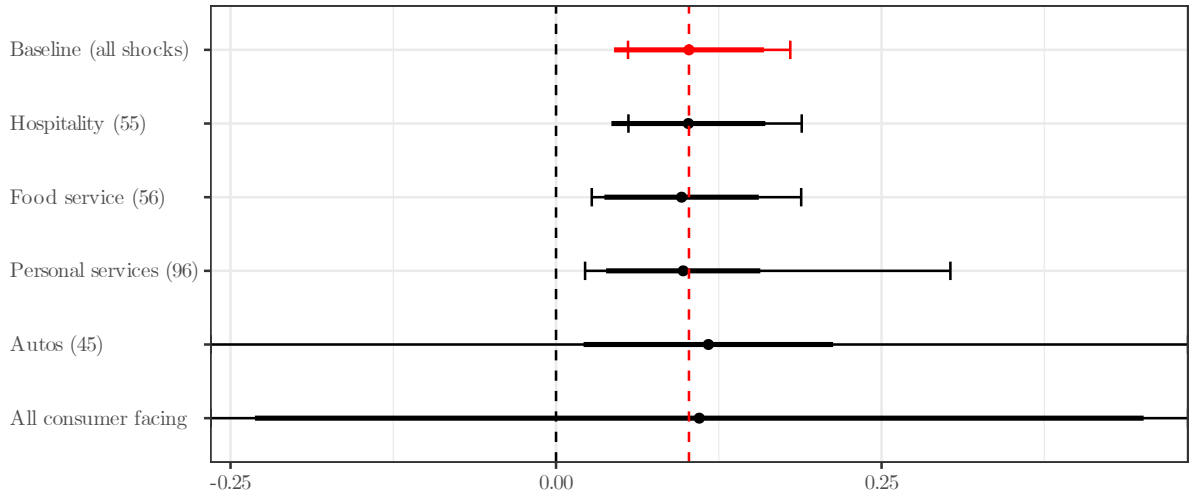


Figure E4: Leave-One-Out 2SLS Estimate for Each Important Group

Note: This figure displays the main 2SLS estimate (in red) as well as a leave-one-out 2SLS re-estimated when each of four substantively important groups of industries (that also receive large Rotemberg weights) is dropped. The thick confidence intervals use Conley standard errors and the thin confidence intervals are the AKM0 asymmetric confidence intervals (Adão, Kolesár, and Morales 2019, p. 20) The latter is valid when the instrument is weak, which it becomes when dropping some of the shocks.

Note that the Rotemberg weights of a significant number of industries (38/87 representing about 29.7% of the summed absolute weights) are negative. The components of the shift-share instrument coming from these industries (which include hairdressing, the second largest weight in absolute value) have a first-stage relationship with the treatment whose sign is opposite to the net first-stage. While positive Rotemberg weights are not necessary for the Borusyak, Hull, and Jaravel (2022) approach to be valid, they can raise concerns about sensitivity to a few shocks. This is addressed by the leave-one-out test in Figure E3. In addition, the existence of negative Rotemberg weights indicates that the main estimate should not be interpreted as a convex average of the industry-specific 2SLS estimates.[27]

27. The industry-specific instruments would be each individual shock, weighted for each municipality by the corresponding share of that municipality's businesses in this particular industry.

F Instrumental Variable: Robustness Tests

F.1 Alternative Approaches to Inference

Conley Inference

Table F1 displays the main estimates with Conley standard errors. In this context, because the number of industry shocks is modest, AKM0 exposure-robust inference tends to be more conservative than Conley inference. The standard errors and confidence intervals displayed in Table 1 of the main text are therefore preferred.

Randomization Inference

The asymptotic properties of the AKM0 inference procedure from Adão, Kolesár, and Morales (2019), accounting for the shock-level dependence, rely on the number of shocks growing large with sufficiently dispersed exposure. As the number of effective shocks in this case is finite and moderate (the exposure-weighted effective number is 31.7), I also implement a finite-sample-robust procedure using randomization inference based on the permutation approach introduced by Borusyak and Hull (2023) and adapted by Alvarez, Ferman, and Oliveira (2022).

I compute the test statistic from Alvarez, Ferman, and Oliveira (2022, p. 6) but use outcome $\tilde{Y}$, treatment $\tilde{X}$, and share matrix $\tilde{S}$ respectively residualized on the main specification covariates. I also incorporate the population weights. With $p = \tilde{S}'\tilde{Y}$, $q = \tilde{S}'\tilde{X}$, and $r_b = p - bq$, where b is the value of the effect under the null hypothesis, we have for industry k: $r_b, k = p_k - bq_k$. The adapted test statistic is:

$$|T_1(g, b)| = \frac{\left|\sum_k g_k r_{b,k}\right|}{\sqrt{\sum_k g_k^2 r_{b,k}^2}},$$

where g_k is the shock for industry k. I test the sharp null of $b = 0$. I then randomly permute the industry shocks within the size strata used to recenter the shocks (Borusyak, Hull, and Jaravel 2022).

As Figure F1 displays, I find a 0.03 randomization inference p-value. The significance of the main finding is therefore robust to a finite-sample procedure that complements the exposure-

Table F1: Complete Estimates (Conley Inference)

	Δ Exposure (log)	Δ RN Vote Share	
	First stage	Reduced form	2SLS
	(1)	(2)	(3)
instrument	0.917***	0.094***	
	(0.143)	(0.021)	
Δ Exposure (log)			0.102***
			(0.029)
Foreign pop. (log, 2009)	0.015***	−0.004***	−0.005***
	(0.005)	(0.001)	(0.001)
Foreign pop. around (log, 2009)	0.012	−0.003***	−0.004**
	(0.009)	(0.001)	(0.002)
Population (log, 2009)	−0.015	−0.002	−0.000
	(0.015)	(0.004)	(0.003)
Population around (log, 2009)	0.003	0.002*	0.002
	(0.019)	(0.001)	(0.002)
Prop. unemployed (2009)	−1.201***	0.208***	0.330***
	(0.331)	(0.080)	(0.081)
Prop. college graduate (2009)	−0.103	−0.317***	−0.307***
	(0.105)	(0.036)	(0.041)
Prop. HS graduate (2009)	0.817***	−0.189***	−0.272***
	(0.223)	(0.038)	(0.063)
Native businesses (log, 2009)	−0.002	−0.004	−0.004
	(0.014)	(0.004)	(0.003)
Dist. to highway (log)	−0.016***	−0.001	0.001
	(0.004)	(0.001)	(0.001)
Per capita income (log, 2009)	−0.039	0.017	0.021
	(0.040)	(0.015)	(0.017)
Vote share (2009)	0.021	0.105	0.103
	(0.170)	(0.067)	(0.068)
Pre-period change (2004-2009)	−0.397***	−0.371***	−0.331***
	(0.136)	(0.037)	(0.043)
Controls	✓	✓	✓
Department FE	✓	✓	✓
Municipality-type FE	✓	✓	✓
Population weights	✓	✓	✓
Outcome mean	0.635	0.205	0.205
Observations	34659	34659	34659

Note: Conley standard errors (50km). The instrument refers to the shift-share instrument, as described in Section 5.1. Exposure (log) refers to exposure to immigrant-associated storefront businesses, as described in Section 4.2. AKM exposure-robust standard errors are reported in Table 1 of the main text.

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

robust asymptotics of Adão, Kolesár, and Morales (2019).

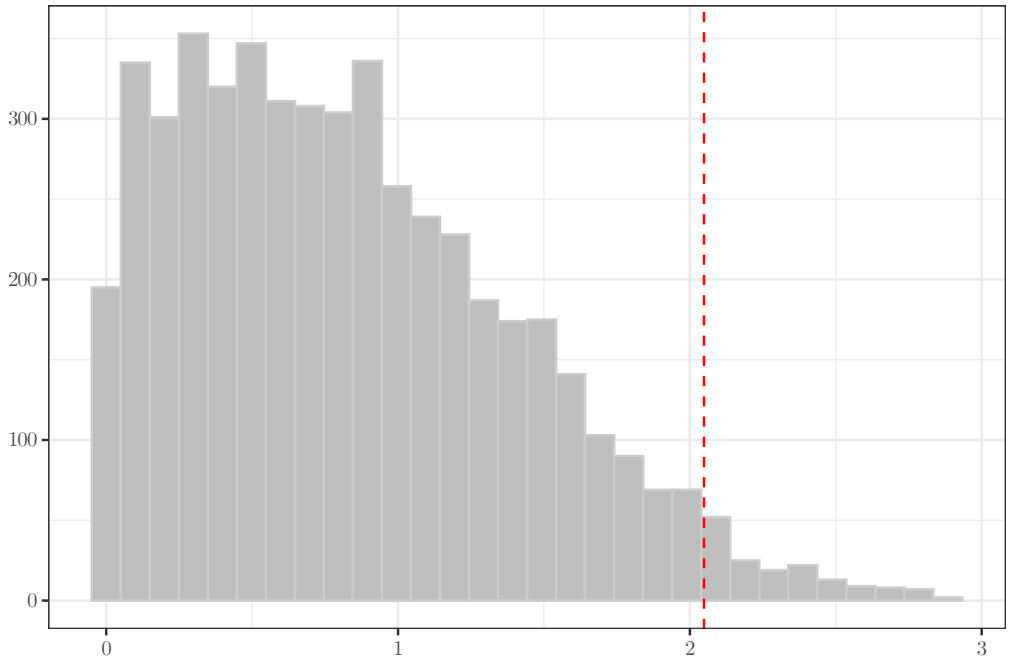


Figure F1: Randomization Inference: Distribution of the Test Statistic

Note: This figure displays the null distribution of the test statistic in the randomization inference test inspired by Alvarez, Ferman, and Oliveira (2022) for the 2SLS estimate, for 5,000 permutations. Randomization inference through an industry shock-permutation procedure is used because the asymptotic properties of AKM0 inference rely on a large number of shocks, whereas the effective number of shocks here is relatively small (31.7). The resulting p-value of 0.03 is consistent with the results from AKM0 and Conley inference.

F.2 Other Elections

Table F2 displays the results of the main IV specification using the 2007 and 2017 presidential elections instead of the 2009 and 2019 European elections. The magnitude and significance of the results are not meaningfully different. While the presidential election in France is the race that determines the executive power, and therefore usually sees higher turnout and receives more attention, the main analysis uses the 2009 and 2019 European elections to optimize covariate availability. The descriptive panel analysis in Section 6 uses all available elections between 2008 and 2021.

Table F2: Main Estimates Using the 2007 and 2017 Presidential Elections

	Δ Exposure (log)	Δ RN Vote Share	
	First stage	Reduced form	2SLS
	(1)	(2)	(3)
Instrument	1.026***	0.095***	
	(0.221)	(0.029)	
Δ Exposure (log)			0.092***
			(0.023)
Foreign pop. (log, 2008)	0.013**	−0.003***	−0.004***
	(0.005)	(0.000)	(0.001)
Foreign pop. around (log, 2008)	0.008	−0.002**	−0.003**
	(0.010)	(0.001)	(0.001)
Population (log, 2008)	−0.009	−0.001	−0.000
	(0.013)	(0.003)	(0.003)
Population around (log, 2008)	0.005	0.001	0.001
	(0.020)	(0.001)	(0.002)
Prop. unemployed (2008)	−1.115***	−0.038	0.065
	(0.400)	(0.057)	(0.067)
Prop. college graduate (2008)	−0.250**	−0.172***	−0.149***
	(0.112)	(0.025)	(0.028)
Prop. HS graduate (2008)	0.829***	−0.017	−0.093**
	(0.216)	(0.029)	(0.039)
Native businesses (log, 2007)	−0.000	−0.004	−0.004
	(0.010)	(0.004)	(0.004)
Dist. to highway (log)	−0.010***	−0.001**	0.000
	(0.004)	(0.000)	(0.000)
Per capita income (log, 2007)	−0.025	−0.008	−0.006
	(0.039)	(0.010)	(0.012)
Vote share (2007)	−0.052	0.262***	0.267***
	(0.214)	(0.059)	(0.061)
Pre-period change (2002-2007)	−0.509***	−0.166***	−0.119***
	(0.153)	(0.032)	(0.039)
AKM0 95% confidence set		[0.033, 0.161]	[0.042, 0.172]
Controls	✓	✓	✓
Department FE	✓	✓	✓
Municipality-type FE	✓	✓	✓
Population weights	✓	✓	✓
Outcome mean	0.654	0.138	0.138
Observations	34653	34653	34653
AKM first-stage F	21.5		

Note: For the coefficients on "Instrument" and "Δ Exposure (log)", parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). In Columns 2 and 3, stars are from AKM0 tests. For the other coefficients, parentheses report Conley (50 km) standard errors. Δ Exposure (log) is the change, between 2007 and 2017, in the exposure variable defined in Equation 1. Most of the baseline covariates are measured in 2008 and not 2007 because of data availability limits. This is why the main specification uses the 2009 and 2019 European elections, for which all covariates are available at both ends of the period.

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

F.3 Alternative Distance Decay

I replicate the main instrument specification but change the unit of distance δ used to construct the measure of exposure defined in Equation 1. I use 1000 meters instead of 100 meters in the denominator, making decay considerably slower. In practice, when measuring exposure to businesses for a given municipality, the 1,000-meter scale allocates more weight to businesses located away from that municipality's downtown area than the 100-meter version. In the main specification ($\delta = 100$), a business located one kilometer away ($d = 1000$ meters) from the municipality receives a decay weight of $\frac{1}{1+1000/\delta} = \frac{1}{1+1000/100} = \frac{1}{11}$. In this $\delta = 1000$ specification, the weight is $\frac{1}{2}$. In practice, the default specification implies that a business located in the center of the municipality receives eleven times the weight of businesses a kilometer away. In contrast, in the $\delta = 1000$ specification, it only receives twice as much. Considering that these businesses on the outskirts are likely to affect daily life and municipal identity considerably less, I select the $\delta = 100$ specification as the main one.

Table F3 shows that the 2SLS estimates with the two decay specifications are similar in sign, magnitude, and significance. The estimated effect has a marginally larger magnitude but the difference with the main specification is not significant.

Table F3: Robustness to Variation in Distance Decay

	Δ Exposure (log)	Δ RN Vote Share		Δ Exposure (log)	Δ RN Vote Share	
	First stage	Reduced form	2SLS	First stage	Reduced form	2SLS
	(1)	(2)	(3)	(4)	(5)	(6)
Instrument	0.917***	0.094**		1.741***	0.210***	
	(0.244)	(0.029)		(0.371)	(0.063)	
Δ Exposure (log)			0.102**			0.121***
			(0.020)			(0.023)
AKM0 95% conf. set		[0.029, 0.157]	[0.055, 0.180]		[0.068, 0.344]	[0.062, 0.187]
Decay δ in meters	100	100	100	1000	1000	1000
Controls	✓	✓	✓	✓	✓	✓
Department FE	✓	✓	✓	✓	✓	✓
Municipality-type FE	✓	✓	✓	✓	✓	✓
Population weights	✓	✓	✓	✓	✓	✓
Outcome mean	0.635	0.205	0.205	0.773	0.205	0.205
Observations	34659	34659	34659	34659	34659	34659
AKM first-stage F	14.07			22.08		

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). In Columns 2, 3, 5, and 6, stars are from AKM0 tests. Columns 1 to 3 are a reminder of the estimates from the main specification, whereas Columns 4 to 6 present an alternative specification where Equation 1 uses a slower distance decay parameter.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

F.4 Exposure: Logs with Zeros

The exposure measure described in Equation 1, as indicated in Section 4.2, uses logarithms of the weighted count. The logarithm function is not defined at zero. As is commonly done, I therefore add a small offset c so that these observations are not dropped. 1,599 municipalities are exposed to no immigrant businesses in 2009, and 457 in 2019. These are primarily cases of first-immigrant-business entry (3.6% of cases), with a few instances of last exit (0.3% of cases). The entry observations make up a non-negligible part of the sample, so the value of the offset parameter c deserves discussion.[28]

Ignoring the offset for illustration, going from one to two "effective" businesses[29] produces a treatment value (change in exposure) of $\Delta = \log(2) - \log(1) = \log\left(\frac{2}{1}\right) \approx 0.693$. Assume a municipality goes from 0 to 1 business. The offset will determine the magnitude of the treatment for these entry municipalities. For $c = \{0.001, 0.01, 0.1\}$, the treatment value will be:

$$\Delta_{0.001} = \log(1 + 0.001) - \log(0 + 0.001) = \log\left(\frac{1.001}{0.001}\right) \approx 6.91$$
$$\Delta_{0.01} = \log(1 + 0.01) - \log(0 + 0.01) = \log\left(\frac{1.01}{0.01}\right) \approx 4.62$$
$$\Delta_{0.1} = \log(1 + 0.1) - \log(0 + 0.1) = \log\left(\frac{1.1}{0.1}\right) \approx 2.40$$

These numbers, compared to the treatment value associated with going from 1 to 2 businesses, show that too small an offset value will give a disproportionate treatment value to the municipalities at the lower end of the distribution. At the same time, selecting a c that is larger compared to the true number of businesses will distort the value of exposure more: for instance, 1 and 2 can reasonably be approximated with 1.1 and 2.1, using $c = 0.1$, but not by replacing them with 2 and 3 when $c = 1$. In practice, in my case, values of c of 0.5 compress a significant portion of the distribution of exposure values at the lower end to $\log(c)$, and the power of the first stage collapses as a result, although the 2SLS estimates remain positive. I

28. There are also 336 municipalities that are zero in both years, but the value of c for them will not affect the value of the treatment: no change.

29. I use the effective number of businesses here illustratively. The true measure of exposure uses a logged, weighted count of businesses in and around each municipality.

therefore select 0.1 as the primary value used in the analysis, but show in Table F4 that the results remain positive and significant when I use $c = 0.001$ or $c = 0.01$.

Table F4: 2SLS Estimates for Other Values of the Log Offset Parameter c

	Δ RN Vote Share				
	2SLS (1)	2SLS (2)	2SLS (3)	2SLS (4)	2SLS (5)
Δ Exposure (log)	0.072**	0.078**	0.102**	0.144**	0.174**
	(0.018)	(0.017)	(0.020)	(0.044)	(0.066)
AKM0 95% conf. set	[0.027, 0.114]	[0.034, 0.117]	[0.055, 0.180]	[0.077, 8.689]	[unbounded]
Log offset c	0.001	0.010	0.100	0.500	1.000
Controls	✓	✓	✓	✓	✓
Department FE	✓	✓	✓	✓	✓
Municipality-type FE	✓	✓	✓	✓	✓
Population weights	✓	✓	✓	✓	✓
Outcome mean	0.205	0.205	0.205	0.205	0.205
Observations	34659	34659	34659	34659	34659
AKM first-stage F	29.44	27.06	14.07	6.41	4.71

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. This table displays the main 2SLS estimates from the pooled instrument under different values of the "offset parameter" (the constant c that ensures that an observation is not dropped when unlogged exposure is equal to zero businesses). The estimates are positive and significant across the board but the first stage is weak for values larger than 0.1.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

F.5 Leave-One-Department-Out Instrument

I reconstruct the national industry shocks separately for each observation. For each municipality, I exclude all businesses located in that municipality's department. This version of the instrument mechanically cannot be driven by immigrant-business growth in or around the municipality itself.[30] This is particularly important to make sure that the value of the instrument in large cities, which receive greater weight in the IV estimates, is not driven by observed business-creation in these cities themselves. The results are displayed in Table F5 and are similar in substance and significance.

Table F5: Leave-One-Department-Out Estimates

	Δ Exposure (log)		Δ RN Vote Share	
	First stage	First stage	2SLS	2SLS
	(1)	(2)	(3)	(4)
Instrument	0.917***	0.905**		
	(0.244)	(0.246)		
Δ Exposure (log)			0.102**	0.100**
			(0.020)	(0.020)
AKM0 95% conf. set			[0.055, 0.180]	[0.051, 0.178]
Department LOO		✓		✓
Controls	✓	✓	✓	✓
Department FE	✓	✓	✓	✓
Municipality-type FE	✓	✓	✓	✓
Population weights	✓	✓	✓	✓
Outcome mean	0.635	0.635	0.205	0.205
Observations	34659	34659	34659	34659
AKM first-stage F	14.07	13.59		

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. Columns 1 and 3 show the main estimates for the pooled instrument, identical to those in Table 1. Columns 2 and 4 show the estimates with the instrument reconstructed such that only businesses outside of a municipality's own department are used in the construction of the shocks (the *shift* in the shift-share instrument).
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

30. Note that there are boundary cases in which the businesses near a municipality are located in a different department. This test does not remove these businesses, but the lack of sensitivity of the estimate does not suggest that a more detailed examination is necessary.

F.6 Alternative Proxy for Immigrant-Origin Presence

In the main specification, I adjust for the number of foreign-born residents to ensure that the effect estimated is specific to businesses. In the tables reported so far, I do so by adjusting for the number of foreign-born residents because I do not have data on the number of second-generation immigrants, as explained in Section 4.1. The number of foreign-born residents is likely strongly correlated with the latent, true number of residents who are perceived immigrants in the sense that I defined immigrant businesses.[31]

Table F6: Adjustment for Imputed Immigration (AKM Inference)

	Δ RN Vote Share
	2SLS
	(1)
Δ exposure (log)	0.095**
	(0.031)
AKM0 95% confidence set	[0.044, 0.167]
Controls	✓
Department FE	✓
Municipality-type FE	✓
Population weights	✓
Outcome mean	0.206
Observations	33057

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. This replicates the main specification but, recognizing that the foreign-born population covariate might not account for all immigrants, adjusts for the proportion of individuals deceased in a municipality around 2009 classified as immigrants using the procedure described in Section C of the appendix.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

However, even under this assumption, one may be worried that what the variation in the number of immigrant businesses truly captures is the residual variation in immigrant residents not captured by the foreign-born population. In such a scenario, the estimated effect would be confounded by this latent residual variation. To address it, I use a dataset reported annually by the French government providing the full list of deceased individuals in the pre-

31. I.e., the number of residents, including second-generation immigrants and beyond, likely perceived as immigrants by natives because of their origins.

vious calendar year, including their names and municipality of death.[32] I first classify these names with the algorithm used to classify business owners. Then, I estimate the share of individuals whose deaths were recorded in the municipality and whose names are classified as immigrant-associated using the same rules as above for each municipality. I use the four years before and after each point in time to increase the sample size, under the assumption that the population composition is sticky. I thus estimate the proportion of immigrants among deceased individuals between 2005 and 2013, which I assign to the year 2009. Tables F6 and F7 replicate the main 2SLS specification (respectively with AKM and Conley inference) with an additional control for this variable: the imputed share of immigrants among deceased individuals around 2009. The results and conclusions are robust to this change, with no notable difference in magnitude or significance.

32. Note that this is the place of death, not the residence at the time of death, which is not recorded. Some municipalities (for instance where a hospital operates) might therefore record the death of residents from other municipalities.

Table F7: Adjustment for Imputed Immigration (Conley Inference)

	Δ RN Vote Share
	2SLS
	(1)
Δ exposure (log)	0.095***
	(0.026)
Foreign pop. (log, 2009)	−0.005***
	(0.001)
Foreign pop. around (log, 2009)	−0.004**
	(0.001)
Imputed immigrant population (share, 2009)	−0.138***
	(0.033)
Population (log, 2009)	−0.002
	(0.004)
Population around (log, 2009)	0.002
	(0.002)
Prop. unemployed (2009)	0.358***
	(0.077)
Prop. college graduate (2009)	−0.310***
	(0.041)
Prop. HS graduate (2009)	−0.252***
	(0.058)
Native businesses (log, 2009)	−0.003
	(0.004)
Dist. to highway (log)	0.001
	(0.001)
Per capita income (log, 2009)	0.014
	(0.014)
Vote share (2009)	0.108*
	(0.063)
Pre-period change (2004-2009)	−0.335***
	(0.042)
Controls	✓
Department FE	✓
Municipality-type FE	✓
Population weights	✓
Outcome mean	0.206
Observations	33057

Note: Conley standard errors (50km). This replicates the main specification but, recognizing that the foreign-born population covariate might not account for all immigrants, adjusts for the proportion of individuals deceased in a municipality around 2009 classified as immigrants using the procedure described in Section C of the appendix.

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

G Instrumental Variable: Muslim-Oriented Instrument

G.1 Construction

I construct a Muslim-oriented instrument to test whether the results hold when the first stage gives more weight to the businesses associated with the groups at the center of cultural conflict, as discussed in the theory. This instrument is constructed similarly to the main instrument described in Section 5, but uses only Muslim-associated businesses in the calculation of the national "shift" instead of all immigrant-associated businesses. I first residualize national Muslim-immigrant-business growth, defined as the Davis-Haltiwanger-Schuh symmetric growth rate for businesses whose owner's predicted origin is classified as likely coming from a country whose population is majority Muslim, on national native-business growth and national non-Muslim immigrant business growth as follows:

$$G_j^{muslim} = \alpha + \beta^{native} G_j^{native} + \beta^{non-muslim} G_j^{non-muslim} + u_j. \tag{9}$$

I recenter these residual shocks within quintiles of baseline industry size as I do for the main instrument:

$$\widetilde{G}_j^{muslim} = \widehat{u}_j^{muslim} - \overline{\widehat{u}^{muslim}}_{q(j)}, \tag{10}$$

The resulting shocks $\widetilde{G}_j^{muslim}$ therefore draw identifying variation from Muslim-origin-specific business growth across industries of similar initial size. The estimation and inference approaches are identical to the ones described in Section 5 of the main text. Table G1 displays the main estimates for this instrument with the exposure-robust standard errors, and Table G2 displays the full tables, with covariates, with Conley standard errors allowing spatial dependencies up to 50 km.

G.2 Imbalance and Conditioning

As described in Section E.1.1 of the appendix, shock-level balance tests for the main instrument reveal imbalance in the baseline share of immigrant ownership: industries with higher

Table G1: Main Estimates for the Muslim-Oriented Instrument

	Δ Exposure (log)				Δ RN Vote Share
	First stage	First stage	First stage	First stage	2SLS
	(1)	(2)	(3)	(4)	(5)
Muslim-oriented instrument	1.342***	1.458***	0.693**	0.116	
	(0.247)	(0.269)	(0.314)	(0.092)	
Δ exposure (log)					0.071**
					(0.017)
AKM0 95% confidence set					[0.013, 0.102]
First stage outcome	Immigrant	Muslim	Non-Muslim	Native	
Controls	✓	✓	✓	✓	✓
Department FE	✓	✓	✓	✓	✓
Municipality-type FE	✓	✓	✓	✓	✓
Population weights	✓	✓	✓	✓	✓
Outcome mean	0.635	0.597	0.432	0.407	0.205
Observations	34659	34659	34659	34659	34659
AKM first-stage F	29.6				

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. Main estimates of the first- and second-stage regressions for Muslim-oriented shift-share design. This instrument uses only Muslim-associated business growth in the "shift." Column 1 shows that the instrument strongly predicts the main treatment (exposure). Columns 2 and 3 show that the instrument predicts local growth in exposure to both Muslim and non-Muslim businesses. The estimate on Muslim businesses is considerably stronger and more precise, but the one on non-Muslim businesses is also significant. This breakdown of the main instrument therefore cannot be used to isolate the effect of Muslim exposure to test Hypothesis 3. Column 4 shows that the instrument does not predict native growth, and Column 5 displays the 2SLS estimate, which is also positive and significant.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

immigrant shares are less likely to see high growth between 2009 and 2019, consistent with a regression to the mean pattern. Moreover, adjusting for this share in the main 2SLS specification reduces the power of the first stage, suggesting that some of the first stage relationship may have captured the explanatory power of this component. Because of this first-stage loss of power, the plausibility of the conditional exogeneity assumption cannot be tested directly in the second stage: although the 2SLS estimate remains positive, the AKM0 confidence set becomes very large and disjointed.

Table G3 shows that this imbalance marginally remains in the case of the Muslim-oriented instrument. However, the additional first-stage power of the Muslim-oriented instrument allows me to adjust for the baseline share of immigrant-associated names, or Muslim-associated names, by industry in the 2SLS regression. The resulting estimates are displayed in Table G4.

Column 2 of Table G4 shows that the main instrument loses too much power in the first stage to be able to produce estimates with a bounded AKM0 confidence set. The last three columns show that the Muslim-oriented instrument, despite losing some first-stage strength,

Table G2: Complete Estimates for the Muslim-Oriented Instrument (Conley Inference)

	Δ Exposure (log)				Δ RN Vote Share
	First stage	First stage	First stage	First stage	2SLS
	(1)	(2)	(3)	(4)	(5)
Muslim-oriented instrument	1.342***	1.458***	0.693***	0.116*	
	(0.160)	(0.182)	(0.191)	(0.070)	
Δ exposure (log)					0.071***
					(0.021)
Foreign pop. (log, 2009)	0.016***	0.014***	0.027***	0.000	−0.005***
	(0.005)	(0.005)	(0.006)	(0.002)	(0.001)
Foreign pop. around (log, 2009)	0.012	0.010	0.038***	0.014***	−0.004**
	(0.009)	(0.010)	(0.014)	(0.003)	(0.001)
Population (log, 2009)	−0.017	−0.023	−0.013	0.040**	−0.000
	(0.014)	(0.017)	(0.014)	(0.018)	(0.004)
Population around (log, 2009)	0.003	0.006	−0.008	−0.010	0.002
	(0.017)	(0.020)	(0.027)	(0.006)	(0.002)
Prop. unemployed (2009)	−1.020***	−1.056***	−0.010	−0.101	0.285***
	(0.313)	(0.364)	(0.505)	(0.094)	(0.075)
Prop. college graduate (2009)	−0.081	−0.202*	−0.180	0.360***	−0.311***
	(0.101)	(0.120)	(0.132)	(0.071)	(0.039)
Prop. HS graduate (2009)	0.778***	0.957***	0.868***	0.809***	−0.248***
	(0.221)	(0.256)	(0.298)	(0.087)	(0.055)
Native businesses (log, 2009)	−0.001	0.018	0.011	−0.088***	−0.004
	(0.013)	(0.015)	(0.011)	(0.020)	(0.004)
Dist. to highway (log)	−0.015***	−0.015***	−0.011**	−0.007***	0.001
	(0.004)	(0.004)	(0.005)	(0.001)	(0.001)
Per capita income (log, 2009)	−0.052	−0.017	0.043	0.018	0.020
	(0.040)	(0.049)	(0.063)	(0.026)	(0.016)
Vote share (2009)	−0.004	−0.005	−0.233	0.090	0.104
	(0.167)	(0.178)	(0.215)	(0.083)	(0.068)
Pre-period change (2004-2009)	−0.387***	−0.300*	−0.527**	−0.193***	−0.343***
	(0.135)	(0.155)	(0.246)	(0.057)	(0.039)
First stage outcome	Immigrant	Muslim	Non-Muslim	Native	
Controls	✓	✓	✓	✓	✓
Department FE	✓	✓	✓	✓	✓
Municipality-type FE	✓	✓	✓	✓	✓
Population weights	✓	✓	✓	✓	✓
Outcome mean	0.635	0.597	0.432	0.407	0.205
Observations	34659	34659	34659	34659	34659

Note: Conley standard errors (50km). Main estimates of the first-stage and second-stage regressions for the Muslim-oriented shift-share design. This instrument uses only Muslim-associated business growth in the "shift." More details are provided in Table G1. Importantly, the estimate for change in exposure to native businesses in Column 4 still does not reach the 5% significance level, even though it does not allow for dependencies induced by the common shock. The standard error estimates from Table G1 should be preferred.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

produces stable and significant 2SLS estimates even when conditioning on the baseline share of businesses in each industry that are either immigrant- or Muslim-associated. This suggests that while the first stage for both instruments likely captures variation due to some baseline characteristics, these variables are unlikely to be driving second-stage results.

Table G3: Muslim-Oriented Instrument: Shock-Level Balance Tests

	Muslim-oriented shock
	OLS
	(1)
Muslim share	0.079
	(0.107)
Immigrant share	−0.178
	(0.113)
Food service	0.006
	(0.110)
Industry size	0.027
	(0.045)
Native growth	0.045
	(0.034)
Exposure weights	✓
Observations	87
Joint F: all 5 characteristics	4.52 ($p = 0.001$)

Note: HC1 standard errors. This table displays estimates for the regression of the industry-level shocks on the baseline or pre-period characteristics of each industry.No individual covariate coefficient is significant at the conventional level, but the joint Wald test has a p-value of 0.001, largely driven by the imbalance of the baseline share of immigrants in the industries that grow the most. This imbalance is similar to the one found for the main instrument. However, where for the main instrument, the imbalance cannot be adjusted for without a loss of first-stage power, Table G4 shows that conditioning on this imbalance preserves the direction, significance, and order of magnitude of the estimate of the Muslim-oriented instrument.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

G.3 Diagnostic Tests and Robustness

Table G5 displays results for additional diagnostics of the exogeneity assumption and exclusion restriction. Column 1 shows marginally significant positive pretrends in terms of immigrant population change at the municipality level. This does suggest that immigrant businesses could be following a rise in the number of foreign-born residents. The main instrument does not suffer from this limitation. Columns 2 and 3 show that the instrument is balanced on the baseline logged number of foreign-born residents in the population and that the pretrends in terms of total municipality population are not statistically different from zero.

Table G6 shows that there is no significant relationship between the value of the Muslim-oriented instrument and the change in RN vote between the previous European election (2004)

Table G4: Muslim-Oriented Instrument: Adjustment for Imbalance

	Δ RN Vote Share				
	2SLS (1)	2SLS (2)	2SLS (3)	2SLS (4)	2SLS (5)
Δ Exposure (log)	0.102**	0.146**	0.071**	0.092**	0.094**
	(0.020)	(0.040)	(0.017)	(0.025)	(0.021)
AKM0 95% conf. set	[0.055, 0.180]	[unbounded]	[0.013, 0.102]	[0.003, 0.158]	[0.026, 0.148]
Instrument	Main	Main	Muslim	Muslim	Muslim
Adjustment		Immigrant		Immigrant	Muslim
Controls	✓	✓	✓	✓	✓
Department FE	✓	✓	✓	✓	✓
Municipality-type FE	✓	✓	✓	✓	✓
Population weights	✓	✓	✓	✓	✓
Outcome mean	0.205	0.205	0.205	0.205	0.205
Observations	34659	34659	34659	34659	34659
AKM first-stage F	14.07	5.89	29.62	17.14	19.86

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. This table displays the 2SLS estimates for the main instrument and the Muslim-oriented instrument described in Section G, with and without conditioning on the variables on which the industry shocks are imbalanced. All columns show positive estimates of a similar order of magnitude. Column 2 shows that the main instrument is too weak to produce a bounded AKM0 confidence set. The last three columns show that the Muslim-oriented instrument, despite losing some first-stage strength, produces stable and significant 2SLS estimates even when conditioning on the baseline share of businesses in each industry that are either immigrant- or Muslim-associated.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

and the beginning of the period (2009).

Table G7 shows first-stage specifications in which the endogenous treatment (exposure to immigrant-associated businesses) is replaced by alternative variables that could suggest violations of the exclusion restrictions. The same table for the main instrument, displayed in Section E.2, shows no significant change in any of these variables. Columns 1 and 2 of Table G7 show that the Muslim-oriented instrument does not predict change in exposure to native storefronts or in the number of foreign-born individuals among the municipality's residents. Columns 3 and 4 are consistent with partial violations of the exclusion restriction: the Muslim-oriented instrument predicts change in the foreign-born population around the municipality and in the number of immigrant-associated non-storefront businesses. The magnitude is significantly smaller than the first-stage effect on the treatment of interest, and both of these variables could be affected by a prior change in the endogenous variable itself. However, these results invite caution in the interpretation of the Muslim-oriented instrument's 2SLS results.

Table G5: Muslim-Oriented Instrument: Population Pretrends

	Δ Foreign pop. (2008-09)	Baseline foreign pop. (2009)	Δ Pop. (2006-09)
	OLS	OLS	OLS
	(1)	(2)	(3)
Muslim-oriented instrument	0.133*	−0.662	0.016
	(0.059)	(0.339)	(0.022)
AKM0 95% conf. set	[-0.010, 0.249]	[-1.288, 0.235]	[-0.037, 0.061]
Controls	✓	✓	✓
Department FE	✓	✓	✓
Municipality-type FE	✓	✓	✓
Population weights	✓	✓	✓
Observations	27665	34659	34659
Outcome mean	0.029	1.990	0.030

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. This table tests whether the instrument predicts baseline and pre-period changes in population characteristics at the municipality level. The years are chosen based on data availability: 2008 is the earliest available year for number of foreign-born residents in each municipality, and 2006 for population. The first column reports a drop of some observations for which this pre-period covariate is missing. Columns 2 and 3 show that the instrument is balanced on the baseline logged number of foreign-born residents in the population and that the pretrends in terms of total municipality population are not statistically different from zero. Column 1 shows a slightly positive pretrend that is marginally significant in terms of immigrant population change at the municipality level.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Finally, Table G8 reports 2SLS estimates for the Muslim-oriented instrument with industry groups excluded one at a time from the instrument construction, to test sensitivity to specific industries. The results show that unsurprisingly, the first stage loses power completely when all consumer-facing industries are excluded. However, the point estimates are stable when dropping any single specific industry set, though the automobile industry here again weakens the first stage and precision significantly. The 5% significance is marginally lost for personal services and the automobile industry when using AKM0 inference, but remains when using Conley standard errors.

Table G6: Muslim-Oriented Instrument: Pretrend Test

	Δ RN Vote Share (2004-09)
	OLS
	(1)
Muslim-oriented instrument	−0.014
	(0.015)
AKM0 95% confidence set	[-0.046, 0.019]
Controls	✓
Department FE	✓
Municipality-type FE	✓
Population weights	✓
Outcome mean	−0.085
Observations	34659

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. This table shows that the pre-trends are flat for the instrument: there is no significant relationship between the value of the Muslim-oriented instrument and the change in RN vote between the previous European election (2004) and the beginning of the period (2009).
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table G7: Muslim-Oriented Instrument: Other Causal Channels

	Δ Native storefront exp.	Δ Foreign-born pop. (own)	Δ Foreign-born pop. (around)	Δ Immig. non-storefr. exp.
	OLS	OLS	OLS	OLS
	(1)	(2)	(3)	(4)
Muslim-oriented instrument	0.116	−0.066	0.177**	0.366**
	(0.092)	(0.234)	(0.067)	(0.128)
AKM0 95% confidence set	[-0.078, 0.328]	[-0.497, 0.552]	[0.008, 0.305]	[0.053, 0.623]
Controls	✓	✓	✓	✓
Department FE	✓	✓	✓	✓
Municipality-type FE	✓	✓	✓	✓
Population weights	✓	✓	✓	✓
Outcome mean	0.407	0.009	0.051	0.727
Observations	34659	34659	34659	34659

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. This table shows estimates from alternative first-stage specifications where the endogenous treatment is replaced by variables measuring alternative causal channels that could reveal violations of the exclusion restriction. Columns 1 and 2 show that the Muslim-oriented instrument does not predict change in exposure to native storefronts or in the log count of foreign-born residents in the municipality. Columns 3 and 4 are consistent with partial violations of the exclusion restriction: the Muslim-oriented instrument predicts change in the foreign-born population around the municipality and in the number of immigrant-associated non-storefront businesses.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table G8: Muslim-Oriented Instrument: Industry Group Exclusions

	Δ RN Vote Share					
	2SLS (1)	2SLS (2)	2SLS (3)	2SLS (4)	2SLS (5)	2SLS (6)
Δ Exposure (log)	0.071**	0.070**	0.078**	0.055*	0.070	0.563
	(0.017)	(0.016)	(0.016)	(0.020)	(0.023)	(2.553)
AKM0 95% conf. set	[0.013, 0.102]	[0.025, 0.101]	[0.024, 0.112]	[-0.019, 0.090]	[-0.059, 0.110]	[unbounded]
Industries dropped	None	Hosp.	Food	Pers.	Auto	Cons.
Exposure dropped (%)	0.00	8.61	16.64	11.85	9.08	87.12
Controls	✓	✓	✓	✓	✓	✓
Department FE	✓	✓	✓	✓	✓	✓
Municipality-type FE	✓	✓	✓	✓	✓	✓
Population weights	✓	✓	✓	✓	✓	✓
Outcome mean	0.205	0.205	0.205	0.205	0.205	0.205
Observations	34659	34659	34659	34659	34659	34659
AKM first-stage F	29.6	45.3	24.9	18.7	14.3	0.0

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019). Stars are from AKM0 tests. This table reports the 2SLS estimates for the Muslim-oriented instrument with industry groups dropped one at a time in the construction of the instrument, and then jointly dropped. Column 1 reports the baseline estimate with all industries, Columns 2 to 5 respectively drop the hospitality industries, food industries, personal-services industries, and automobile industries. Column 6 drops all of these, as well as other industries grouped in a "consumer services" set.

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

H Other Voting Outcomes

H.1 Turnout

To examine whether the increase in RN vote share is the result of changes in turnout or changes in vote choice among voters who turn out, I estimate the effect of exposure on turnout and re-estimate the main specification using RN votes as a proportion of all registered voters instead of RN votes as a proportion of valid votes. Results are displayed in Table H1.

A one-unit increase in logged exposure increases the RN's share of registered voters by 0.040, whereas the estimated effect on turnout is precise and close to zero. The registered-voter result is approximately what would be expected from scaling the main vote-share effect by average turnout. These results are jointly more consistent with the observed increase in RN support being a consequence of changes in vote choice than with a net new mobilization of previously abstaining voters. That being said, I cannot rule out offsetting changes in turnout, where some previously abstaining individuals turn out to vote while a similar number of former voters abstain.

Table H1: Other Voting Outcomes: Registered Vote and Turnout

	Δ Registered vote share	Δ Turnout
	2SLS	2SLS
	(1)	(2)
Δ Exposure (log)	0.040**	−0.000
	(0.011)	(0.011)
Foreign pop. (log, 2009)	−0.003***	−0.002***
	(0.001)	(0.001)
Foreign pop. around (log, 2009)	−0.002**	−0.000
	(0.001)	(0.001)
Population (log, 2009)	−0.001	0.002
	(0.002)	(0.005)
Population around (log, 2009)	0.001	0.001
	(0.001)	(0.003)
Prop. unemployed (2009)	0.091**	0.014
	(0.038)	(0.037)
Prop. college graduate (2009)	−0.117***	−0.002
	(0.019)	(0.019)
Prop. HS graduate (2009)	−0.118***	−0.027
	(0.029)	(0.021)
Native businesses (log, 2009)	−0.001	−0.001
	(0.002)	(0.004)
Dist. to highway (log)	0.000	0.000
	(0.000)	(0.000)
Per capita income (log, 2009)	0.021***	0.031***
	(0.008)	(0.006)
Vote share (2009)	0.180***	0.209***
	(0.033)	(0.024)
Pre-period change (2004-2009)	−0.172***	−0.138***
	(0.026)	(0.022)
AKM0 95% confidence set	[0.012, 0.077]	[-0.03, 0.035]
Department FE	✓	✓
Municipality-type FE	✓	✓
Population weights	✓	✓
Outcome mean	0.118	0.110
Observations	34659	34659
AKM first-stage F	14.07	14.07

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019) for the Δ Exposure (log) coefficient, with stars from AKM0 tests. Conley standard errors (50 km) are reported for control variables. This table reports second-stage estimates in which the main outcome is replaced by change in support for the RN as a proportion of registered voters, in Column 1, and turnout in Column 2. Though these results are consistent with a switch to far-right vote by existing voters, I cannot rule out a change in turnout among some voters compensated by an exit from voting among others.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

H.2 Other Parties

I re-estimate the second stage with change in the support for other major parties as the outcome variables. The theory presented in Section 2 indicates that immigrant businesses will primarily affect voting outcomes by triggering native backlash, which could translate into voting behavior changes when vote choice is a function of attitudes toward immigration. The "Far Left" label represents Jean-Luc Mélenchon's *Front de Gauche* in 2009 and *France Insoumise* in 2019. "Republicans" captures *UMP* in 2009 and "Les Républicains" in 2019.

Table H2 displays the results. The results suggest that change in exposure to immigrant businesses does not cause significant change in the electoral performance of other major political parties. Note that the parties that have a negative estimate are the major left-wing parties traditionally promoting more inclusionary policies, while the mainstream right-wing Republican party is unaffected. However, because this interpretation would predict a negative coefficient for the Green party, and because none of the four coefficients are significant, I do not claim that the increased support for the far right is conclusively drawn from these mainstream parties. Because the number of votes received by the RN increases, and because average turnout is not significantly affected, it should be the case that other parties lose voters. The absence of a significant effect here therefore does not mean that these parties are unaffected: the loss of support may be sufficiently diffuse across existing parties that no single effect is precisely estimated, or may come from less established parties.

Table H2: 2SLS Estimates for Selected Other Parties

	Δ Vote share Far Left	Δ Vote share Greens	Δ Vote share Socialists	Δ Vote share Republicans
	2SLS (1)	2SLS (2)	2SLS (3)	2SLS (4)
Δ Exposure (log)	−0.018	0.003	−0.024	0.001
	(0.021)	(0.014)	(0.053)	(0.065)
Foreign pop. (log, 2009)	0.000	−0.001**	−0.001*	0.001
	(0.001)	(0.000)	(0.001)	(0.001)
Foreign pop. around (log, 2009)	0.000	0.000	−0.002	0.004***
	(0.001)	(0.001)	(0.001)	(0.001)
Population (log, 2009)	0.007***	0.001	−0.000	−0.003
	(0.002)	(0.001)	(0.002)	(0.003)
Population around (log, 2009)	−0.002	0.000	0.000	0.001
	(0.002)	(0.001)	(0.002)	(0.002)
Prop. unemployed (2009)	−0.141**	−0.184***	−0.088	0.067
	(0.071)	(0.043)	(0.060)	(0.070)
Prop. college graduate (2009)	0.126***	−0.058***	0.129***	0.050**
	(0.020)	(0.014)	(0.025)	(0.020)
Prop. HS graduate (2009)	0.080**	−0.036***	0.135***	0.059*
	(0.039)	(0.012)	(0.022)	(0.033)
Native businesses (log, 2009)	−0.002	0.001	0.004***	−0.006*
	(0.002)	(0.001)	(0.001)	(0.003)
Dist. to highway (log)	0.001	−0.000	0.000	−0.000
	(0.001)	(0.000)	(0.001)	(0.000)
Per capita income (log, 2009)	−0.017***	−0.001	0.034***	−0.129***
	(0.005)	(0.003)	(0.007)	(0.015)
Vote share (2009)	0.119**	0.018	0.176***	0.018
	(0.051)	(0.015)	(0.024)	(0.057)
Pre-period change (2004-2009)	−0.105*	0.021	−0.084***	0.196***
	(0.054)	(0.013)	(0.026)	(0.023)
AKM0 95% confidence set	[-0.061, 0.021]	[-0.017, 0.037]	[-0.077, 0.132]	[-0.188, 0.068]
Controls	✓	✓	✓	✓
Department FE	✓	✓	✓	✓
Municipality-type FE	✓	✓	✓	✓
Population weights	✓	✓	✓	✓
Outcome mean	0.007	-0.025	-0.101	-0.188
Observations	34659	34659	34659	34659

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019) for the Δ Exposure (log) coefficient, with stars from AKM0 tests. Conley standard errors (50 km) are reported for control variables. This table presents 2SLS estimates for four other major political blocs. None of the exposure coefficients is statistically significant. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

I Mechanisms

I.1 Visibility Mechanisms

Table I1 reports the estimates displayed in Figure 3 of the main text.

I.2 Economic Mechanisms

Data Construction

This section provides more detail on the economic outcomes used to test the economic mechanism in Section 2.3: per capita income, native unemployment, industry diversity, property prices, and exposure to native businesses. All of them are measured at the municipality level. All specifications use first differences, weighted by 2006 population as in the main analysis, with department and municipality-type fixed effects. I use AKM exposure-robust (Adão, Kolesár, and Morales 2019) and Conley (50 km) standard errors. The endogenous treatment across all of them is the same as in the main analysis: change in exposure to immigrant-associated storefronts. The five outcomes are standardized (mean zero and standard deviation one) for comparability of the coefficients.

Change in per capita income, native unemployment, industry diversity, and native businesses is measured like the treatment between 2009 and 2019. This ten-year difference cannot be calculated for property prices because transaction records only go back to 2014. I therefore use change between 2014 and 2019 as an outcome. It is not perfectly aligned with the treatment window, and should therefore be interpreted more cautiously.

Municipality-level income is logged and uses data from Cagé and Piketty (2023). Native unemployment is the share of non-immigrant residents aged 15 or older who are unemployed, using Insee's census classification of immigration status (the denominator includes both active and inactive residents). Exposure to native businesses is the measure of exposure defined in Equation 1 but in which immigrant-associated businesses are replaced by non-immigrant-associated businesses. Property prices are calculated using the *Demandes de valeurs foncières (DVF)* (Requests for Land Values) files from the Ministry of the Economy. These files record the amount of each real estate transaction recorded in each municipality in each year, as well

as characteristics of each transaction. I use the median residential transaction price per square meter in 2014 and 2019. I restrict observations to plausible values between 100€ and 30,000€, and drop municipalities for which fewer than five transactions were recorded. This explains the significantly smaller number of municipalities included in the analysis. Because of this smaller sample size, it is particularly plausible in the case of property values that the lack of detectable effect is a product of insufficient power as opposed to a null result.

I measure *industry diversity* in community m as one minus the sum of the squared distance-weighted industry shares, adapting the Herfindahl–Hirschman Index, with businesses in municipality m indexed as $i \in \{1, \ldots, K_m\}$, across all industries $j \in \{1, \ldots, J\}$:

$$\text{Industry diversity}_m = 1 - \sum_{j=1}^{J} \left(\frac{\sum_{i \in j} w_{m,i}}{\sum_{i}^{K_m} w_{m,i}} \right)^2 \tag{11}$$

The communities m correspond to the 10 km-radius areas around each municipality used to measure exposure, and the businesses are weighted by inverse distance to the municipality. Businesses are classified into industries using their APE code as described in Section 4.1. Industry diversity is a decreasing function of the concentration of businesses into fewer industries. A business reduces industry diversity if it belongs to a relatively more represented industry, and increases industry diversity if it belongs to a new or underrepresented industry.

Detailed Results

Table I2 reports the estimates displayed in Figure 4 of the main text, with exposure-robust standard errors and confidence intervals.

Table I3 reports the estimates with Conley standard errors, as well as all the control coefficients. Note that with less conservative Conley standard errors, the coefficient on industry diversity reaches marginal significance (at the 10% level) and the coefficient on per capita income reaches significance at the 5% level. As discussed in the main analysis, AKM standard errors are more appropriate to reflect the common exposure of municipalities to the smaller number of industry shocks used in the construction of the instrument.

Table I1: Tests of the Visibility Mechanism

	RN Vote Share		
	OLS (1)	OLS (2)	OLS (3)
All exposure (storefront, log)	0.001		
	(0.001)		
All exposure (non-storefront, log)	−0.009**		
	(0.004)		
Exposure - Muslim (storefront, log)		0.003**	
		(0.001)	
Exposure - Muslim, food (storefront, log)			0.003**
			(0.001)
Exposure - Muslim, non-food (storefront, log)			0.001
			(0.001)
Exposure - other (storefront, log)		−0.003**	−0.004**
		(0.001)	(0.001)
Exposure - Muslim (non-storefront, log)		−0.005*	−0.005*
		(0.003)	(0.003)
Exposure - other (non-storefront, log)		−0.006***	−0.006***
		(0.002)	(0.002)
Foreign pop. (log)	−0.001***	−0.001***	−0.001***
	(0.000)	(0.000)	(0.000)
Foreign pop. around (log)	−0.008	−0.007	−0.007
	(0.008)	(0.007)	(0.007)
Population (log)	0.037***	0.036***	0.037***
	(0.009)	(0.009)	(0.009)
Population around (log)	0.039	0.040	0.038
	(0.057)	(0.057)	(0.057)
Prop. unemployed	0.178	0.184	0.184
	(0.124)	(0.125)	(0.125)
Prop. college graduate	−0.231***	−0.230***	−0.229***
	(0.040)	(0.040)	(0.040)
Prop. HS graduate	0.392***	0.388***	0.390***
	(0.067)	(0.066)	(0.066)
Per capita income (log)	0.004**	0.004**	0.004**
	(0.002)	(0.002)	(0.002)
Native businesses (storefront log)	0.023***	0.023***	0.023***
	(0.009)	(0.008)	(0.008)
Native businesses (non-storefront, log)	−0.116**	−0.110**	−0.112**
	(0.048)	(0.046)	(0.046)
Election FE	✓	✓	✓
Municipality FE	✓	✓	✓
Population weights	✓	✓	✓
Outcome mean	0.214	0.214	0.214
Observations	432585	432585	432585
p (Wald null: Storefront = non-storefront)	0.013		
p (Wald null: Muslim = other, storefront)		0.006	

Note: Parentheses report Conley standard errors (50 km). This table reports two-way fixed effects estimates of the relationship between exposure to different categories of businesses and the RN vote share, as a test of the visibility mechanisms. These estimates are displayed graphically in Figure 3 and the specification is described in Equation 8.

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table I2: Economic Outcomes

	Δ Per cap. income	Δ Native unemp.	Δ Industry diversity	Δ Property prices	Δ Native businesses
	2SLS (1)	2SLS (2)	2SLS (3)	2SLS (4)	2SLS (5)
Δ Exposure (log)	−0.573	−0.504**	−0.623	−0.271	0.005
	(0.350)	(0.161)	(0.650)	(0.171)	(0.407)
AKM0 95% conf. set	[-1.284, 0.981]	[-1.141, -0.139]	[-2.248, 1.717]	[-0.763, 0.149]	[-2.092, 0.716]
Department FE	✓	✓	✓	✓	✓
Municipality-type FE	✓	✓	✓	✓	✓
Population weights	✓	✓	✓	✓	✓
Num. obs.	34604	34659	34659	12425	34659
AKM first-stage F	14.07	14.07	14.07	19.57	14.07

Note: Parentheses report AKM exposure-robust standard errors (Adão, Kolesár, and Morales 2019) for the Δ Exposure (log) coefficient, with stars from AKM0 tests. This table reports 2SLS estimates with economic outcomes: per capita income in Column 1, native unemployment in Column 2, industry diversity in Column 3, property prices in Column 4, and exposure to native businesses in Column 5. Industry diversity is defined in Equation 11. Though confidence intervals include values consistent with economic harm, no estimate significantly points in that direction.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table I3: Economic outcomes (Conley inference)

	Δ Per cap. income	Δ Native unemp.	Δ Industry diversity	Δ Property prices	Δ Native businesses
	2SLS	2SLS	2SLS	2SLS	2SLS
	(1)	(2)	(3)	(4)	(5)
Δ Exposure (log)	−0.573**	−0.504***	−0.623*	−0.271	0.005
	(0.270)	(0.159)	(0.367)	(0.235)	(0.295)
Foreign pop. (log, 2009)	0.039***	0.058***	0.009	0.010	0.000
	(0.015)	(0.009)	(0.007)	(0.013)	(0.007)
Foreign pop. around (log, 2009)	−0.015	0.021	−0.018**	−0.007	0.054***
	(0.016)	(0.013)	(0.009)	(0.012)	(0.014)
Population (log, 2009)	−0.119***	−0.054***	−0.013	0.027	0.158**
	(0.025)	(0.020)	(0.013)	(0.028)	(0.070)
Population around (log, 2009)	0.029	−0.023	0.017	0.033	−0.039
	(0.037)	(0.021)	(0.013)	(0.033)	(0.026)
Prop. unemployed (2009)	−4.181***	−16.633***	−0.679	−0.425	−0.536
	(0.867)	(1.297)	(0.551)	(0.930)	(0.442)
Prop. college graduate (2009)	1.918***	−0.666***	−0.092	0.907***	1.404***
	(0.225)	(0.140)	(0.107)	(0.197)	(0.297)
Prop. HS graduate (2009)	0.100	−1.210***	0.716**	1.838***	3.188***
	(0.438)	(0.263)	(0.359)	(0.439)	(0.427)
Native businesses (log, 2009)	0.041*	0.069***	−0.021*	−0.032	−0.351***
	(0.021)	(0.023)	(0.012)	(0.026)	(0.079)
Dist. to highway (log)	−0.003	0.001	−0.004	−0.008	−0.030***
	(0.009)	(0.007)	(0.005)	(0.006)	(0.006)
Per capita income (log, 2009)	−0.371***	−0.835***	−0.141***	0.071	0.075
	(0.126)	(0.094)	(0.039)	(0.073)	(0.096)
Vote share (2009)	0.144	0.023	−0.137	0.362	0.373
	(0.275)	(0.267)	(0.153)	(0.307)	(0.327)
Pre-period change (2004-2009)	−0.669***	−0.106	−0.090	−0.474	−0.766***
	(0.255)	(0.235)	(0.198)	(0.299)	(0.243)
Department FE	✓	✓	✓	✓	✓
Municipality-type FE	✓	✓	✓	✓	✓
Population weights	✓	✓	✓	✓	✓
Observations	34604	34659	34659	12425	34659

Note: Parentheses report Conley standard errors (50 km). This table reports 2SLS estimates with economic outcomes: per capita income in Column 1, native unemployment in Column 2, industry diversity in Column 3, property prices in Column 4, and exposure to native businesses in Column 5. Industry diversity is defined in Equation 11.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

J Survey Experiment

To further understand the aggregate effect and test some of the mechanisms presented in Section 6, I analyze the results of a pre-registered online survey experiment conducted in France in August 2025. Using Qualtrics, I surveyed an online convenience sample of French respondents recruited through CloudResearch's Prime Panels platform (Chandler et al. 2019). My final sample includes 1,470 respondents.[33] The experiment was pre-registered: an anonymized copy of the pre-analysis plan is available here.

J.1 Treatment Conditions

Respondents were randomly allocated to one of four conditions by receiving none, one, or both of the following treatments.

Media treatment. The first treatment asked respondents to read a one-page newspaper article about local economic development in the town of Perpignan. The article shown to subjects in the control condition described the closure of a local bakery, and its upcoming replacement by a classic French brasserie. The article shown to subjects in the treatment condition described the upcoming opening of "halal restaurants", and briefly reported the mayor of Perpignan's opposition to these restaurants. The mayor of Perpignan, Louis Aliot, was also vice president of the National Rally from 2011 to 2018 and has remained a recognizable face. The two articles are shown in Section J.5.

Businesses treatment. The second treatment presented respondents with five restaurants in or around their municipality.[34] They were asked whether they knew each restaurant and, if they did, whether they would recommend it to friends or family members. The population of restaurants was established by scraping data from restaurants with a Google Maps presence in France. One or more business categories are associated with each Google Maps restaurant,

33. That number corresponds to the set of completed responses that pass the pre-registered filter (completion must have lasted at least 300 seconds). Among the 2,361 respondents randomized to the four conditions, 1,499 completed the survey and 1,470 remained after the speed filter.

34. Respondents were asked to identify their place of residence earlier in the survey.

for example, “Lebanese Takeout” or “Japanese Restaurant”. Respondents in the control condition were shown five stereotypically French restaurants whose Google Maps labels included categories like “French Brasserie” or “French Bistrot”. Respondents in the treatment condition were shown restaurants typically associated with immigration from Muslim-majority countries, with labels such as “Kebab Shop” or “Halal Restaurant”. An example of a business from each condition is displayed in Appendix J.6. Conditional on the treatment condition, the five restaurants displayed were those closest to the center of the respondent’s municipality of residence. Each restaurant was shown with the first photo provided for it by the Google Maps API.

Immigrant population vignette. The experiment was meant to estimate an effect specifically associated with businesses and not individuals. A potential interpretation concern could be that the presence of immigrant businesses is simply interpreted by respondents as a proxy for the presence of immigrants. To avoid this, I informed each respondent, before treatment, of the evolution in the number of foreign residents in their municipality between 2011 and 2021, according to national statistics, and of the share and number of immigrants present in their municipality in 2021. This holds fixed perceptions of the local presence of immigrants in expectation and helps interpret the estimated treatment effects as specific to businesses. All respondents, across treatment conditions, were shown this pre-treatment vignette.

J.2 Experimental Design

The eligible population for the study is the set of people registered with CloudResearch’s Prime Panels platform who live in France and speak French. CloudResearch has been widely used by researchers and, unlike many competing online platforms, includes a large number of potential French respondents. This is a convenience sample that may not be nationally representative. In particular, respondents are likely to be younger and more digitally proficient on average than the French population. While this may affect the generalizability of the findings, some studies have shown that convenience samples can provide findings consistent with nationally representative samples (Berinsky, Huber, and Lenz 2012; Mullinix et al. 2015;

Coppock, Leeper, and Mullinix 2018).

Treatment was assigned with complete block randomization to ensure balance within strata defined by whether a respondent's municipality of residence is rural or urban, and whether they supported Emmanuel Macron, Marine Le Pen, or neither in the runoff of the 2022 presidential election.

I collected several pre-registered outcomes. The closest outcome to the observational analysis described in the main text was respondents' trust in the National Rally. To capture the local political consequences of exposure, respondents were asked to indicate with what probability they would vote again for their incumbent mayor. Finally, respondents were asked two series of questions using seven-point Likert scales: one about their attitudes toward immigrant-associated businesses specifically, and one about their attitudes toward immigration and immigrants more broadly. These answers were aggregated into indexes, in which higher values represent less pro-immigrant attitudes. I only pre-registered hypotheses about these four outcomes (trust in RN, support for mayor, and two indexes), but also tested exploratory hypotheses described below.

My estimands are the Average Treatment Effects (ATEs) of the two treatments as well as their interaction effect. For each outcome and each treatment (or interaction), I estimate a simple OLS regression of the outcome on the treatment and a model with demographic controls.[35] The two indexes are averages of eight items each, given equal weight (on a seven-point scale), with every item first standardized by its mean and standard deviation in the control group. Because I test four pre-registered outcomes across two treatments and their interaction, I adjust for multiple testing using the Benjamini-Hochberg procedure. I report both unadjusted and adjusted p-values.

J.3 Main Results

Figure J1 reports the ATE for four outcomes: 1) respondents' trust in the National Rally (pre-registered), 2) the level of support for their incumbent mayor (pre-registered), 3) whether

35. The covariates controlled for are: pre-treatment trust in Marine Le Pen, self-location on a left-right scale, change in the number of foreign-born residents in the respondent's municipality between 2011 and 2021, years of education, age, gender, income, and whether the respondent identifies as an immigrant themselves.

respondents select immigration as one of the top four topics they find concerning (not pre-registered), and 4) post-treatment belief about the recent trend in immigrant presence on a one-to-five scale, where one means "has decreased a lot" and five means "has increased a lot" (not pre-registered).

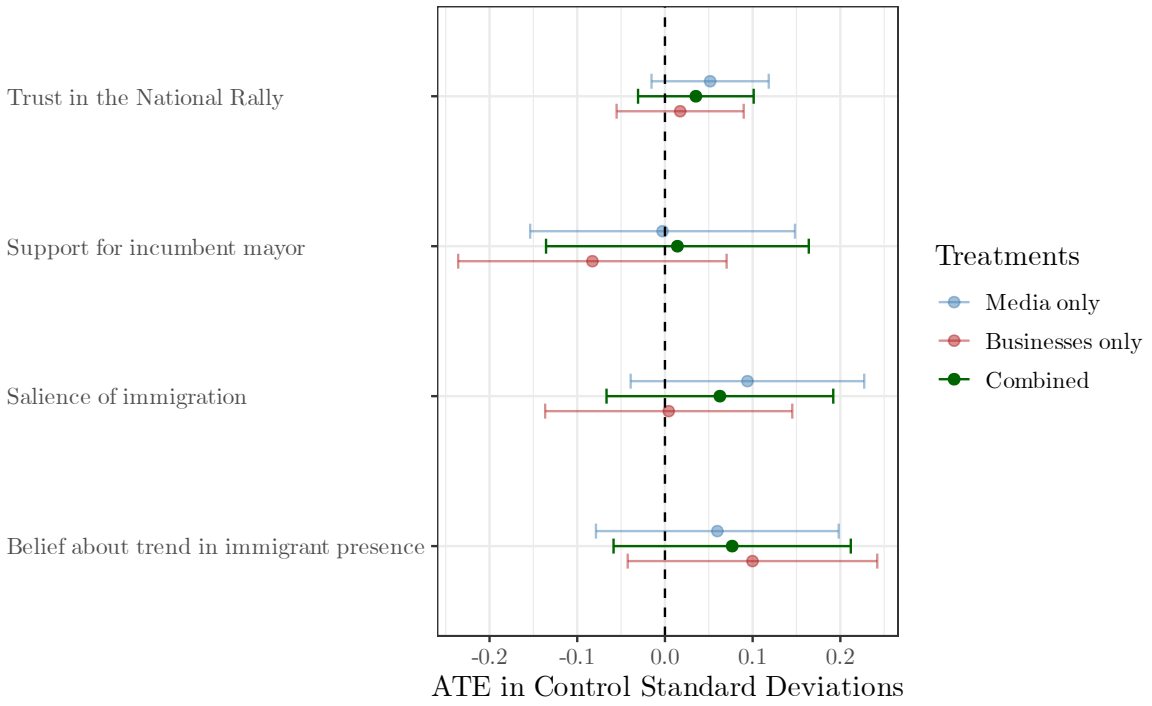


Figure J1: Main Experimental Estimates

Note: This figure displays the average treatment effect (ATE) for the two treatments (media and businesses) and for their combined effect, for four different outcomes. The bars display 95% confidence intervals calculated from heteroskedasticity-robust standard errors. Each coefficient represents a difference from the control group, which is represented by the zero line.

Table J1 reports all four pre-registered outcomes numerically. The businesses treatment, the manipulation closest to the observational design, is not statistically distinguishable from zero for any pre-registered outcome. The media treatment significantly raises the business-attitudes index by 0.24 control-group standard deviations. This is the only pre-registered test that passes multiple-testing adjustment. Its effects on the immigration index and on trust in the National Rally are positive and significant before multiple-comparison adjustment.

Among the four outcomes in Figure J1, no estimate survives multiple-testing adjustment. The absence of a businesses-treatment effect may be due to the weakness of the intervention: brief images and primes embedded in a short survey, as opposed to repeated exposure, may not shift strong priors and entrenched partisanship. Moreover, respondents in the treatment

Table J1: Pre-Registered Outcomes: Adjusted Treatment Effects

	Business idx.	Immigration idx.	Trust RN	Mayor
Businesses treatment (β_B)	0.055	−0.001	0.050	−0.132
	(0.051)	(0.051)	(0.086)	(0.159)
Media treatment (β_M)	0.175***	0.108**	0.163**	0.015
	(0.052)	(0.050)	(0.082)	(0.155)
Businesses × Media (β_{BM})	−0.044	−0.029	−0.141	0.168
	(0.072)	(0.069)	(0.114)	(0.220)
p_{BH}: Businesses	0.630	0.977	0.687	0.687
p_{BH}: Media	0.009	0.169	0.174	–
p_{BH}: Interaction	0.687	0.743	0.600	0.687
Lin covariates (HC2)	✓	✓	✓	✓
Num. obs.	1470	1470	1470	1470

Note: Adjusted (Lin 2013, fully interacted) OLS estimates with HC2 standard errors (in parentheses) on the N = 1,470 analyzed respondents. Columns are the four pre-registered outcomes; the two indexes are equal-weight averages of control-standardized items, and trust and mayoral support are 1–7 scales, with higher values denoting less pro-immigrant attitudes, more trust, or higher re-election probability. The p_{BH} rows report Benjamini–Hochberg-adjusted p-values across the eleven pre-registered contrasts (the media contrast for mayoral support was not pre-registered). For reference, the business (immigration) index has a control-group standard deviation of 0.72 (0.79), so the media coefficient of 0.175 corresponds to 0.24 control-group SD.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$ (unadjusted).

group may already have been aware of the existence of local immigrant businesses: on average, treated respondents indicated that they already knew 1.27 of the immigrant businesses presented to them. Furthermore, because all respondents are given information about true immigrant presence before treatment, the true causal effect of exposure to businesses on support for the RN may be reduced significantly if updating about immigrant presence is a mediator.

Though they are not consistently significant and should therefore not be interpreted as conclusive, the estimates are consistent with the backlash direction identified in the observational analysis. Respondents in both treatment conditions report marginally higher trust in the National Rally, though this coefficient is only significant for the media treatment, and not after multiple-testing adjustment. Similarly, estimates for the second outcome are consistent with respondents reporting lower support for their incumbent government. Because a majority of mayors come from mainstream parties (only a handful are RN members at the time of the survey), this may benefit the far right electorally at both local and national levels. The sign of the third outcome shows a non-significant marginal increase in the salience of immigration, and estimates for the last outcome indicate that respondents shown nearby immigrant

businesses report believing that local immigrant presence has increased more. Note that this is true even though all respondents were shown the same facts about immigrant presence before treatment. One possible interpretation is that even after receiving factual information, exposure to visual symbols of immigration distorts perceptions of immigrant presence. The coefficient on interacted treatments is in the same direction but not significant, and the business-only treatment estimate does not survive multiple-testing adjustment.

Visibility

The businesses treatment does not affect the pre-registered outcomes. The media treatment significantly worsens business-related attitudes (including after adjustment) and negatively affects general immigration attitudes, though this effect does not survive multiple-testing adjustment. Figure J2 decomposes these indexes into components related to identity and to culture, the traditional channels through which immigration attitudes are examined (Hainmueller and Hopkins 2014). These estimates are exploratory as they were not pre-registered and are not corrected for multiple testing. The businesses treatment is never significant, but its point estimates suggest worse attitudes toward immigrants and immigrant businesses. The media treatment increases respondents' tendency to perceive immigrants as an identity threat and as less integrated in French society, and to prefer typically French businesses.

Economic Mechanisms

Figure J3 reports experimental estimates related to individuals' economic perceptions. As with the cultural items, these tests are exploratory since they were not pre-registered. Respondents exposed to the media treatment agree more with statements characterizing immigrant-associated businesses as an economic threat. In particular, treated respondents are more likely to believe that immigrant businesses pose a risk to other businesses and to the value of real estate. This is true despite the fact that the far-right politician quoted in the media treatment article does not characterize immigrant businesses as an economic threat but as an identity threat. Here again, the businesses treatment alone does not shift attitudes significantly. However, the direction of the point estimates suggests that priming about local immigrant-owned

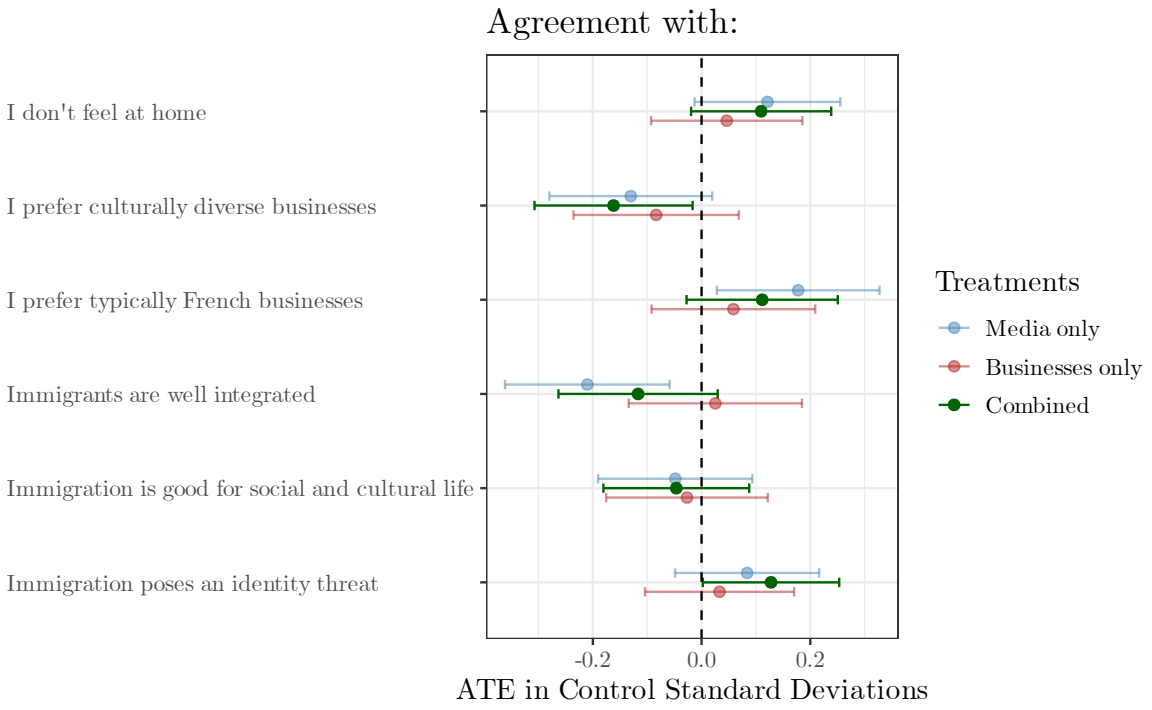


Figure J2: Treatment Effect on Cultural Attitudes

Note: This figure displays the ATE for individual index components relevant to the symbolic mechanisms. The bars display 95% confidence intervals calculated from heteroskedasticity-robust standard errors. Each coefficient represents a difference from the control group, which is represented by the zero line.

businesses may also have a similar negative effect on beliefs about their economic impact.

J.4 Differential Attrition

Importantly, I find evidence of differential attrition by treatment arm. Table J2 displays the reduction in sample size from recruitment to the final analysis sample.

Out of the 2,361 initial respondents randomly assigned to the four treatment conditions, 1,499 completed the survey and 1,470 passed the pre-registered minimum-duration filter. Completion is not affected by the businesses treatment, but it is affected by the media treatment: 58% of respondents exposed to the control article complete the survey, compared to 69% of respondents exposed to the treatment article. The difference is statistically significant at the 1% level (see Table J3). The timing of the differential attrition does not seem to interact with the outcome questions: instead, the completion rate is comparable across the two arms until the questions that follow the businesses-treatment page. I also do not find evidence that attrition is correlated with pre-treatment characteristics.

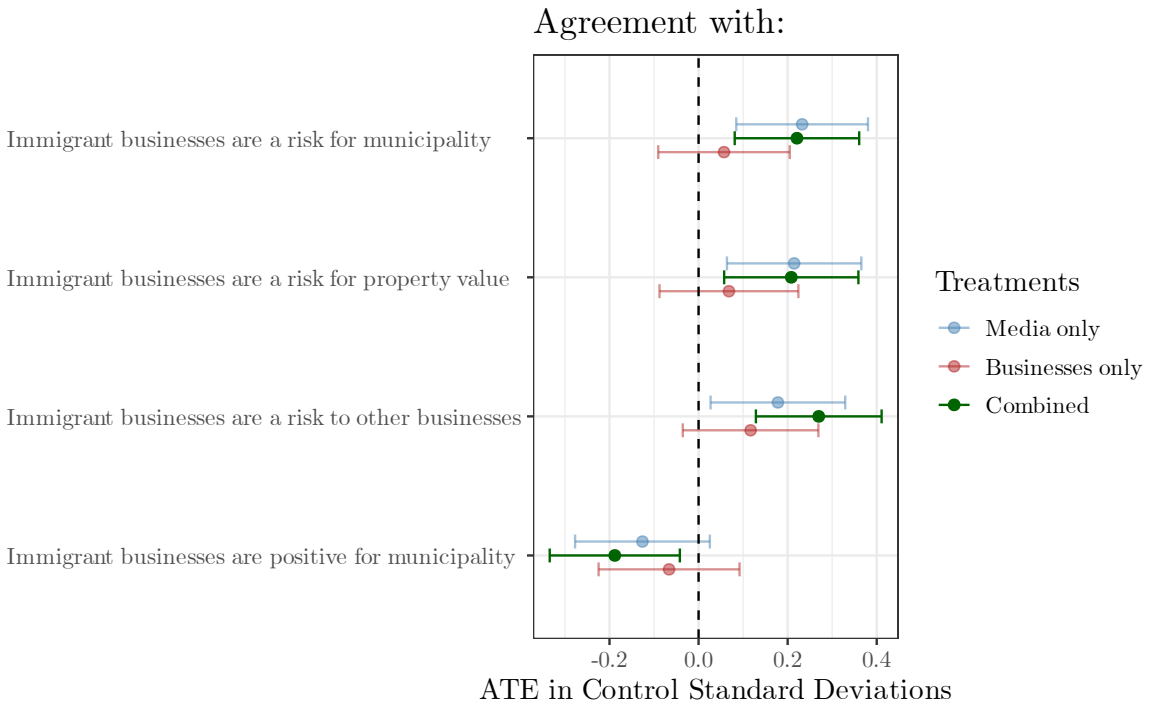


Figure J3: Treatment Effect on Economic Attitudes

Note: This figure displays the ATE for index components relevant to the economic mechanisms. The bars display 95% confidence intervals calculated from heteroskedasticity-robust standard errors. Each coefficient represents a difference from the control group, which is represented by the zero line.

Table J2: Participant Flow by Arm

		Media treatment		Businesses treatment	
Stage	All	Control	Treated	Control	Treated
Raw responses	2609	—	—	—	—
Genuine respondents (real fielding window)	2489	—	—	—	—
Randomized to a condition	2361	1184	1177	1184	1177
Completed the survey	1499	685	814	746	753
Analyzed (duration ≥ 300 s)	1470	671	799	730	740

Note: "Genuine respondents" excludes an on-platform test account and responses outside the two fielding dates (12 and 14 August 2025). Every retained response is an anonymous record, so there are no survey previews or test responses. Arm columns are shown from randomization onward. The 128 respondents lost between the second and third rows were not randomized: most left before reaching treatment assignment, while a handful reached the end through a branch that bypassed the compulsory municipality question and randomization. The larger drop from randomization to completion (2,361 to 1,499) is survey non-completion, i.e. attrition, which is analyzed in the text below; it is unrelated to the municipality question.

Because the media treatment predicts completion, I evaluate the robustness of its effect on business attitudes to attrition. I construct Lee (2009) bounds by trimming the treatment arm that saw a larger completion rate to equalize completion between the two arms. I find bounds running from −0.13 to 0.53 control-group standard deviations. Sharp bounds for the one effect whose point estimate was statistically different from zero therefore include zero,

Table J3: Attrition by Arm

	Completed (0/1)
Intercept (control cell)	0.572***
	(0.020)
Media treatment	0.117***
	(0.028)
Businesses treatment	0.013
	(0.029)
Businesses × Media	−0.007
	(0.039)
Num. obs.	2361

Note: Linear-probability (OLS, HC2) model of a survey-completion indicator on the two treatments and their interaction, estimated on the 2,361 randomized respondents. The intercept is the completion rate in the pure control cell (57.2%). Completion by cell is 57.2% and 58.5% in the two control-article cells and 68.9% and 69.4% in the two treated-article cells.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

consistent with the other experimental results. I therefore read the experiment as suggestive only: the businesses treatment is null for every pre-registered outcome, and I cannot exclude the possibility that the media treatment's effect, even though it is pre-registered and survived multiple-comparison adjustment, is caused by compositional differences due to differential attrition.

J.5 Newspaper Articles (Translations)

In Perpignan, the bakery-pastry shop is leaving the Halles Vauban for good: find out what concept will replace it

Published on November 5th, 2024 at 5:48 PM

By Diane Sabouraud

The Halles Vauban pastry shop ceased all operations on October 31st. Within its walls, a brasserie project is being spearheaded by two young Catalan entrepreneurs.

The Halles pastry shop has closed its doors for good. It's the most visible brand at the Halles Vauban, as it was the first to introduce the twenty or so corners located in the former eponymous passageway. It's therefore difficult to walk past without noticing that the stalls have been emptied of their delicacies, the breads from their baskets, and the coffees from their cups, which still sit proudly on the back counter. However, the establishment ceased all operations on the evening of Thursday, October 31, 2024. "They've been trying to sell for months, as business hasn't been going as planned," they say at the Halles Vauban.

But such a location obviously doesn't stay on the market for long. According to information from L'Indépendant, a buyer is already in the running. While they wish to remain discreet while the administrative process is finalized, two young entrepreneurs have launched procedures to take over the business. But they will completely change the nature of the place. Out with the pastry shop, in with a brasserie. The project requires major renovations to redesign the premises. Ultimately, 60 to 70 seats are expected inside and around 40 on the terrace.

...

A new bakery coming soon?

We'll have to wait a while before discovering this new culinary offering in town. While the transaction is finalized and the renovations are completed, the opening is not expected to take place before the end of the year.

Figure J4: Control Group Article (English Translation)

Note: Source: Sabouraud (2024)

Perpignan City Council: The establishment of "halal restaurants" on Rue de la Cloche-d'Or raises eyebrows among Louis Aliot and the opposition

Published on June 22nd, 2022 at 6:52 PM, updated on June 23rd, 2022 at 10:34 AM

By Diane Sabouraud

During the city council meeting this Wednesday, June 22, Louis Aliot returned to the issue of the purchase of the eleven empty businesses on Rue de la Cloche d'Or. He asserted, among other things, that the buyers were companies specializing in halal catering. And he protested.

...

In the Perpignan assembly, the RN mayor, Louis Aliot, raised the issue before beginning the agenda. For the record, lindependant.fr revealed the day before that the eleven businesses purchased by entrepreneur Abi-Nader Chukri in October 2020 to create a gourmet street had been resold. The buyer is believed to be a group of investors, represented by a spokesperson. "This is a rather serious matter," Louis Aliot announced solemnly. "Our services have received a declaration of intent to sell (DIA) concerning these eleven businesses. We met with the selling owner this morning, who confirmed our concerns. He plans to sell to companies that share the common goal of providing halal restaurants."

...

"This also raises a question of overall identity."

The RN mayor insists that this plan for halal restaurants is in "total opposition to our plan for the city center, which is to make it a high-quality commercial hub. And this also raises a question of overall identity."

Two proposals were submitted to the city council: buying back the businesses or pre-empting them. On behalf of the opposition LR party, Pierre Parrat advocated for pre-emption, assuring that "given the state of the shops, it won't cost you a lot. And whatever the cost, you must act. We share your concerns and, in this regard, we will be fully in line with the mayor's decision."

Figure J5: Treatment Group Article (English Translation)

Note: Source: Sabouraud (2022)

J.6 Examples of Businesses Shown in the Second Treatment



Figure J6: Qualtrics Screenshot: Example of Business Shown in the Control Condition

Note: The top panel displays a French restaurant, with the following categories: "caterer," "barbecue restaurant," "beer shop," "pastry shop," "cocktail bar," "French restaurant," "burger restaurant," and "wine bar." It also shows its address. The bottom Qualtrics question asks the respondents whether they know the business. The options are "Yes," "No," "I am not sure," and "This business does not exist any more."

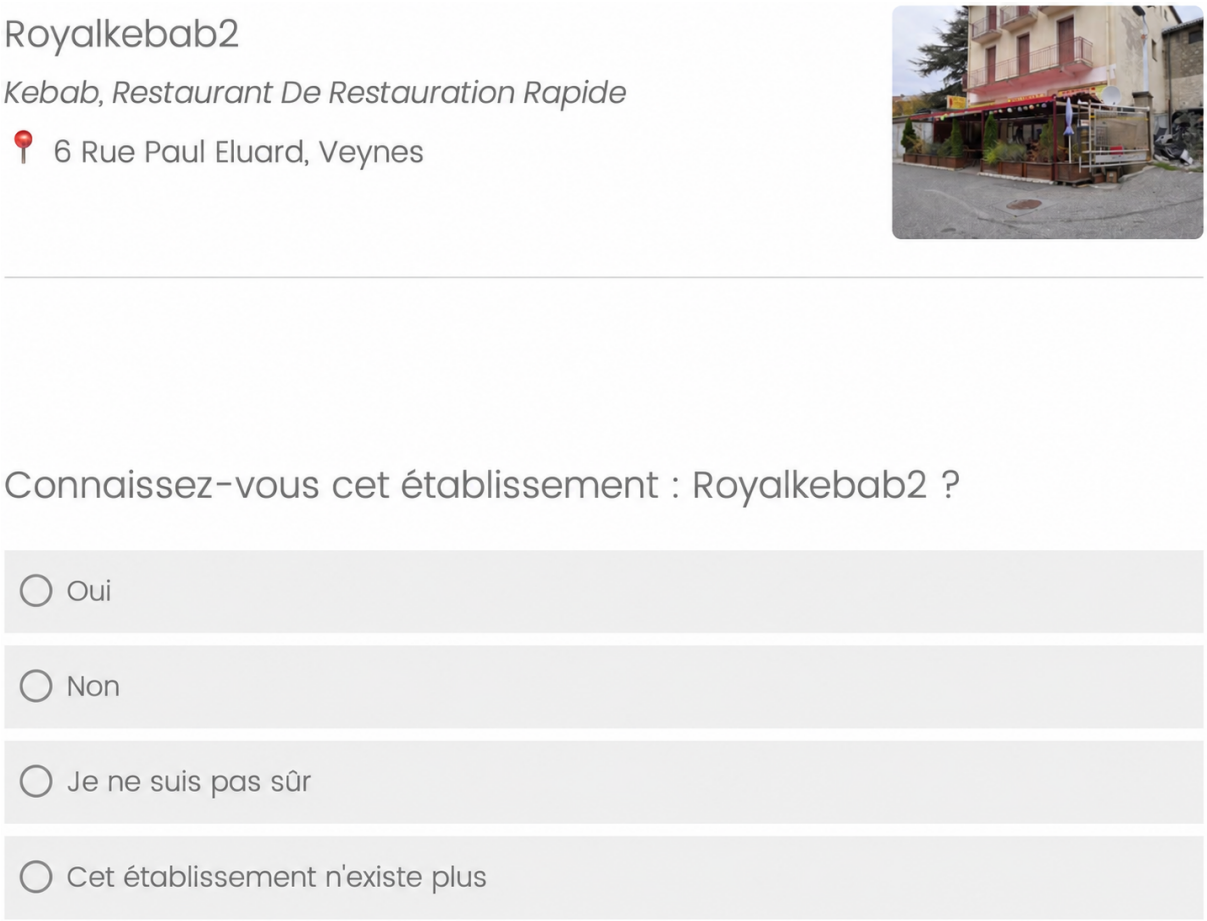


Figure J7: Qualtrics Screenshot: Example of Business Shown in the Treatment Condition

Note: The top panel displays a kebab shop, with the "fast food" category, and its address. The bottom Qualtrics question asks the respondents whether they know the business. The options are "Yes," "No," "I am not sure," and "This business does not exist any more."